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Report No: PAD4142

INTERNATIONAL BANK FOR RECONSTRUCTION AND DEVELOPMENT

PROGRAM APPRAISAL DOCUMENT

ON A

PROPOSED LOAN

IN THE AMOUNT OF US\$300 MILLION

TO THE

ARGENTINE REPUBLIC

FOR THE

BUENOS AIRES WATER SUPPLY AND SANITATION WITH A FOCUS ON VULNERABLE AREAS PROGRAM

(PROGRAMA DE AGUA Y SANEAMIENTO CON ATENCIÓN A ZONAS VULNERABLES)

February 3, 2021

Water Global Practice
Latin America and the Caribbean Region

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CURRENCY EQUIVALENTS

(Exchange Rate Effective January 20, 2021)

Currency Unit = Argentine Peso (ARS)

ARS 1 = US\$ 0.012

US\$1 = ARS 86.35

FISCAL YEAR

January 1 – December 31

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ABBREVIATIONS AND ACRONYMS

ABC	Areas of Low Consolidation (<i>Áreas de Baja Consolidación</i>)
ACGs	Anti-corruption Guidelines
AFERAS	Federal Association of Water and Sanitation Regulatory Agencies (<i>Asociación Federal de Entes Reguladores de Agua y Saneamiento</i>)
AGN	Supreme Audit Institution (<i>Auditoría General de la Nación</i>)
A+T	Water and Jobs (<i>Agua más Trabajo</i>)
ARS	Argentine Pesos
AySA	Agua y Saneamientos Argentinos S.A.
APLA	National Planning Agency (<i>Agencia de Planificación</i>)
BABE	Burst and Background Estimates
C+T	Sewage and Jobs (<i>Cloacas más Trabajo</i>)
CABA	Autonomous City of Buenos Aires (<i>Ciudad Autónoma de Buenos Aires</i>)
CAF	Development Bank of Latin America
CAIU	Customer Care Center (<i>Centro de Atención Integral al Usuario</i>)
COVID-19	Coronavirus Disease 2019
CPF	Country Partnership Framework
CRM	Citizen Engagement and Customer Relationship Management System
DiGePPSE	General Directorate for Special and Sectoral Programs and Projects (<i>Dirección General de Programas y Proyectos Sectoriales y Especiales</i>),
DLI	Disbursement-Linked Indicator
DLR	Disbursement-Linked Result
EAHU	Annual Urban Household Survey (<i>Encuesta Anual de Hogares Urbanos</i>)
EIA	Environmental Impact Assessment
EIAP	Efficiency Improvement Action Plan
EIRR	Economic Internal Rate of Return
EPH	Permanent Household Survey (<i>Encuesta Permanente de Hogares</i>)
ERAS	Water and Sanitation Regulator (<i>Ente Regulador de Agua y Saneamiento</i>)
e-SIDF	Integrated Financial Management Information System (<i>Sistema Integrado de Información Financiera</i>)
ESSA	Environmental and Social System Assessment
ETAS	Environmental and Social Technical Specifications (<i>Especificaciones Técnicas Ambientales y Sociales</i>)
FM	Financial Management
GBA	Greater Buenos Aires (<i>Gran Buenos Aires</i>)
GDP	Gross Domestic Product
GEMS	Geo-Enabling initiative for Monitoring and Supervision
GHG	Greenhouse Gas
GoA	Government of Argentina
GP	Global Practice
GRS	Grievance Redress Service
HACCP	Hazard Analysis and Critical Control Points
HH	Household
IADB	Interamerican Development Bank
IAU	Internal Audit Unit
INDEC	National Institute of Statistics and Census (<i>Instituto Nacional de Estadísticas y Censo</i>)
INT	Institutional Integrity Vice-Presidency
IPF	Investment Project Financing
KPI	Key Performance Indicator
LAC	Latin America and the Caribbean
LMICs	Low- and middle-income countries



M&E	Monitoring and Evaluation
NDC	Nationally Determined Contribution
NPV	Net Present Value
NRW	Non-revenue water
OHS	Occupational Health and Safety
MOPW	Ministry of Public Works (<i>Ministerio de Obras Públicas</i>)
O&M	Operation and Maintenance
OM	Operations Manual
ONP	National Budget Office (<i>Oficina Nacional de Presupuesto</i>)
OPRC	Operations Procurement Review Committee
PAP	Program Action Plan
PBA	Province of Buenos Aires
PCU	Program Coordination Unit
PDO	Program Development Objectives
PforR	Program for Results
PMOEM	Service Improvement, Expansion, Operation and Maintenance Program (<i>Plan de Mejoras, Operación, Expansión y Mantenimiento de los Servicios</i>)
PNAPyS	National Plan for Drinking Water and Sanitation (<i>Plan Nacional de Agua Potable y Saneamiento</i>)
PPE	Prevention and Emergency Plan (<i>Plan de Prevención y Emergencias</i>)
PQRS	Digital user- question-complaint-claim feedback platform (<i>Peticiones, Quejas, Reclamos y Sugerencias</i>)
RENAP	National Registry of Vulnerable Neighborhoods (<i>Registro Nacional de Barrios Populares</i>)
SCD	Systematic Country Diagnostic
SDGs	Sustainable Development Goals
	General Syndicate of the Nation (<i>Sindicatura General de la Nación</i>)
SIPH	Secretariat of Infrastructure and Water Policy (<i>Secretaría de Infraestructura y Política Hídrica</i>)
SSO	AySA's Monitoring and Evaluation System (<i>Sistema de Seguimiento de Obras</i>)
ToRs	Terms of Reference
UPEX	System of Financial Administration and Control for Agencies Executing Foreign Loans (<i>Sistema de Administración y Control Financiero para las Unidades Ejecutoras de Préstamos Externos</i>)
UREM	Emergent Urbanizations (<i>Urbanizaciones Emergentes</i>)
VAT	Value Added Tax
VSD	Variable Speed Drive
WA	Withdrawal Application
WASH	Water, Sanitation and Hygiene
WBE	Wastewater-Based Epidemiology
WBG	World Bank Group
WSS	Water Supply and Sanitation
WTP	Water Treatment Plant
WWTP	Wastewater Treatment Plant



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DATASHEET

BASIC INFORMATION

Country(ies)	Project Name	
Argentina	BUENOS AIRES WATER SUPPLY AND SANITATION WITH A FOCUS ON VULNERABLE AREAS PROGRAM	
Project ID	Financing Instrument	Does this operation have an IPF component?
P172689	Program-for-Results Financing	No

Financing & Implementation Modalities

☐ Multiphase Programmatic Approach (MPA)	☐ Fragile State(s)
☐ Contingent Emergency Response Component (CERC)	☐ Fragile within a non-fragile Country
☐ Small State(s)	☐ Conflict
☐ Alternate Procurement Arrangements (APA)	☐ Responding to Natural or Man-made Disaster
☐ Hands-on Enhanced Implementation Support (HEIS)	
Expected Project Approval Date	Expected Closing Date
25-Feb-2021	26-May-2026

Bank/IFC Collaboration

No

Proposed Program Development Objective(s)

The Program Development Objectives are to increase access to safely managed water and sanitation services with a focus on vulnerable areas of Buenos Aires, and improve AySA's efficiency and capacity to respond to water and sanitation emergency needs

Organizations

Borrower :	Argentine Republic
Implementing Agency :	Agua y Saneamientos Argentinos S.A. (AySA)
Contact:	Normando Birolo
Title:	Director



Telephone No: 63195865
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COST & FINANCING

SUMMARY

Government program Cost	4,865.72
Total Operation Cost	576.61
Total Program Cost	575.86
Other Costs	0.75
Total Financing	576.61
Financing Gap	0.00

Financing (USD Millions)

Counterpart Funding	227.26
Borrower/Recipient	227.26
International Bank for Reconstruction and Development (IBRD)	300.00
Cofinancing - Other Sources (IFIs, Bilaterals, Foundations)	49.34
Inter-American Development Bank	49.34

Expected Disbursements (USD Millions)

Fiscal Year	2021	2022	2023	2024	2025	2026
Absolute	34.68	61.32	96.59	65.24	41.37	0.77
Cumulative	34.68	96.01	192.60	257.84	299.22	300.00

INSTITUTIONAL DATA



Practice Area (Lead)

Water

Contributing Practice Areas

Climate Change and Disaster Screening

This operation has been screened for short and long-term climate change and disaster risks

SYSTEMATIC OPERATIONS RISK-RATING TOOL (SORT)

Risk Category	Rating
1. Political and Governance	Moderate
2. Macroeconomic	High
3. Sector Strategies and Policies	Moderate
4. Technical Design of Project or Program	Moderate
5. Institutional Capacity for Implementation and Sustainability	Moderate
6. Fiduciary	Substantial
7. Environment and Social	Substantial
8. Stakeholders	Moderate
9. Other	
10. Overall	Substantial

COMPLIANCE

Policy

Does the program depart from the CPF in content or in other significant respects?

☐ Yes ☒ No

Does the program require any waivers of Bank policies?

☐ Yes ☒ No

Legal Operational Policies

	Triggered
Projects on International Waterways OP/BP 7.50	Yes
Projects in Disputed Areas OP/BP 7.60	No



Legal Covenants

Sections and Description

Section 3.01. The Borrower, through the MOP, shall cause the Program to be carried out by AySA in cooperation with Selected Municipalities, and with the assistance of APLA, and ERAS, all in accordance with the provisions of Article V of the General Conditions and Schedule 2 to the Loan Agreement.

Sections and Description

Schedule 2, Section I.A.1. The Borrower, through the MOP, shall cause AySA to maintain, throughout Program implementation, a Program Coordination Unit within its General Administrative Directorate (the “AySA CU”), with a structure, functions, and responsibilities acceptable to the Bank, as set forth in the Operations Manual.

Sections and Description

Schedule 2, Section I.B.1. To facilitate the carrying out of the Program, the Borrower, through the MOP, shall make the proceeds of the Loan available to AySA on a grant basis under a transfer agreement (“Transfer Agreement”) between the Borrower and AySA, under terms and conditions approved by the Bank.

Sections and Description

Schedule 2, Section I.C.1. Prior to carrying out activities with a Selected Municipality under Results Area 1 of the Program, the Borrower shall cause AySA to make part of the proceeds of the Loan available to the Selected Municipality on a grant basis under an agreement (“Municipality Agreement”) between AySA and each Selected Municipality, under terms and conditions approved by the Bank.

Sections and Description

Schedule 2, Section I.F.

1. The Borrower shall cause AySA to appoint a verification agent (“Verification Agent”) for the Program that is acceptable to the Bank, in accordance with terms of reference acceptable to the Bank.
2. The Borrower shall cause AySA to ensure that the verification agent referred to in the preceding paragraph shall: (a) verify the data and other evidence supporting the achievement of one or more DLRs and recommend corresponding payments to be made, as applicable, in accordance with the Verification Protocol agreed with the Bank; and (b) submit to AySA and the Borrower the corresponding verification reports in a timely manner and in form and substance satisfactory to the Bank.

Conditions

Type

Effectiveness

Description

Section 5.01. The Transfer Agreement has been executed and delivered by the



	Borrower, through the MOP, and AySA and any conditions precedent to its effectiveness have been fulfilled
Type Disbursement	Description Section IV.B.1 (a) of Schedule 2. No withdrawal shall be made on the basis of DLRs achieved prior to the Signature Date, except that withdrawals up to an aggregate amount not to exceed \$15,000,000 may be made on the basis of DLRs achieved prior to this date but on or after July 16, 2020 under Category 5(a)
Type Disbursement	Description Section IV.B.1 (b) of Schedule 2. No withdrawal shall be made for any DLR under Categories (1), (2), (3), (4), (5), (6), or (7), until and unless the Borrower has furnished evidence satisfactory to the Bank that said DLR has been achieved.
Type Disbursement	Description Section IV.B.2 of Schedule 2. The Borrower may withdraw: (i) an amount not to exceed \$75,000,000 as an advance; provided, however, that if the DLRs in the opinion of the Bank, are not achieved (or only partially achieved) by the Closing Date, the Borrower shall refund such advance (or portion of such advance as determined by the Bank in accordance with the provisions of paragraph 3 of this Part B) to the Bank promptly upon notice thereof by the Bank. Except as otherwise agreed with the Borrower, the Bank shall cancel the amount so refunded. Any further withdrawals requested as an advance under any Category shall be permitted only on such terms and conditions as the Bank shall specify by notice to the Borrower.
Type Disbursement	Description Section IV.B.3 of Schedule 2. In the case of Scalable DLRs, the Bank may, by notice to the Borrower: (a) authorize the withdrawal of such lesser amount of the unwithdrawn proceeds of the Loan then allocated to said Category which, in the opinion of the Bank, corresponds to the extent of achievement of said DLR during any given Verification Period, said lesser amount to be calculated in accordance with the formula set out in the fourth column of the table in Section IV.A.2. of Schedule 2 to this Agreement and further detailed in the Verification Protocol; and/or (b) cancel all or a portion of the proceeds of the Loan then allocated to a DLR that is not fully achieved by the Closing Date, as the case may be.



I. STRATEGIC CONTEXT

A. Country Context

1. **The COVID-19 outbreak has hit Argentina's economy at a moment of significant macroeconomic imbalances and social stress.** Following a two-year recession, high inflation, and lack of access to capital markets, the strict lockdown to contain the spread of the pandemic triggered the sharpest Gross Domestic Product (GDP) decline ever recorded, in March-April of 2020. The gradual easing of confinement measures has allowed economic activity to slowly pick up pace since the third quarter but has also set off a rapid rise in COVID-19 cases. The implementation of a fiscal stimulus package – worth 3.5 percent of GDP¹ - coupled with an abrupt decline in revenues resulted in a central government deficit of nearly 7 percent of GDP in 2020², the largest in more than 30 years. Lack of market access mandated a full monetization of the deficit, which is exacerbating macroeconomic imbalances. The gap between official and parallel foreign exchange rates is at historic high levels and the Central Bank has rapidly lost reserves as it attempted to maintain the official exchange rate stable, despite a trade surplus and tight currency controls.

2. **It is estimated that the economy has declined by over 10 percent in 2020³ and the unprecedented economic contraction has had a severe impact on employment and poverty.** Economic recovery is gradually picking up as containment measures are progressively lifted, building on the economy's ample idle capacity. However, high uncertainty, price and capital controls limit investment growth, while the imperative to reduce the large fiscal deficit will strongly curtail demand stimulus. As a result, only a partial rebound is expected in 2021, and the economy is not projected to reach 2019 GDP levels before 2023. Labor market statistics point to a sharp 9 percentage points decline in formal employment between the first and the second quarter to 33.4 percent, with a partial recovery in the third quarter to 37.4 percent. Unemployment has increased only by 2.5 percentage points in the second quarter as an important share of workers ceased looking for jobs and receded to 11.7 in the third quarter⁴.

3. **As a result of higher inactivity rates and decreasing purchasing power, Argentina's poverty rate increased from 35 to 41 percent in the first half of FY2020, with 11.7 million people now living below the poverty line and 3 million in extreme poverty.**⁵ More than half of Argentina's poor live in Greater Buenos Aires (*Gran Buenos Aires, GBA*), by far the greatest concentration of urban and peri-urban poverty in the Southern Cone of Latin America.⁶ With 15.4 million inhabitants, GBA is one of Latin America's urban giants, accounting for one third of Argentina's population and nearly half of its GDP.⁷ Although urbanization has contributed to economic growth and improved living standards for millions of people, it has not always been accompanied by adequate planning and investment in infrastructure and services, leading to "pockets of poverty." At the close of second semester 2019, the percentage of people living in poverty and indigence in GBA was already considerable, standing at 35.2 and 9.3 percent, respectively. The outbreak of the COVID-19 pandemic raised these figures to 41.6 percent and 11.7 percent.⁸

¹ Estimates of the Congressional Budget Office: <https://www.opc.gob.ar/covid-19/impacto-financiero-del-covid-19-al-5-de-octubre-2020/>

² Data from Ministry of Finance: <https://www.argentina.gob.ar/noticias/resultado-fiscal-diciembre-2020>

³ According to the latest World Bank projections: <http://pubdocs.worldbank.org/en/983991599838744787/Global-Economic-Prospects-January-2021-Analysis-LAC.pdf>

⁴ INDEC, "Mercado de Trabajo. Tasas e indicadores socioeconómicos, Tercer Trimestre 2020". December

⁵ INDEC, "Incidencia de la Pobreza y la Indigencia en 31 conglomerados urbanos, Primer Semestre 2020." September 2020.

⁶ GBA comprises the capital of Buenos Aires and 24 municipalities of the Province of Buenos Aires: Avellaneda, General San Martín, Hurlingham, Ituzaingó, José C. Paz, Lanús, Lomas de Zamora, Malvinas Argentinas, Morón, Quilmes, San Isidro, Tres de Febrero, Vicente López, Almirante Brown, Berazategui, Esteban Echeverría, Ezeiza, Florencio Varela, La Matanza, Merlo, San Fernando, San Miguel, and Tigre.

⁷ Source: National Institute of Statistics and Census (*Instituto Nacional de Estadísticas y Censo, INDEC*), 2010 census (the latest official data).

⁸ INDEC, "Incidencia de la Pobreza y la Indigencia en 31 conglomerados urbanos, Primer Semestre 2020." September 2020.



4. **In GBA, which is ground zero for the COVID-19 pandemic in Argentina, low-income neighborhoods are being hit the hardest by the sanitary and economic crises.** There are three main factors driving the high vulnerability of GBA's poor: crowded living conditions; high dependence on informal economic activities, which undermines compliance with lockdown policies; and lack of access to safely managed water supply and sanitation (WSS), all of which are key to preventing virus transmission. Over 30 percent of these neighborhoods are in flood-prone areas, rendering existing informal and often unsafe WSS systems even more prone to failure.⁹ Furthermore, a methodology prepared by the World Bank to predict contagion hotspots for COVID-19 in cities uses access to WSS and population density as the main identification criteria¹⁰.

B. Sectoral and Institutional Context

Water Supply and Sanitation Sector Performance Challenges at the National Level

5. **Argentina faces a considerable gap in access to piped water, sewerage and wastewater treatment.** An estimated 7.5 million inhabitants (17 percent of the population) lack access to safely managed piped water systems, and 21 million (48 percent) are not connected to sewerage networks. The combined economic impact of these service gaps is estimated at 1.32 percent of GDP,¹¹ including harm to health and education — the essentials for building Argentina's human capital. Furthermore, only 11 percent of total wastewater generated is treated¹², which poses serious environmental and health impacts, especially on populations living near polluted urban waterways.

Table 1: Access to piped water and sewerage services in Argentina (2018)

Service	Urban (90 percent or 40.4 million people)	Rural (10 percent or 4.1 million people)	Total
Water coverage	86.3 percent (34.9 million)	52.8 percent (2.1 million)	83.3 percent (37.0 million)
Sewerage coverage	57.3 percent (23.1 million)	8.5 percent (345,000)	52.8 percent (23.4 million)

Source: National Institute of Statistics and Census data and projections.

6. **Although the main provincial WSS operators provide acceptable service levels in terms of continuity and water quality, their efficiency is low compared with regional peers.**¹³ Critical indicators are the low level of micro-metering (25 percent compared to 80 percent average for main regional utilities) and the high level of daily per capita water production (560 liters compared to a regional average of 330 liters and an international reference value of 200 liters). Without micro-metering, real volumetric tariffs cannot be applied, making it difficult to link operating costs and tariff revenues and to control physical and commercial losses (estimated at 45 percent compared to 36 percent average for main regional utilities). The financial sustainability of Argentina's main WSS service providers is also low. On average, tariff revenues cover 83 percent⁵ of operation and maintenance (O&M) costs, with the rest provided as subsidies by the provinces and, in the case of the GBA operator, *Agua y Saneamientos Argentinos S.A.* (AySA), by the National Government.

7. **Under Argentina's federal system of government, provinces retain legal jurisdiction and responsibility for the management of water resources and provision of water services.** However, the National Government is responsible for guaranteeing the fulfillment of national objectives related to socio-economic development and environmental protection — where water plays a key role. The national objectives laid out in the first National Plan for

⁹ Results of Fathom flood modeling at 90 m resolution, using location of Buenos Aires peri-urban slums from the National Registry of Vulnerable Neighborhoods (*Registro Nacional de Barrios Populares, RENABAP*) registry.

¹⁰ [Cities, crowding and the coronavirus: Predicting contagion risk hotspots \(2020\)](#).

¹¹ Economic analysis using costs avoided, consumer surplus and hedonic prices, conducted from service coverage levels estimated from INDEC, 2010, updated to 2018 by the World Bank team.

¹² Data derived from the National Water and Sanitation Plan (2017), and projections from the *INDEC*, Permanent Household Survey (*Encuesta Permanente de Hogares, EPH*) and Annual Urban Household Survey (*Encuesta Anual de Hogares Urbanos, EAHU*)

¹³ Regional Benchmarking Report. ERAS (2017). Data from 15 utilities in Colombia, Brazil, Chile, Peru, Panama, Costa Rica, Ecuador and Uruguay.



Drinking Water and Sanitation (*Plan Nacional de Agua Potable y Saneamiento, PNAPyS*) set ambitious targets to secure universal access to water and 75 percent access to sewerage service throughout Argentina's urban areas by 2030. Although the PNAPyS represents a commitment to achieve the Sustainable Development Goals (SDGs), lack of federal funding, operational inefficiencies of service providers, and the insufficient enforcement of regulations undermine its implementation.

Sectoral Context in GBA

8. **AySA is the concessionaire responsible for providing WSS services to most of GBA¹⁴.** AySA is one of the largest WSS utilities in the world, and the largest in the country, with 14.4 million residents (96 percent of the population in GBA) living in its service area. It is estimated that 66 percent of the population without access to piped water in Argentina (3.7 million) and 39 percent without access to sewerage (6.8 million), live in this area. The concession area is divided into two clearly differentiated zones, the Autonomous City of Buenos Aires – with high levels of piped water and sewerage coverage - and the municipalities and peri-urban communities of the Province of Buenos Aires (PBA) with lower service coverage (as described in Table 1).

9. **The regulatory and institutional frameworks at AySA's level are well established, although they still present some challenges.** AySA is 90 percent owned by the National Government through the Ministry of Public Works, which is responsible for tariff setting, and 10 percent by its 7,800 employees through the "Participatory Ownership Program". The WSS Regulator (*Ente Regulador de Agua y Saneamiento, ERAS*) is an experienced autonomous public body created in 1993 and as such, it has gained important experience in asset management, service quality monitoring and development of tariff models. Moreover, ERAS has incorporated into its decision-making process two committees representing municipalities in its concession area and civil society organizations representing the users. ERAS is also responsible for monitoring AySA's compliance with its legal obligations under the concession contract. The National Planning Agency (*Agencia de Planificación, APLA*) is an autonomous public body in charge of reviewing and coordinating investments in expansion and service improvement carried out by AySA. The main challenges facing the regulation of WSS service in GBA relate to the need to adapt regulation to the service delivery model of a state-owned company, strengthen capacities in these entities, and establish a clear tariff setting methodology.

AySA's Specific Challenges

10. **As AySA has expanded coverage to underserved municipalities, its average access levels have decreased proportionately.** Overall access levels are 75 percent for water and 53 percent for sewerage. There are, however, considerable differences in coverage among municipalities. For instance, while 97.2 percent of residents in the capital have access to water and 96.2 percent to sewerage, rates in the PBA municipalities are below 20 percent. Many of these unserved residents live in fragile neighborhoods¹⁵ and the incorporation of nine new municipalities since 2016 has lowered AySA's average access rates. Whereas the 17 municipalities in AySA's original coverage area have connection rates of 80 percent for water and 49 percent for sewerage, the rates for municipalities included in AySA

¹⁴ Almirante Brown, Avellaneda, Escobar, Esteban Echeverría, Ezeiza, Florencio Varela, Hurlingham, Ituzaingó, José C. Paz, La Matanza, Lanús, Lomas de Zamora, Malvinas Argentinas, Merlo, Moreno, Morón, Pilar, Presidente Perón, Quilmes, San Fernando, San Isidro, San Martín, San Miguel, Tigre, Tres de Febrero, y Vicente López.

¹⁵ There are more than 1,000 fragile neighborhoods in AySA's service area, commonly referred to as informal settlements or slums. These are classified by AySA as Emergent Urbanizations (*Urbanizaciones Emergentes, UREM*), which include neighborhoods recognized under the National Registry of Vulnerable Neighborhoods (*Registro Nacional de Barrios Populares, RENAP*) (Decree 358/2017). These UREMs are characterized by varying degrees of land tenure, overcrowding, and overriding deficiencies in access to basic public services. Furthermore, some peripheral areas, identified as Areas of Low Consolidation (*Áreas de Baja Consolidación, ABC*) have also been identified as fragile neighborhoods, as they also present social vulnerabilities associated with low educational levels, high levels of dependency and low insertion in the formal labor market that lead to an overall socio-economic fragility.



since 2016, average 40 and 23 percent, respectively.¹⁶ The inclusion of the new municipalities also introduced operational challenges. As illustrated in Table 2, the evolution of coverage levels in the last 15 years has been limited.

Table 2: Evolution of AySA's service area coverage rates between 2006 and 2018.

Department	Population		Water Coverage (%)			Sewerage Coverage (%)		
	2010	2018	2006	2010	2018	2006	2010	2018
AySA's initial service area	10,390,843	11,160,191	74.9	82.8	84.7	54.0	56.4	61.8
Capital of Buenos Aires	3,028,481	3,068,043		97.3	97.2		93.9	96.2
17 initial PBA's municipalities	7,362,362	8,092,148		76.8	79.9		41.0	48.8
9 new PBA's municipalities	2,878,386	3,281,231		38.3	40.2		15.4	23.3
AySA's current service area	13,269,229	14,441,422		73.1	74.6		47.5	53.1

Sources: AySA Annual Report (2017) and water coverage by network and sewerage estimated by the World Bank team at the first quarter of 2018 based on INDEC, EPH and EAHU data.

11. **Since 2006, AySA has focused considerable attention on expanding WSS services in the peri-urban municipalities, however connectivity rates are low among poor households.** Between 2015 and 2019, AySA installed 68 secondary drinking water and 144 secondary sewage networks throughout 16 municipalities of its concession area, benefitting 166,880 and 635,089 residents, respectively.¹⁷ However, recent surveys carried out by the utility reveal that only 83.2 percent have effectively connected to the water networks. Roughly 50 percent of these connections are either incomplete or poorly installed.¹⁸ Survey findings further reveal that insufficient household income, time, or the existence of alternative, often unsafe water sources (such as wells) are the primary reasons for not connecting to the water network. Similarly, only 77.2 percent of households have connected to the sewerage network. Safely connecting these households to the networks would yield public health, human capital, and environmental benefits as indicated earlier, and increase AySA's operational and financial efficiency. Expanding WSS services would also prove timely and impactful given their critical role as a first line of defense in slowing down the transmission of COVID-19.

12. **Needs are particularly dire throughout fragile urban settlements of AySA's concession area.** According to the latest data, an estimated 384,810 households or 2.82 million people inhabit 982 *fragile* neighborhoods classified as "Emergent Urbanizations" (*Urbanizaciones Emergentes*, UREM).¹⁹ In particular, 57 percent of UREMs do not have access to drinking water and 80 percent lack access to sewerage.²⁰ In all, an estimated 1.54 million UREM inhabitants lack access to an official water supply, obligating them to seek out other coping mechanisms including receiving water from informal neighboring networks or collecting water from wells. Roughly 2.32 million do not have access to sewerage networks, and instead resort to precarious, informal connections or septic tanks, which are often poorly constructed and/or managed.²¹ The municipalities with the highest numbers of UREMs are La Matanza (151), Quilmes (82), Lomas de Zamora, (82), Autonomous city of Buenos Aires (76), Avellaneda (68) and San Martín (64).

13. **AySA has a strong track record of supporting "last mile connectivity" in poor and vulnerable households.**²² Since 2010, AySA has partnered with municipalities and municipal labor cooperatives in implementing its Water and

¹⁶ Service coverage levels estimated from INDEC, 2010, and updated to 2018 (first trimester) by the World Bank team.

¹⁷ Internal report on finished works: period 2015-2019 (AySA, 2020).

¹⁸ Community Promotion Management Unit, Directorate of Community Development (2016). *Relevamiento conexiónado a la Red de Agua Potable, Informe Final 2014*. AySA. Extrapolated based on random sampling data from three municipalities: Almirante Brown, Morón y Lomas de Zamora.

¹⁹ Relevamiento Urbanizaciones Emergentes. Resultados generales 2013-2016 (AySA, 2016).

²⁰ Ibid.

²¹ Values according to *Relevamiento Urbanizaciones Emergentes. Resultados generales 2013-2016* (AySA, 2016). These values correspond to existing UREMs in AySA's concession area by 2015.

²² For the purposes of the Program, the term "vulnerable" is grounded in AySA's tariff calculations which are based on coefficients (called Z coefficients) that reflect the socio-economic situation of users, which ranges between 1.10 and 3.50. Users with a Z coefficient less than or equal to 1.45 are considered living below the poverty line as defined by INDEC. (AySA's annual report, 2018).



Jobs (*Agua más Trabajo*, A+T) and Sewerage and Jobs (*Cloaca más Trabajo*, C+T) programs. These programs have been successful in increasing intra-household connectivity rates, particularly in the vulnerable peri-urban districts of GBA. The A+T and C+T programs not only improve the quality of life and reduce water, sanitation, and hygiene (WASH)-related health risks, but reduce local unemployment through participation in municipal cooperatives, generating income opportunities and promoting social inclusion.²³

14. **Untreated wastewater continues to pose significant environmental and health concerns.** According to AySA, the utility has a capacity to treat 2.8 million m³/day of wastewater, corresponding to the share of the population connected to sewerage. Of the treated effluent, only 26 percent of the wastewater receives secondary treatment. The remaining wastewater is only pretreated and predominantly released into the Rio de la Plata²⁴. Effluents from more than 6 million people not connected to sewers are often inadequately disposed, polluting aquifers or discharged in urban waterways, creating public health concerns for those living along the riverbanks.

15. **The risk to public health is compounded by floods of increasing frequency and intensity due to climate change.** During the period of 2000-2011, flooding events affected 5.5 million people and caused nearly US\$4.5 billion in losses. Although this trend is reversing due to different interventions being implemented, the risk remains significant, particularly in peri-urban areas.²⁵ More than one million people in the metropolitan area of Buenos Aires are exposed to floods with a return period of 5 years, and more than 30 percent of fragile neighborhoods are exposed to 100-year return period floods.²⁶

16. **Public health concerns can also be traced to a widescale lack of sewerage connections and the use of unsafe groundwater.** Numerous peri-urban areas use groundwater rendered unsuitable to serve as a sole drinking water source due to years of unchecked pollution – primarily associated with households relying on suboptimal onsite sanitation solutions. As a result, piped water tapping into groundwater sources sometimes surpass national regulatory limits on nitrate concentrations. Such is the case for the estimated 100,000 residents of Florencio Varela, most of whom are vulnerable, who are exposed to unsafe water, underscoring the urgency to increase the quality and safety of water services in this area. Similarly, there is a pressing need for AySA to provide residents of the Escobar, Esteban Echeverría, Ezeiza, San Fernando, San Isidro, San Martín, and Tigre municipalities access to safely managed sanitation services by ensuring sewage treatment and that associated discharges are in compliance with regulatory standards.

17. **AySA faces substantial financial constraints and relies heavily on government subsidies.** Although the company began to increase tariffs in 2012, following several years of frozen rates, increases remained below the rate of inflation until 2015. By 2019, revenues only covered 75 percent of O&M costs. These gaps, together with new infrastructure investment needs, were financed by short-term debt and transfers from the National Government (US\$450 million), equivalent to 80 percent of AySA's total revenues.²⁷

18. **There is considerable scope to reduce AySA's O&M costs and increase its operational efficiency.** AySA posts 35 percent more network breakages due to the aged water network, 8 percent more water losses and produces

²³ As of December 2019, a total of 905 works comprising 3,321 kms of pipelines and 307,903 household connections were completed generating over 912 jobs in municipal cooperatives. Source: *AySA Plan Agua + Trabajo y Cloaca + Trabajo*.

²⁴ AySA is in the process of significantly increasing its wastewater treatment capacity through the construction of large infrastructure works in the Riachuelo system (collector and WWTP with subfluvial outfall) and in Berazategui (subfluvial outfall from the Bicentenario WWTP).

²⁵ AR Flood risk management support project for the city of Buenos Aires (P145686).

²⁶ Results of Fathom flood modeling at 90 m resolution (simulating fluvial and pluvial floods), using location of Buenos Aires peri-urban slums from the RENABAP registry.

²⁷ Starting in 2018, AySA presents inflation-adjusted financial statements; in order to carry out an analysis of historical evolution, the unadjusted values have been used. However, if inflation-adjusted data is used, there are no relevant changes in the percentages and amounts indicated.



74 percent more water per person²⁸ than similar utilities in Latin America and the Caribbean (LAC).²⁹ This may be addressed in large by through the reduction in non-revenue water (NRW), estimated at 41 percent³⁰ (almost all of which is attributable to physical losses) through network rehabilitation and upgrading. These indicators are also associated with high levels of energy consumption given the need to pump higher than needed volumes of water.³¹ Similarly, on the sewerage front, AySA posts 361 percent more network blockages and 194 percent more breakages than regional comparators.³² In addition, the inclusion of the nine additional municipalities and of their corresponding utilities since 2016 resulted in a 95 percent staff increase (from 4,058 to 7,841 employees by 2018), far surpassing the 53 percent increase in new connections, and translating into a workforce of 3.51 employees per 1,000 water users, higher than the regional average of 1.8, thereby decreasing labor productivity and overall efficiency. AySA plans to increase the number of connections at a higher annual rate of 8.2 percent to offset the rate of staff change to further increase operational efficiency and quality of services.

19. **Low use of micro-metering and a tariff structure that does not provide appropriate incentives also undermine AySA's commercial and operational efficiencies.** Micro-metering throughout AySA's concession area stands at approximately 17 percent, a quarter of the level of other large utilities throughout the region.³³ Since most users do not have metered service, they are charged a fixed amount regardless of consumption levels. AySA recently changed the tariff structure applicable to metered connections by reducing the fixed charge and increasing the share that corresponds to metered consumption. This together with increased micro-metering is considered central to improving AySA's commercial, financial and operational efficiencies. With regard to subsidies, although the scheme provides some tariff differentiation criteria based on location, area, quality and age of property, in 2015, roughly 93 percent of residential users received a tariff discount (compared to the coefficient that determines tariff increases) and about 48 percent received a discount of 50 percent or above compared to this coefficient.³⁴ Water subsidies in Argentina have been shown to be progressive in relative terms, since the poorest 20 percent receives 4.7 percent of subsidies while having only 3.6 percent of income; this is not the case, however, in absolute terms.³⁵ Therefore, additional mechanisms are necessary to ensure that subsidies are more efficiently targeted, avoiding errors of inclusion and exclusion that undermine the financial sustainability and effectiveness of the system.

20. **AySA has expanded a targeted social tariff scheme to support low-income populations in its concession area.** This is provided to households that request the tariff discount and meet the eligibility criteria. In 2018, approximately 60,000 vulnerable users benefitted from the social tariff, up from 23,500 in 2015. AySA also supports debt cancellation for low-income households unable to pay outstanding bills and provides subsidies to incentivize low-income households to connect to the water and sewerage networks, including a 50 percent discount in the connection charge and additional subsidies of up to 100 percent of the tariff and free network connections to households in extreme poverty.

21. **The COVID-19 crisis has imposed significant cash flow constraints, further straining AySA's financial challenges.** Widescale job loss has sharply curtailed consumer ability to pay water and sewerage bills, thereby

²⁸ This translates into a production of 550 L/person/day. PMOEM 2019-2023, Plan Director de Mejora y Mantenimiento (AySA, Volume II).

²⁹ Based on a set of 35 regional utilities included in benchmarking reports produced by ERAS.

³⁰ This is an estimate provided by AySA, given the low macro and micro metering rates.

³¹ Energy costs represent 10 percent of AySA's O&M or about 50 percent of O&M if the cost of personnel is excluded.

³² ERAS 2018 Report, using 2017 data.

³³ Micro-metered rates in main regional WSS providers: AySA-Buenos Aires (23 percent), Acueducto-Bogotá (45 percent), IDAAN-Panamá (52 percent), SEDAPAL-Lima (87 percent), Aguas Andinas-Santiago de Chile (96 percent), EPMAPS-Quito (99 percent), SABESP-Sao Paulo (99 percent).

³⁴ Federal Association of Water and Sanitation Regulatory Agencies (*Asociación Federal de Entes Reguladores de Agua y Saneamiento, AFERAS*) (2018) *Sistemas Tarifarios y Tarifas en Argentina. Serie de Publicaciones sobre Tarifas* N° 3.

³⁵ Lombardi, M., Mongan, J. C., Puig, J. y Salim, L. (2014) *Una aproximación a la focalización de los subsidios a los servicios públicos en Argentina*. DPEPE N°09/2014, Ministerio de Economía de la Provincia de Buenos Aires. 2014.



reducing revenues. During the first two months of the pandemic, AySA reported a 30-40 percent decrease in revenues, while emergency mandates require it to continue providing services. Concurrently, increased emergency response expenses associated with, *inter alia*, delivering drinking water in trucks and boats, investments in safety gear and measures for frontline operators and staff at risk of contagion, remote work and other adaptation measures, have further increased the utility's already high O&M costs. The combination of these factors coupled with a freeze on tariffs following the outbreak, has resulted in cash liquidity issues and is expected to decrease the cost recovery rate to 47 percent in 2020. Although AySA has introduced measures to tackle these challenges, i.e., promoting online payment, and is receiving extra support from the Government, more robust actions are needed to reduce the cashflow gap.

22. **To capitalize on experience gained through its COVID-19 response, AySA needs to incorporate innovative digital solutions to further improve emergency preparedness and resilience.** Despite relying on an established Prevention and Emergency Plan (*Plan de Prevención y Emergencias, PPE*), and having responded swiftly to the COVID-19 crisis with a package of measures, including wastewater-based epidemiology (WBE), AySA is consolidating this knowledge and incorporating it into its PPE to strengthen preparedness to deal with this pandemic and other potential future epidemiological disruptive events. In addition, AySA has yet to tap into the full potential of digital solutions in the context of COVID-19 recovery and seek opportunities to enhance its ability to apply data science and augmented intelligence techniques to its business, commercial and technical challenges.

23. **AySA presents low levels of digitalization of commercial processes³⁶ and is one of the few large utilities in the region today that lacks the ability to interact with its customers through a digital platform.** For example, 90 percent of claims are currently received by telephone, 54 percent of payments are made through in-person interactions, and most management and procedures are conducted using traditional paper files. These approaches are associated with higher bureaucracy, slow processing times, and reduced agility in responding to crises. Bridging the significant digital divide is not only considered a necessity but an opportunity for AySA to review its outdated but strongly rooted internal processes and to promote organizational changes. This is particularly important to improve customer relations.

24. **AySA also lags behind in terms of female participation in executive positions, which is widely recognized as a key contributor to improved financial performance³⁷ and is particularly crucial in times of crisis.** Currently, 22 percent of managers at AySA are women, which is well below the regional average of 35 percent.³⁸ Contributing to this gap is the fact that women are not promoted at higher levels – only 18.5 percent of promotions to managerial positions were granted to female staff in the past year. A 2013 study of 345 companies in six LAC countries revealed that companies with women on their executive committees reported over 40 percent higher average return on equity and EBIT margins than those without female participation.³⁹ It also bears highlighting that during crises, female representation in leadership positions is critical to better respond to the differential impacts on women and men. During the Ebola pandemic, for example, insufficient female representation in decision making bodies undermined efforts to respond to the specific challenges faced by women, underscoring the importance of ensuring equal female

³⁶ Regional peer utilities such as SABESP in Sao Paulo or COPASA in Minas Gerais (Brazil), EPM in Medellin (Colombia), Aguas Andinas in Santiago de Chile (Chile), and SEDAPAL in Lima (Perú), all show higher levels of digitalization.

³⁷ World Bank. 2019. *Women in Water Utilities: Breaking Barriers*. World Bank, Washington, DC. © World Bank. <https://openknowledge.worldbank.org/handle/10986/32319>.

³⁸ World Bank Water Data. <https://wbwaterdata.org/breakingbarriers/tool/>. December 17, 2020. (Based on data from 9 LAC water utilities collected through the World Bank's HR Utility Survey including data submitted by AySA.

³⁹ McKinsey & Company. 2013. "Women Matter: A Latin American Perspective. Unlocking women's potential to enhance corporate performance". McKinsey & Company. <https://www.femtech.at/sites/default/files/Women%20Matter%20Latin%20America.pdf>



representation in sectors directly involved in the COVID-19 response.⁴⁰ Recognizing these shortcomings, AySA adopted a variety of measures during the pandemic, under its newly appointed female President, to address gender gaps, and is committed to promoting women's career advancement opportunities to improve its operational and financial performance, and crisis response capacities going forward.

C. Relationship to COVID Recovery, the Country Partnership Framework, and Rationale for Use of Instrument

25. **The proposed operation addresses the COVID-response strategy of the World Bank in Argentina and is consistent with the FY19-22 Country Partnership Framework (CPF).**⁴¹ The COVID-response approach of the World Bank provides for financing of urgent health-sector responses, investment in the expansion of infrastructure and improvement in basic services for the poor, and support to well-targeted and essential social protection measures. The proposed Program provides a critical public-health related service that is closely correlated with the spread of communicable diseases, is targeted at the most vulnerable urban communities, and is job-creating as a works project in a sub-sector with extraordinarily high employment multipliers—expansion of water and sanitation networks. The Program will specifically support Objectives 4 and 5, which call for strengthening transparency and improving service delivery, respectively, and are at the core of the proposed operation.

26. **The Systematic Country Diagnostic⁴² endorses closing basic infrastructure gaps to reduce inequality.** The Program will specifically support Objectives 4 and 5 of the SCD, which call for strengthening transparency and improving service delivery, respectively, and are at the core of the proposed operation. Given that there are more than 4,000 informal settlements in the country, a more equal provision of WSS services will be key for the convergence of living standards. The proposed Program is in line with the SCD recommendation that Argentina will benefit from a comprehensive and strategic planning approach that covers both expansion plans and maintenance and improvement of existing infrastructure.⁴³

27. **The Argentina Nationally Determined Contribution (NDC) to the Paris agreement sets new goals for carbon dioxide emission resulting from mitigation measures.** According to the NDC, Argentina will not exceed 483 million tons of carbon dioxide equivalent (tCO₂eq) by 2030 and this would require strong mitigation measures. The proposed Program contributes to mitigation by implementing energy efficiency measures and by reducing greenhouse gas (GHG) emissions such as fugitive methane emissions from sewage treatment.

28. **The Government of Argentina (GoA) has trusted the WBG as a long-term partner in helping improve complex water issues.** Bank support to the sector has a long track record of bold interventions including over 20 years of engagement in mitigating flood risks, support in addressing the poor environmental conditions of the Matanza-Riachuelo Basin and expanding WSS services in low-income Northern provinces. The Bank is a trusted partner in the sector, helping the government overcome technical, institutional, financial, legal and other challenges. The Bank's emphasis goes beyond supporting infrastructure investments and contributing global technical expertise, to also encompass institutional development, environmental management, and citizen engagement, supporting the Government of Argentina (GoA) on its path towards the 2030 agenda and pursuing the SDGs. The Bank is committed to supporting this Program through relevant global knowledge and by mobilizing a wide range of experts as needed.

29. **The Program for Results (PforR) was chosen as the most appropriate financing instrument to support the GoA's expected program results within the current context.** The proposed Program is being requested at a moment

⁴⁰ Comisión Interamericana de Mujeres. COVID-19 en la vida de las mujeres: Razones para reconocer los impactos diferenciados. 2020. OEA/Ser.L/II.6.25.

⁴¹ Report No. 131971-AR. Discussed by the Board of Executive Directors on April 25, 2019.

⁴² Report Reference: SecM2018-280. Discussed by the Board of Executive Directors on September 9, 2018

⁴³ Argentina: Escaping Crises, Sustaining Growth, Sharing Prosperity (World Bank Group, 2019).



when the country faces dual challenges: a weak macroeconomic situation and the threat to health and environmental conditions posed by COVID-19 in the nation's largest hotspot. These challenges demand urgent and effective solutions. The GoA recognizes the need to accelerate implementation of AySA's Service Improvement, Expansion, Operation and Maintenance Program (*Plan de Mejoras, Operación, Expansión y Mantenimiento de los Servicios, PMOEM*) to expand access to WSS services quickly, particularly to low-income areas, while reducing AySA's losses and O&M costs, as well as its dependence on government subsidies, and making it more resilient to climate (e.g., floods) and non-climate shocks, while reducing energy related GHG emissions.

30. In this sense, the PforR is considered an adequate financing instrument for the following reasons:

- (i) **By designing a strong system of incentives through well-defined Disbursement-Linked Indicators (DLIs), the utility will be incentivized to achieve critical results** such as expanding connectivity through community participation, increasing operational efficiency, streamlining managerial processes, improving the agility of internal procedures while at the same time promoting innovation and improving transparency.
- (ii) **The PforR will acknowledge AySA's ongoing efforts to mitigate COVID-19 impacts** – including providing water and disinfectant to unserved areas to mitigate the outbreak, and expand services in the most vulnerable population, implementing transmission prevention measures for staff at risk, enabling telework for employees and promoting digital payment methods.
- (iii) **It will be possible to strengthen both the Government's and AySA's own systems**, by addressing gaps identified in the technical, social and environmental, and fiduciary assessments, to be included in the corresponding Program Action Plan (PAP).⁴⁴
- (iv) **The Program will serve as an example to promote innovative approaches in the water sector in the country and in the LAC region** by focusing on results and promoting institutional strengthening in a most cost-effective manner.

II. PROGRAM DESCRIPTION

A. Government Program

31. **The proposed interventions are part of AySA's Program for Service Improvement, Expansion, Operations and Maintenance (PMOEM), which is well aligned with the goals of Argentina's National Water and Sanitation Plan.**

⁴⁵ As a program designed through national commitments to achieving the SDGs in the areas of water and sanitation, the PMOEM serves as AySA's Master Plan. It is revised every five years based on progress made and evolving priority considerations and is approved by the National Planning Agency. The PMOEM includes interventions to expand and improve water and sanitation services and increase AySA's efficiency and quality of operations throughout the 27 districts it serves.

32. The current iteration of the PMOEM (2019-2023) builds on the goals of the first two programs (2010 and 2015). It comprises three subprograms and operational expenses totaling an estimated US\$4.86 billion as follows:

- (i) Sub-program 1: *Improvements and Expansion*. This Sub-program has a 30-year planning horizon, which includes the analysis and scheduling of works needed to satisfy current and future demand, with a focus on the newly incorporated municipalities. It is complemented by the concurrent strategies outlined in the Improvement and Maintenance Master Plan (Sub-program 2), so infrastructure incorporated in the

⁴⁴ PforR Policy, paragraph 10.

⁴⁵ RESOL-2019-29-APN-SIPH#MI, May 9, 2019.



Plan considers both the proper functioning of existing facilities and compatibility with new works in order to maintain and progressively improve the quality of the service.

- (ii) Sub-program 2: *Improvements and Maintenance*. This Sub-program has a strong focus on efficient management of water distribution through the systematic implementation of technical strategies to reduce physical losses that make it possible to optimize the supply chain, ensuring fulfillment of current and future demand for drinking water. Efficiency improvements are achieved by network renovation; demand management through installation of consumption meters and sectorization; and energy efficiency through infrastructure rehabilitation and replacement. Benefits include a reduction of physical losses, lower operating costs and a decrease in future network repairs, thus increasing AySA's resilience.

AySA's 2018 Service Levels Report shows that, although installations have been closely monitored and maintained, a large proportion of the existing infrastructure is aging. Consequently, this Sub-program outlines infrastructure and asset diagnostics and management actions aimed at: ensuring the reliability of the systems and guaranteeing their availability for the provision of WSS; achieving greater versatility and operability; substantially improving service levels; and positively influencing service indicators that drive efficiency.

- (iii) Sub-program 3: *Operations*. This Sub-program seeks to improve the efficiency of internal processes through systems integration and upgrades, and increase transparency through improvements in communication by providing the general public and stakeholders access to information. It presents the strategies and resources necessary to ensure quality service provision, in compliance with the provisions of the Regulatory Framework and responding to current and projected operational challenges.

- 33. The next phase of the PMOEM will entail the period from 2024-2028 and will give continuity to the activities planned in the current version.

B. PforR Program Scope

- 34. **The proposed Buenos Aires Water Supply and Sanitation with a Focus on Vulnerable Areas Program (the Program), will support the implementation of AySA's 2019-2023 Program for Service Improvement, Expansion, Operations and Maintenance (PMOEM) and its continuation in the period 2024-2028⁴⁶, including operational expenditures of infrastructure related to the Program.** The Program is defined by selecting areas of the PMOEM that are a priority for AySA, in terms of the expansion and improvement of infrastructure to serve vulnerable population, target clear efficiency gains to lower operational costs, and modernize internal systems to become more agile and resilient to shocks. This priority is evidenced by the amount of investments that AySA has been making in these areas in the past and will continue do in the future beyond this operation.⁴⁷

- 35. **The Program is firmly aligned with the PMOEM and is structured around three key Results Areas:** (i) Expansion of WSS services, with a focus on vulnerable areas; (ii) Improvements in AySA's efficiency; and (iii) Increasing AySA's capacity to respond to water supply and sanitation emergency needs. The Program has an estimated cost of US\$576.6 million, including US\$227.3 million in counterpart funding, US\$300 million in World Bank financing, and US\$49.3 million in co-financing from the Inter-American Development Bank (IADB).

⁴⁶ The Program includes activities under the PMOEMs carried out from July 16, 2020 through May 26, 2026 and operational expenditures associated with newly built infrastructure under Results Area 1 of the Program.

⁴⁷ AySA has projected to spend more than US\$850 million to further expand WSS infrastructure in the Program Areas beyond 2026



36. **In terms of access to water and sanitation services, the Program has a focus on vulnerable areas of Greater Buenos Aires.** These areas belong to peri-urban municipalities within AySA's jurisdiction where households do not have access to safely managed water and sanitation services. The inhabitants of these areas are defined as living below the poverty line.⁴⁸ Within these vulnerable areas, there are also socially *fragile neighborhoods*, where most of the people live below the *extreme poverty* line.⁴⁹ These neighborhoods are adequately registered by AySA in national databases that are periodically updated.⁵⁰

37. The scope of each Results Area is described as follows:

Results Area I - Expansion of WSS services, with focus on vulnerable areas

38. The Program will contribute to PMOEM's Sub-program 1 by improving access to safely managed water and sewerage through: (i) extending WSS infrastructure in selected peri-urban Buenos Aires areas that are currently not adequately covered by these services; and (ii) building intra-household connections for water and sewerage in fragile neighborhoods, where secondary networks already exist.

39. **Under Results Area 1.1, the Program will expand safely managed water supply services in selected municipalities of peri-urban Buenos Aires.** Approximately 140,000 people will benefit from the activities under this sub-result area, of which 100,000⁵¹ are located in the municipality of Florencio Varela. This area is characterized by water quality problems due to the presence of nitrates in the groundwater source which feeds the distribution network. Under the Program, AySA aims to improve the quality of water provided to the residents of this municipality by bringing treated surface water sourced from the Belgrano water treatment plant (WTP) to the area and mixing it with existing groundwater, thereby lowering nitrate concentrations to safe drinking standards. This action will furthermore reduce the energy requirements related to groundwater pumping. To this end, AySA plans to build a new 4,000 m³/h pumping station, a 1,600 m long and 1200 mm diameter reinforcement pipeline, and a 4,550 m long and 700 mm diameter conveyance pipeline to the "Cruce" cistern, where the clean water will be mixed with existing groundwater. Of the total beneficiaries from these interventions, 14,500 people live in fragile neighborhoods, where household connections already exist.

40. **Another 40,000 beneficiaries are located in fragile neighborhoods of 16 selected municipalities of peri-urban Buenos Aires who will benefit from the installation of intra-household water connections, where primary and secondary networks already exist.** This intervention will be implemented through AySA's A+T model consisting of a series of technical-social works coordinated between each municipality, municipal labor cooperatives, and professionals selected to execute and supervise the works.

41. **Under Results Area 1.2, the Program will expand safely managed sewerage services in selected municipalities of peri-urban Buenos Aires.** Approximately 305,000 people will benefit from the activities under this sub-result area, of which 225,000⁵² will benefit from the expansion of sewerage networks and of the capacity of corresponding wastewater treatment plants (WWTPs). For this purpose, AySA will intervene in two WWTPs⁵³ and

⁴⁸ The poverty line is here defined using national standards (INDEC). It considers minimum food consumption plus other basic consumption. https://www.indec.gov.ar/ftp/cuadros/sociedad/EPH_metodologia_22_pobreza.pdf.

⁴⁹ The concept of *extreme poverty* is defined using national standards (INDEC). It is equivalent to the "Indigence Line" which seeks to establish whether households have sufficient income to cover a food basket capable of satisfying a minimum threshold of energy and protein needs, called the Basic Food Basket. In this way, households that do not exceed that threshold or line are considered indigent.

⁵⁰ *Fragile neighborhoods* are defined, for the purpose of the Program, as (i) UREM, (ii) areas included in the RENABAP, and (iii) ABC areas.

⁵¹ In total, 92 percent are classified by AySA as "vulnerable" (with a Z coefficient less than or equal to 1.45), and 15 percent live in slums in Florencio Varela.

⁵² All these users with water and without sewerage have a Z coefficient less than or equal to 1.45, and therefore live below the poverty line and are considered vulnerable under the Program.

⁵³ Both WWTPs are flaring the captured methane, contributing to the reduction of GHG emissions.



their respective catchment areas, including associated operational expenditures. The key interventions are the following:

- (i) **Norte WWTP.** Optimization of the two existing modules, and construction of a third one, including an upgrade in the treatment capacity for nutrient removal. This will increase the total capacity of the plant from 1.8 m³/s to 3.2 m³/s. In parallel, the sewerage network in its catchment area will also increase with 410 km of secondary sewers. It is expected that 168,000 people will be connected to the system by the end of the Program, out of which 20,000 live in fragile neighborhoods.
- (ii) **Jagüel WWTP.** Upgrading of the currently decommissioned Module I to increase the capacity of the plant from 0.5 m³/s to 0.8 m³/s. The Program will also support the construction of 5 km of primary networks and 78 km of secondary sewerage networks in the catchment area. It is expected that 57,000 people will be connected to this system, out of which 10,000 live in fragile neighborhoods.

42. **The remaining 80,000 people are located in fragile neighborhoods of 16 selected municipalities of peri-urban Buenos Aires and will benefit from intra-household sewerage connections, where primary and secondary networks already exist and are connected to functioning WWTPs.** This intervention will be implemented through AySA's C+T model which works like the A+T model. The Program will also acknowledge operational expenditures associated with all new infrastructure (WTPs and WWTPs) to ensure water and wastewater meet the regulatory standards.

Results Area II – Improve AySA's efficiency

43. **The Program will contribute to PMOEM's Sub-programs 2 and 3 by improving AySA's efficiency through the design and implementation of an Efficiency Improvement Action Plan (EIAP).**⁵⁴ This Plan entails key interventions included in PMOEM's Sub-programs 2 and 3 (Energy Efficiency Plan and Non-Revenue Water Reduction Plan), and will incorporate measures to improve the efficiency of commercial operations and processes. The introduction of the EIAP seeks to promote a planning methodology that goes beyond just solving operational problems and contributes to assess and improve AySA's financial sustainability going forward. The EIAP will measure the respective impacts of such activities on AySA's financial standing, thus providing a clear rationale underlying the long-term strategic approach and specific five-year service improvement targets including the financial impacts of the proposed interventions to achieve set targets. Therefore, the two areas where this Program will support the implementation of the EIAP are the following:

44. **Under Results Area 2.1, the EIAP will promote operational efficiency via:**

- (i) **The reduction of leakages through the rehabilitation and replacement of critical water networks.** Investments will focus on scaling up AySA's replacement program of critical aging water networks throughout the service area. Priority interventions will include, inter alia, sectorization and pressure control, increased micro-metering, and the replacement of reticulation mains, pipelines and water connections. These interventions are expected to generate savings of 123,000 m³/day.
- (ii) **Improvements in energy efficiency.** This will include improving monitoring and energy efficiency of pumping stations and deep wells through the installation of telemetry systems, variable speed drives and upgrading of electrical installations to meet national regulations. These actions will increase the reliability of facilities while improving consumption management with power utilities. AySA will also

⁵⁴ This plan will summarize all interventions necessary to improve AySA's operational and commercial performance and measure respective impacts on its financial standing. The EIAP will provide a clear rationale of the long-term strategic approach and specific five-year service improvements targets including the financial impacts of the proposed projects to achieve the set targets.



install photovoltaic solar panels for in-situ power generation. Estimated savings from these interventions amount 34 GWh/year.

45. **Under Results Area 2.2, the EIAP will promote commercial management efficiency by expanding the menu of digital payment options available to AySA's customers.** AySA will follow a strategy to roll-out: automatic debit,⁵⁵ internet, and digitized phone (*Pagophone*) payment options, which are expected to lower transaction costs, including time spent on phone- or in person user interactions with customer care representatives, for the consumer and utility alike. The move towards electronic billing and payment will allow AySA to reach digitalization levels comparable to other large regional providers, and to increase the number of users who pay through digital channels by 30 percent, freeing up staff in commercial offices from highly transactional “manual” tasks to more productive duties.

Results Area III – Increase AySA’s capacity to respond to water supply and sanitation emergency needs

46. The Program will contribute to PMOEM’s Sub-program 3 by scaling up and expanding AySA’s efforts to mitigate the impacts of COVID-19 on AySA’s operations and by strengthening its systems to render them more resilient. The Program will strengthen AySA’s Prevention and Emergency Plan (PPE) and support its strategy to streamline internal and managerial processes to become more agile in responding to shocks (i.e., epidemiological, climate related or financial). At the same time, this Results Area will enhance transparency, citizen engagement, and promote gender diversity within AySA, as proven approaches for a utility to interact closely with its users, better understand their needs, and, respond faster and in a more tailored manner to the different challenges that may arise from crises/shocks. As such, this Results Area will include the following:

47. **Under Results Area 3.1 AySA will Implement and Strengthen its Prevention and Emergency Plan by:**

- (i) **Supporting AySA’s efforts and initiatives to confront the COVID-19 outbreak**, which have aimed at: (i) ensuring continuity and efficiency in the safe provision of services to the population; and (ii) preserving and caring for the health and physical integrity of its employees. Beneficiaries include AySA’s own employees, its providers, as well as the general population in AySA’s service area. This strategy included, among others, the delivery of water and bleach to underserved areas, the facilitation of remote working arrangements for AySA staff, the development of awareness campaigns for AySA’s employees on precautions to be taken during the COVID-19 pandemic outbreak, the implementation of safety protocols for AySA’s A+T and C+T programs; and the use of WBE to complement the epidemiological surveillance system and support the development of an early warning signal for future outbreaks.
- (ii) **Strengthening of AySA’s PPE**, which comprises a set of guidelines aimed at preventing or reducing the possibility of a risk occurring (Proactive Stage) or providing a quick and efficient response to an ongoing crisis (Reactive Stage). The PPE applies to all activities related to the production and distribution of water, quality control, collection and treatment of sewage, as well as the different processes supporting these activities. The PPE will be reinforced by capturing the lessons learnt from dealing with the ongoing crisis, to be embodied in a contingency plan to respond to epidemiological risks, including a specific “lessons learned” document; and a seminar to share with other regional service providers experiences and advances on managing the COVID19 crisis and more in particular on molecular biology for the control of other potential epidemiological risks.

48. **Under Results Area 3.2 AySA will implement a series of digital solutions to streamline internal processes and enhance transparency and citizen engagement:**

⁵⁵ Refers to automatic withdrawal of the amount of the service bill from the customer’s bank account (without user intervention).



- (i) **Implementation of Digital Solutions for Internal Systems.** AySA is pursuing a series of innovative technological enhancements that are critical to anticipate, mitigate, prepare for and recover from the effects of a hazard in a timely and efficient manner. These include, among others: (i) the consolidation of human resources systems to enable the integration of critical information on the company's human capital; (ii) solutions to enable the remote work modality on a continuous basis; (iii) predictive maintenance of main assets, incorporating solutions such as advanced predictive analytics and artificial intelligence, digital twins and dynamic 3D visualization; (iv) digitalization of commercial processes in AySA's network of commercial offices; and (v) optimization of the customer complaint management system through the adoption of mobile and digital tools. Overall, these digital solutions will contribute to increasing AySA's resilience to climate and non-climate shocks, by enabling it to be better prepared to its changing environment.
- (ii) **Strengthening AySA's capacity to engage citizens** through the phased implementation of a digital strategy that facilitates a two-way engagement between user/citizen and the utility. The strategy will focus on: (i) incorporating a new digital channel – mobile app – for community engagement; (ii) the development of a solid and integrated Customer Relationship Management System (CRM); (iii) the development of a digital user- question-complaint-claim feedback platform (*Peticiones, Quejas, Reclamos y Sugerencias* – PQRS) and average response times; and (iv) the reduction of the top 10 PQRS response times. The incorporation of these new technologies will allow for rationalized customer data management, improved customer communication, and timely two-way engagement among others.

49. **Under Results Area 3.3 AySA will promote gender diversity and inclusion.** As one of the largest utilities in the region, AySA's ability to address gender inequality in its programs will be strengthened by improving its own gender diversity. The Program will support AySA in its Inclusion, Gender, Equal Opportunities and Treatment and Workplace Violence Program, particularly actions envisioned under the 2020-2023 action plan. This action plan constitutes a crosscutting approach applied to all AySA programs, including the PMOEM. It will focus on improving working conditions for AySA employees (by extending care leave for women and men, including flexible work options for recent mothers, as well as family friendly facilities, designing protocols to address gender-based violence and sexual harassment at the work place – see details in Annex 3) and particularly on promoting equal career advancement opportunities for women, with the goal of increasing female representation in decision making positions.

50. **To accomplish its gender and diversity goals related to career advancement, AySA will introduce targets for gender diversity in promotions, particularly to management positions.** AySA has found that strong gender norms on the type of work that is considered appropriate for women tend to deter them from applying to technical and managerial positions. For this reason, it is launching targeted communications campaigns to motivate women to apply to such opportunities, and also offering targeted training to fill existing capacity gaps in women's existing technical and leadership skills. Through these actions, AySA aims to increase the percentage of women in managerial positions. Given the large number of employees, and taking into account the current recruitment rates, as well as new promotions targets, the utility intends to achieve a gradual but steady increase in women's representation in managerial positions – from 22 to 28 percent – which in turn, is expected to contribute to improving AySA's capacity to respond to the differential impacts and needs of women and men in its emergency response efforts.

51. **Role of Development Partners.** The Interamerican Development Bank (IADB) is supporting infrastructure investments centered on *Results Area 1* of the Program. In particular, this funding is focused on the expansion of sewerage networks, including primary and secondary collectors, to deliver wastewater to the Norte WWTP. These networks will connect to the WWTP to be upgraded and expanded with AySA and World Bank funds. IADB funding for these investments is largely secured as all envisaged contracts are either under implementation or in the bidding process.



52. **Beneficiaries.** By Program closing, it is expected that an estimated 445,000 people (see Table 3) will have directly benefitted from access to safely managed water and sanitation services (Results Area 1)⁵⁶. The main benefits are related to improved health and life quality in the Program areas, which lead as well to increased labor productivity due to the reduction in the incidence of waterborne and communicable diseases. The Program will also have substantial environmental benefits derived from the reduction of groundwater pollution and improved water quality of urban rivers due to the increased connectivity to sewers and volume of wastewater treated. Indirectly, 312,000 people will also benefit from the reduction in water losses. Additionally, the entire population currently served by AySA (10.7 million people) is expected to indirectly benefit from more efficient, sustainable, and resilient services derived from the results achieved under Results Areas 2 and 3. The Program is also expected to generate around 6,000 jobs (direct and indirect).

Table 3: Summary of Beneficiaries

Result Area	Theme	Beneficiaries		
		Fragile Neighborhoods	Non-Fragile Neighborhoods	Total
1 Access	Connections	164,500	280,500	445,000
1.1 Water Supply				
	HH connections	40,000	0	40,000
	Florencio Varela	14,500	85,500	100,000
1.2 Sewerage				
	HH connections	80,000	0	80,000
	Norte	20,000	148,000	168,000
	Jagüel	10,000	47,000	57,000
2 Efficiency	Leak reduction	-	-	312,000
Total				757,000

53. **Program Boundaries.** Eligible expenditures are defined by the Program Expenditure Framework, included in Annex 3 and further detailed in the Operations Manual⁵⁷. The typology of activities corresponding to such expenditures are aligned with the description of the Results Areas and are further detailed in the Technical Assessment (Annex 3) and in the Operations Manual. For Results Area 1 on the expansion of WSS services, the Program will finance investments in 18 municipalities⁵⁸, out of the 26 municipalities of PMOEM's Sub-program 1, where AySA made efforts to extend WSS infrastructure during the period 2015-19. These municipalities have low levels of water supply and sewerage coverage, and high socio-economic vulnerability. For Results Area 2, with activities such as network renovation and energy efficiency measures, the geographical scope is AySA's entire service area.

⁵⁶ 98 percent of which are considered vulnerable (living below the poverty line) and 165,500 living in fragile neighborhoods

⁵⁷ The Operations Manual has been formally adopted on January 20, 2020.

⁵⁸ Almirante Brown, Avellaneda, Esteban Echeverría, Ezeiza, Hurlingham, Ituzaingó, La Matanza, Lanús, Lomas de Zamora, Morón, Tres de Febrero, Escobar, San Fernando, San Isidro, San Martín, Tigre, Florencio Varela and Quilmes. There are additional eligibility criteria on the type of infrastructure (household connections or bulk WSS infrastructure), according to the municipality. Details are included in the Operations Manual.



Table 4: Program Financing

PMOEM Sub-Program / Result Area	Total PMOEM	Total Program	AySA / Public Financing	IADB	World Bank
Sub Program 1: Improvements and Expansion of Services / RA1	\$ 1,622,596,000	\$ 312,005,000	\$ 98,994,000	\$ 49,342,000	\$ 163,669,000
Sub Program 2: Improvements and Maintenance / RA2	\$ 552,983,000	\$ 123,441,000	\$ 21,438,000	\$ -	\$ 102,003,000
Sub Program 3: Operations / RA3	\$ 157,522,000	\$ 35,409,000	\$ 1,831,000	\$ -	\$ 33,578,000
PMOEM operational expenditures	\$ 2,532,623,000	\$ -	\$ -	\$ -	\$ -
Other Expenditures					
Operation and Maintenance Costs for Plants		\$ 100,000,000	\$ 100,000,000	\$ -	\$ -
Evaluation and Verification costs		\$ 5,000,000	\$ 5,000,000	\$ -	\$ -
Front-end fees IBRD		\$ 750,000	\$ -	\$ -	\$ 750,000
Total	\$ 4,865,724,000	\$ 576,605,000	\$ 227,263,000	\$ 49,342,000	\$ 300,000,000

*PMOEM 2019-2023. Budget from the next phase is unknown at the time of Program preparation. A more detailed table is included in Annex 3.

C. Program Development Objective(s) (PDO) and PDO Level Results Indicators

54. **The Program Development Objectives** are to increase access to safely managed⁵⁹ water and sanitation services with a focus on vulnerable areas⁶⁰ of Buenos Aires⁶¹, and improve AySA's efficiency and capacity to respond to water and sanitation emergency needs.

55. **The PDO-Level Results Indicators are the following:**

- (i) Number of households gaining access to safely managed drinking water services (Access)
- (ii) Number of households gaining access to safely managed sanitation services (Access)
- (iii) Annual savings from operational and commercial efficiency measures (Efficiency)
- (iv) Number of AySA's emergency response mechanisms strengthened⁶²(Capacity to respond)

⁵⁹ Safely managed drinking water is here defined as the use of an improved drinking water source which is located on premises, available when needed, and free of fecal chemical contamination; safely managed sanitation is here defined as the use of an improved sanitation facility which is not shared with other households and where the excreta is safely disposed in situ or transported to a treatment plant.

⁶⁰ It refers to people living under the poverty line this figure is 98 percent of the beneficiaries.

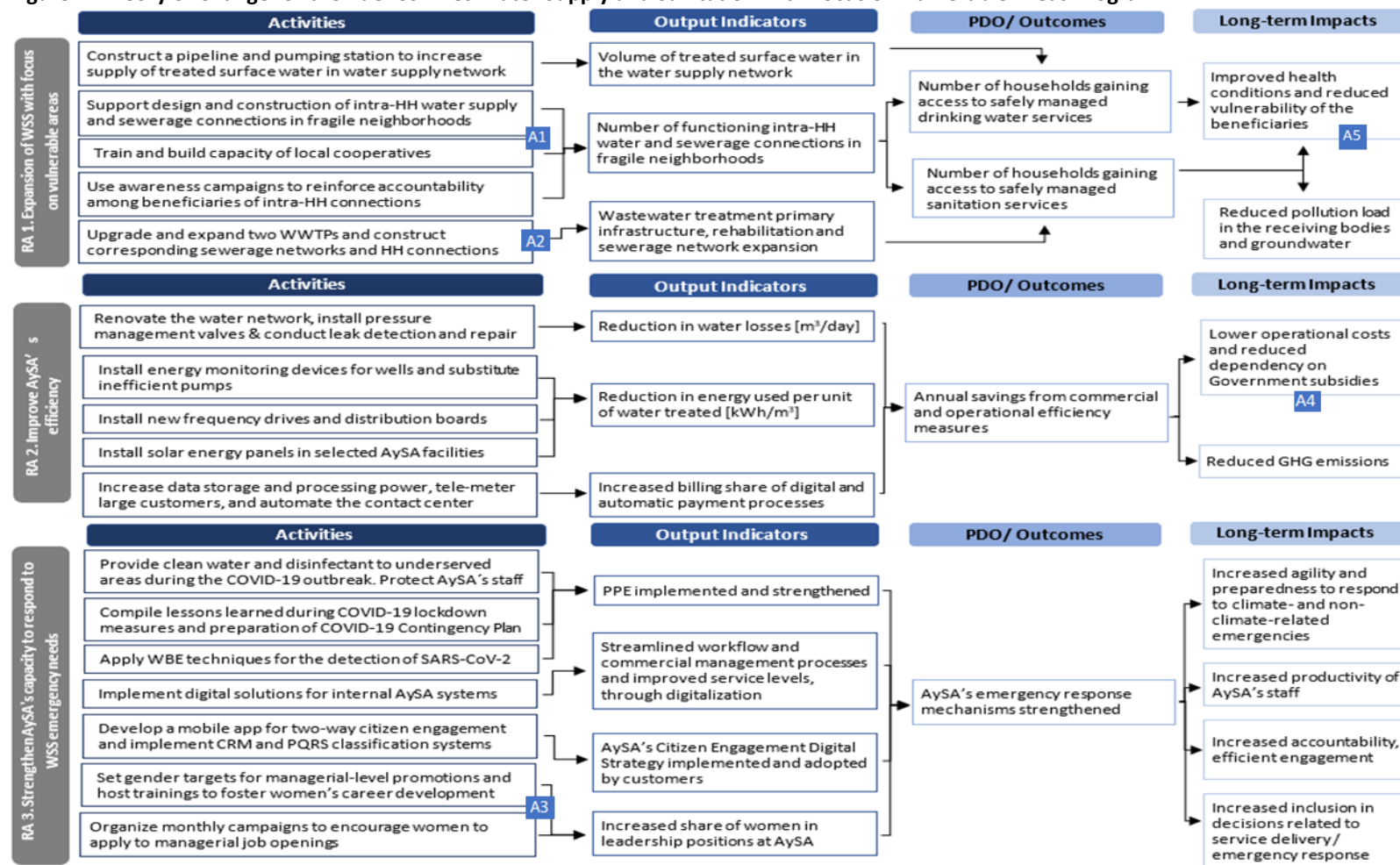
⁶¹ "Buenos Aires" means, for purposes of the Program, the following twenty-six municipalities in the PBA and CABA, which is AySA's service area according to the Regulatory Framework: Almirante Brown; Avellaneda; Escobar; Esteban Echeverría; Ezeiza; Florencio Varela; Hurlingham; Ituzaingó; José C. Paz; La Matanza; Lanús; Lomas de Zamora; Malvinas Argentinas; Merlo; Moreno; Morón; Pilar; Presidente Perón; Quilmes; San Fernando; San Isidro; San Martín; San Miguel; Tigre; Tres de Febrero; and Vicente López

⁶² These are 2 mechanisms : (i) the strengthening of AySA's Prevention and Emergency Plan with the COVID contingency action plan; and (ii) the implementation of AySA's digitalization strategy including the citizen engagement digital platform, as per the standards defined in the corresponding DLIs and verification protocol in the Operations Manual.



D. Theory of Change

Figure 1: Theory of Change for the Buenos Aires Water Supply and Sanitation with Focus on Vulnerable Areas Program.



A1: AySA is able to efficiently implement COVID-19 safety and preventive measures for workers building household connections, so that construction rate keeps a constant pace.

A2: Procurement processes for the extension of the WWTPs are successful, with no significant delays in the awarding of the contracts

A3: Women staff at AySA respond to the trainings and apply to managerial positions

A4: The magnitude of the reduction of government subsidies also depends on the size of tariff increases expected for the years 2022-2025

A5: Other risk factors and exposures that might affect the health conditions in beneficiary communities do not increase



E. Disbursement Linked Indicators and Verification Protocols

56. **AySA and the World Bank have agreed on twelve DLIs that will trigger disbursements under the Program.** They were selected based on their relevance to the Program objectives and structured around the three Results Areas.

57. **The DLIs have been designed to introduce clear incentives to achieve results, emphasizing areas where AySA is also making strong efforts.** That is the case of the connectivity in *fragile* neighborhoods, where the most vulnerable population live, and who, in turn, are most impacted by the lack of access to safely managed water and sanitation services, particularly during the COVID-19 pandemic. The DLI matrix design also puts emphasis on the efficiency agenda, where AySA needs to make tangible progress to lower operational costs; and introduce strong incentives to achieve the important goals of Results Area 3, such as emergency response, digitalization, citizen engagement and gender inclusion, which are crucial in increasing the company's resilience towards shocks. DLI 5.1 on COVID-19 response has been achieved and the verification process is expected to be completed before signing of the Legal Agreement as a prior result.

58. Table 5 below summarizes the DLIs with their corresponding definitions.

Table 5: DLIs definition

DLIs	Definition of DLIs	Target	Allocated Amount, Million US\$
Results Area I - Expansion of WSS services, with focus on vulnerable areas			
DLI 1. Number of HH gaining access to safely managed water services. Scalable			
DLI 1.1. Number of HH* gaining access to safely managed water services living in <u>fragile neighborhoods</u>	Households in fragile neighborhoods** that will gain access to safely managed water supply either (i) through the construction of the intrahousehold connections in areas of 16 municipalities of peri-urban Buenos Aires where the water network and service pipes already exist, or (ii) by improving water quality in the municipality of Florencio Varela.	13,625 (equivalent to 54,500 beneficiaries)	16.87
DLI 1.2. Number of HH* gaining access to safely managed water services living in <u>non-fragile/formal neighborhoods</u>	Households in non-fragile/formal neighborhoods** that will gain access to safely managed water supply by improving water quality in the municipality of Florencio Varela.	21,375 (equivalent to 85,500 beneficiaries)	19.13
DLI 2. Number of HH gaining access to safely managed sanitation services. Scalable			
DLI 2.1. Number of HH* gaining access to safely managed sanitation services living in <u>fragile neighborhoods</u>	Households in <u>fragile neighborhoods</u> ** that will gain access to safely managed sanitation through the construction of intrahousehold connections in areas of 16 municipalities of peri-urban Buenos Aires where the sewerage network and service pipes already exist and are connected to a WWTP. In 7 of those municipalities additional works to expand primary wastewater infrastructure will also be needed.	27,400 (equivalent to 110,000 beneficiaries)	37.84
DLI 2.2. Number of HH* gaining access to safely managed sanitation services living in <u>non-fragile/formal neighborhoods</u>	Households in non-fragile/formal neighborhoods that will gain access to safely managed sanitation through the expansion of primary wastewater infrastructure (wastewater treatment and sewerage***) in 7 municipalities of peri-urban Buenos Aires.	48,959 (equivalent to 195,000 beneficiaries)	49.16



DLIs	Definition of DLIs	Target	Allocated Amount, Million US\$
Results Area II – Improve AySA’s efficiency			
DLI 3. Preparation of AySA’s Efficiency Improvement Action Plan (EIAP)	The EIAP will include a minimum content acceptable to the Bank entailing the identification of efficiency improvement initiatives and goals including the expected effect in AySA’s financial performance. It will cover, at least (i) a diagnostic of factors influencing AySA’s financial performance (NRW, Energy Efficiency, Commercial Efficiency); (ii) identification and prioritization of activities to improve financial performance; and (iii) a collection improvement study. Non scalable.	1	20.00
DLI 4. Implementation of AySA’s EIAP. The EIAP has two main Key Performance Indicators (KPIs) that measure progress in its implementation. These are represented by sub-DLIs 4.1 and 4.2			
DLI 4.1 Increase in Operational Efficiency (OE-KPI) (US\$/day)	Measures AySA’s improvements in NRW (in saved m ³ /day) and energy efficiency (measured in saved kWh/m ³). The KPI is normalized in order to homogenize savings of different nature, measuring them in US\$/day. Scalable.	20,078	90.25
DLI 4.2 Increase in Digital Payment share (DP-KPI)	Percentage increase in the use in AySA of Efficient (Digital) Payment Channels compared to less efficient payment channels. Efficient payment channels are those that imply lower commissions charged to AySA, lower costs in logistics resources and shorter times and costs for users. Scalable.	51%	10.00
Results Area III – Increase AySA’s capacity to respond to water supply and sanitation emergency needs			
DLI 5. Implement and Strengthen AySA’s PPE			
DLI 5.1 COVID-19 Contingency measures implemented	This sub-DLI will monitor the implementation of measures related to the COVID-19 response, to be recognized as prior results, including: (i) the operationalization of the Crisis committee; (ii) the delivery of 60,163 m ³ of water and 370 m ³ of disinfectant delivered to 23 municipalities within the PBA; (iii) the facilitation of remote working arrangements for AySA staff; (iv) the development of awareness campaigns for AySA employees about precautions to be taken during the COVID-19 pandemic outbreak; (v) the implementation of safety protocols for AySA’s A+T and C+T programs; and (vi) the use of the WBE method to monitor the presence of COVID-19 in sewage, and (vii) preventive measures for COVID-19 transmission implemented for AySA’s staff, and reorganization of human resources in AySA’s operations department, while maintaining service levels. Non Scalable. A lump sum disbursement will be made once the implementation of all these measures is verified according to the verification protocol.	7	15.00
DLI 5.2 Preparation of the epidemiological risk contingency plan	PPE updated with a Prepare a Contingency Plan to deal with epidemiological risks, using lessons learned from the response to the COVID-19 pandemic. This plan will boost the effectiveness of AySA’s response to incidents and help minimize the impacts of a wider variety of potential epidemiological hazards including the COVID-19 epidemic. Organization and delivery of an international seminar on COVID-19 to share lessons learned on dealing with epidemiological risks. The features of the Contingency Plan are included in the Verification Protocol. Non scalable.	1	5.00
DLI 6. Implementation of digital solutions			



DLIs	Definition of DLIs	Target	Allocated Amount, Million US\$
DLI 6.1. Implementation of digital solutions for internal systems	Implementation of 5 digital and innovative technology solutions to better anticipate, mitigate, prepare for and recover from the effects of a hazard in a timely and efficient manner, as well as continue with critical activities during emergency situations. The list of verifiable measures is included in the Verification Protocol. Scalability proportional to the number of measures achieved.	5	26.00
DLI 6.2. Implementation of a Digital Strategy for Citizen Engagement including mobile app and customer relationship management system (CRM)	Implementation of mobile app and CRM system to improve AySA's customer relations and citizen engagement, by broadening the digital communication channels, incorporating feedback from users, providing tailored information and improving reception and management of claims, questions and suggestions, ending in the improvement of response times. The disbursement is scalable related to 5 milestones of the strategy detailed in the Verification Protocol.	5	7.00
DLI 7. Increase in share of women in leadership positions in AySA	The DLI will measure the increase in percentage points of women's representation at management level positions, thus reflecting efforts made by AySA in promoting equal career advancement opportunities for women and improving working conditions for all AySA employees. Scalable.	6 perc. Points (28% from a baseline of 22%)	3.00
Front-end Fee			0.75
Total			300.00

* An average number of 4 people per household was considered.

**Fragile neighborhoods are ABC, UREM and RENABAP⁶³, which are well defined geographically by AySA. Non-fragile or formal neighborhoods are traditional residential areas, with regular urban land title and access to at least two of the basic services (running water network, electricity network with a household meter and/or sewage network).

*** The Program will not finance intra-household connections in these neighborhoods but rely on natural connectivity ratios, which were estimated for each location based on AySA's experience.

59. **Verification Protocols of the Program.** A credible Verification Agent (VA) will be a firm, with proven experience in technical auditing in the three Results Areas of the Program. It will be competitively selected and contracted by AySA, under Terms of Reference (ToRs) acceptable to the Bank. The VA will be responsible for the credible verification reports. The VA will be responsible for: (i) the confirmation of DLI baselines; and (ii) biannual verification of DLIs. The Program Results will be consolidated and notified in a Monitoring Results Report by AySA to the VA, with a copy to the Bank. The VA will undertake the verification of such results and prepare a draft verification

⁶³ ABC areas are determined based on several indicators from the 2010 Census (the most recently available data): land tenure, quality of household construction, and the material deprivation index. The population of the ABC typology live on the periphery in areas of low urban consolidation and low real estate valuation. They present social vulnerability associated with low educational levels, high levels of dependency and low insertion in the formal labor market.

UREM is AySA's classification referred to those informal neighborhoods that have emerged as a result of the housing crisis, when people were unable to access the formal land market. They have groups of at least eight contiguous houses, where more than half do not have formal land tenure, nor formal access to at least two basic services (potable water network, gas network, electricity, or sewerage network). There are three types of UREM: villas, settlements, and housing complexes.

RENABAP refers to the similar typology of neighborhoods than UREM, but in this case is defined by Decree 358/2017. This database has been elaborated by the National Government instead, but it is also included in AySA's UREM database.



report, submitted simultaneously for review to AySA and the Bank. Neither party can modify such reports except to correct factual errors. For disbursement purposes, the Bank will make the final decision on whether Disbursement-Linked Results (DLRs) have been achieved, as provided for in Bank Policy -Program-for-Results Financing. A summary of the verification protocols for each DLR are presented in Annex 2. The full Verification Protocol is part of the Operations Manual. Prior Results under DLR 5.1 achieved by AySA between the date of the Program Concept Review, July 16, 2020, and the date of the signing of the Legal Agreement,⁶⁴ will be verified before loan signing by an interim third party verification agent (but not the VA referred to above) agreed by AySA and the World Bank.

III. PROGRAM IMPLEMENTATION

A. Institutional and Implementation Arrangement

60. **AySA will lead the implementation of the Program.** The Directorate for World Bank (WB) and CAF loans, which is housed in AySA's General Administrative Directorate, is in charge of all aspects of Program monitoring and reporting to the World Bank. This Directorate includes a Program Coordination Unit (PCU) which will be maintained the purpose of this Program. This unit currently houses one director, three managers, and 9 technical staff including mainly fiduciary, administrative and monitoring and evaluation (M&E) functions and serves as a coordination unit with all other Departments in AySA in charge of planning, implementing and monitoring of the PMOEM.

61. **The PMOEM is a responsibility of AySA's General Directorate of Planning.** The following departments within this Directorate have functions related to the PMOEM:

- (i) The Directorate of Technical Planning is responsible for the design of the PMOEM at a technical level. It works closely with the Directorate of Engineering and Design, which is responsible for the specific infrastructure projects after the planning stage.
- (ii) The Directorate of Coordination and Monitoring is responsible for monitoring implementation of the PMOEM.
- (iii) The Directorate of Community Development is responsible for community engagement and the development of socio-technical initiatives, such as A+T and C+T, hence leading activities linked to the expansion of WSS in vulnerable settlements. It includes the Community Promotion Office that manages the relationship with project affected people and the local community in general, in coordination with different regional directorates.
- (iv) The Technical Directorate for Technological Development is in charge of the updating of AySA's PPE and the Directorate of Information and Technology is responsible for the development and implementation of the digitalization strategy.
- (v) The Directorate for Sustainability is responsible for implementing a comprehensive Environmental and Social Management System for AySA and carrying out the environmental and social supervision of works.

62. **In addition, there are three other departments that are also related to the Program.** Procurement of contracts for large works is a responsibility of the Directorate of Contracts and Management Control, and the Directorate of Purchases and Storages is in charge of the acquisition of goods and services. The Hygiene and Safety Management Unit, within the Directorate of Human Resources, is responsible for ensuring the integrity of AySA's personnel and the Company's assets, preventing and minimizing occupational health issues and accidents, complying with and enforcing pertinent regulations, and implementing good practices on the matter. Second, the Directorate of Legal Affairs is responsible, among other functions, for the verification and monitoring of compliance with

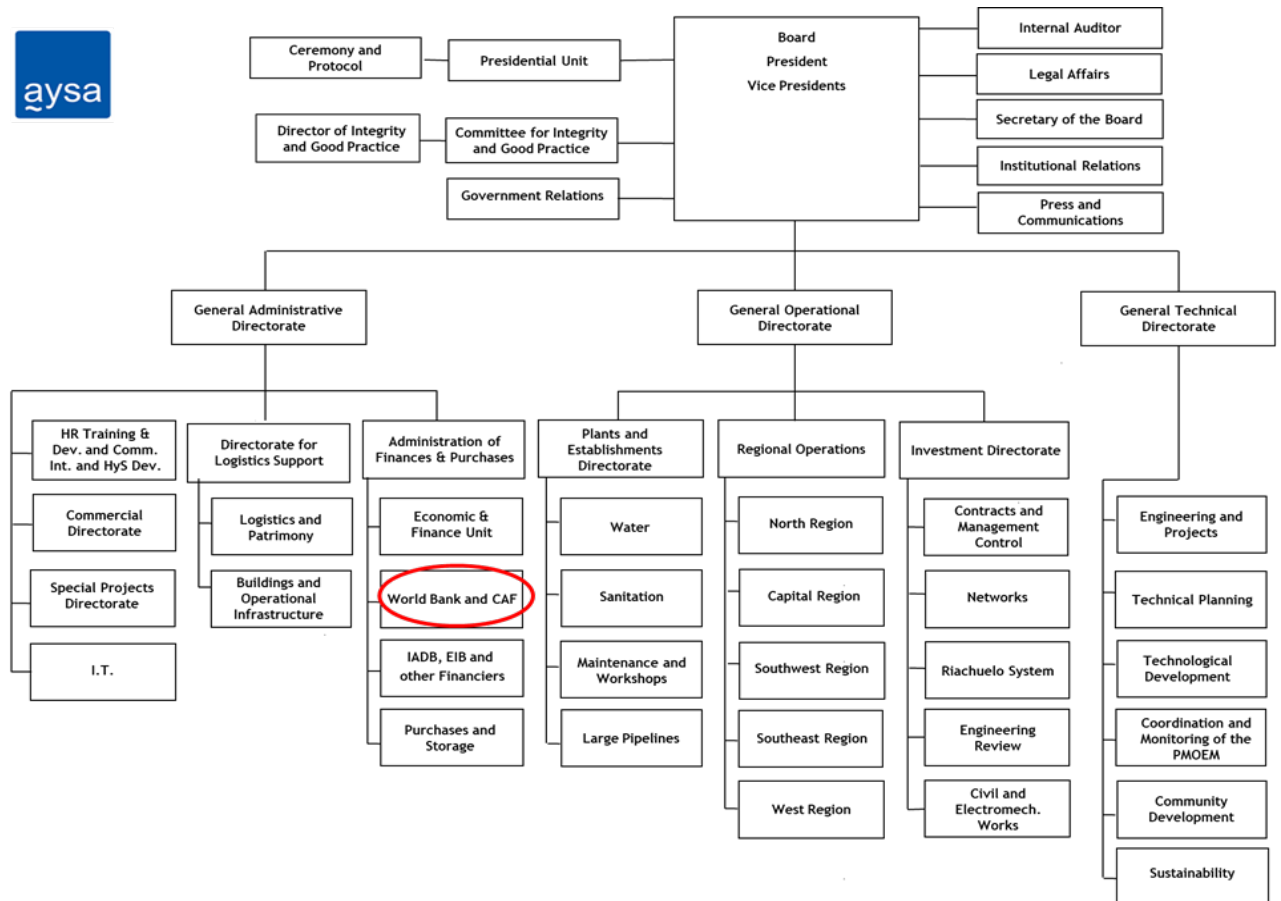
⁶⁴ In accordance with the World Bank Directive-Program for Results, para. 15, September 17, 2020.



environmental and health and safety legal requirements of AySA's facilities. And finally, the Directorate of Human Resources is responsible for human resources issues and will serve as the entity in charge of implementing the gender policies and guidelines.

63. Figure 2 helps elucidate AySA's structure.

Figure 2: AySA's organigram.



64. **The key responsibilities of the PCU include:** (i) coordinating overall implementation of the Program; (ii) preparing periodic monitoring reports on the progress of the different activities and contribution towards DLIs; (iii) monitoring progress of the different procurement processes; (iv) informing AySA's Board and the Secretariat of Strategic Matters under the President's Cabinet on progress in Program implementation; (v) ensuring that the different departments in AySA submit periodic follow-up and evaluation reports; (vi) ensuring that the designs and implementation of activities are performed in accordance with the Program's conditions agreed to by the Bank; (vii) specifying the reasons for plan deviations, if any, and propose remedial actions; (viii) preparing periodic Program Reports for the VA; (ix) preparing progress monitoring reports on compliance with the agreed actions in the PAP; (x) preparing financial statements of the Program; and (xi) preparing requests for disbursement of funds based on VA reports.



65. **The PCU has solid experience in implementing complex projects and is taking the necessary actions for programming and budgeting.** The PCU is the same unit in charge of coordinating all actions related to the Matanza-Riachuelo Basin Sustainable Development project (P105680), financed by the World Bank. It will ensure adequate support for administrative and financial management of the Program facilitating compliance with the agreed objectives; plan the calendar under which disbursements will be made according to the projections; and analyze and present the Program audited financial statements. The PCU will also coordinate with the corresponding departments in charge of Environmental, Health and Safety and Social management. The PCU will verify the degree of compliance with contractual obligations and keep the accounting records permanently updated.

66. **Flow of funds and Transfer Agreement.** AySA will act in close coordination with the Ministry of Public Works and specifically the General Directorate for Special and Sectoral Programs and Projects (*Dirección General de Programas y Proyectos Sectoriales y Especiales, DiGePPSE*), in relation to the release of funds required to finance Program expenditures as specified in Annex 4. A Transfer Agreement between the Borrower (through the Ministry of Public Works) and AySA will be signed to transfer the funds and ensure that the Program is carried out in accordance with all the provisions of the Loan Agreement, including the Operations Manual.

67. **Activities under the A+T and C+T programs are implemented in coordination with the corresponding municipalities.** Prior to carrying out activities with a selected municipality, an agreement (Municipality Agreement) will be signed between AySA and each selected municipality. These agreements include the obligation by each municipality to ensure that their respective activities under the Program are carried out in accordance with the Operations Manual and the Anti-corruption Guidelines.

Table 6: Responsibilities and implementation arrangements for intra household connections

Stakeholder	Main responsibilities
AySA	AySA is responsible for: (i) preparing relevant studies (including social viability analyses); (ii) determining the technical feasibility of works; and (iii) managing the corresponding approvals from municipal authorities. AySA is also in charge of training cooperative members, designating site inspectors and approving the executed works. Once approved, AySA notifies APLA that it is incorporating the works into the concession, then operates the service and bills users according to the approved tariff structure.
Municipalities	In their capacity as executing agencies of the works, they are responsible for: (i) identifying and prioritizing project locations; (ii) presenting a proposal of planned works to AySA and requesting the associated funding; (iii) collaborating on the formation and registration of work cooperatives; (iv) managing the execution of the works, through professionals hired for these tasks; (iv) presenting the executed works to AySA for approval, and collaborating with the service provider to ensure the effective incorporation of the beneficiaries to the network; and (v) assisting in sealing any alternative water sources used by households before they are connected to the network.
Cooperatives	Their core responsibility is the correct performance and execution of works financed by the Program. Furthermore, they must guarantee the attendance of their members in training activities. Cooperatives assume full and exclusive responsibility for compliance with legal, tax, social security and any other regulations that govern their actions.
APLA	Monitors the progress of A+T and C+T Plans, execution of the works, and the level of investments.
ERAS	Responsible for assessing the beneficiaries' eligibility for the Social Tariff, when applicable.



B. Results Monitoring and Evaluation

68. **The Results Framework provides key measures of Program achievements against stated objectives, indicates the Program's contribution to AySA's higher-level objectives, and provides evidence for decision making and policy formulation.** AySA will retain overall responsibility for Program results M&E. For the construction works, AySA is equipped with an M&E system (*Sistema de Seguimiento de Obras - SSO*) housed within its Investments Directorate. In order to track progress under the PMOEM for each investment, AySA designates inspection units in charge of works and contract supervision. Although these inspection units are usually staffed by AySA employees, services for large infrastructure investments can be contracted externally, when in-house capacity is insufficient and/or to ensure compliance with multilateral investment bank requirements. Data collected by these units is recorded in the SSO, which reports on progress and physical advances. The PCU will continue to aggregate M&E data in the SSO providing evidence of progress on the Results Framework indicators, with emphasis on results related to the DLIs.

69. **AySA will produce Program Monitoring Reports** before the start of each verification period, summarizing the progress in achieving results up to the specific verification date. The reports will be submitted to the VA, as the basic document to work with before conducting the actual verification as per the instructions stated in the Verification Protocol. The Program Monitoring Reports will also include information on progress made in complying with the actions in the PAP. A template of the Program Monitoring Report will be included in the Operations Manual.

70. **Incorporation of the Geo-Enabling initiative for Monitoring and Supervision (GEMS) approach.**⁶⁵ Given the current COVID19-related safety measures and mobility restrictions, the Program will adopt the approach and tools developed by GEMS to collect Program baseline data and track achievement of results. During implementation, the focus will be to inform relevant aspects of results verification (collection of data and information needed to inform the verification protocol), operational M&E, supervision, fiduciary, and environmental and social risk monitoring; and to promote the mapping of operational sites. Use of the GEMS tools and methods will enhance the transparency and accountability of Program implementation.

C. Disbursement Arrangements

71. **Disbursement requests or Withdrawal Applications (WA) will be submitted to the Bank by AySA using the Bank's e-disbursement (Client Connection) system.** A copy of the Bank's official communication, confirming DLI achievement, should be attached to the disbursement requests.

72. **The Program intends to use the functionality of PforR Advances.** The Bank may make an advance payment of up to US\$75 million (maximum) for one or more DLIs, which have not yet been achieved. To request an advance, AySA attaches a "request for advance" letter to the WA, as per the sample included in the DFIL. When DLIs are achieved, the amount of the advance is recovered from the amount due to be disbursed under such DLIs. The advanced amount recovered by the Bank is then available for additional advances ("revolving advance"). When an advance has been provided and the DLIs are achieved and verified, the advance is always recovered first. If the amount allocated to a DLI that has been achieved and verified is larger than the advance, the Bank will disburse the amount in excess of the advance through a Reimbursement (DLI Payment). The Bank requires that AySA refund any advances (or portion of) if the DLIs have not been fully achieved by the Program closing date.

73. **Due to exchange rate pressures, there is a risk that the amount of Program Expenditures once converted to United States dollars (US\$) may be insufficient to cover the withdrawn loan balance in US\$ terms.** This risk will be closely monitored during implementation, not only during supervision missions, but also through the review of the

⁶⁵ <http://documents1.worldbank.org/curated/en/271431561153010274/pdf/Geo-Enabling-Initiative-for-Monitoring-and-Supervision-GEMS.pdf>



semesterly Interim Financial Reports (IFRs), together with the flow of funds and accounting arrangements described further. The Bank's loan will be disbursed via the Central Bank of Argentina to a segregated US\$ denominated bank account held by the Ministry of Public Works (*Ministerio de Obras Públicas*, MOPW) at the National Bank (*Banco de la Nación*). The US\$ will then be converted to Argentine pesos (ARS) as required to meet payment obligations and to be deposited in a segregated ARS denominated account held by the MOPW (also held at the *Banco de la Nación*). From this account, funds are then transferred to a separate ARS denominated AySA physical bank account (also held at the *Banco de la Nación*) set up for this specific programmatic/financing purposes. From this separate bank account, payments will be made to suppliers/vendors/contractors (and to the municipalities to pay for the works of the cooperatives under the A+T and C+T programs)⁶⁶. This arrangement will allow AySA to account for and know the exact exchange rates to use to record Program expenses (based on US\$ received), thus eliminating refund problems at the end of the Program, due to possible exchange rate differences.

74. **In addition, an amount of US\$15 million will be disbursed on account of the DLIs met by AySA between the date of the Program Concept Review** (July 16, 2020) and the date of the signature of the Legal Agreement (Prior Results Financing), subject to the corresponding verification.

75. **There will be a total of 12 DLIs.** The financing amount allocated per DLI has been determined based on the relative importance of the indicator to provide the incentive necessary to achieve overall Program goals and outcomes. There are no DLIs defined as being time bound. The DLIs do not have a firm achievement date attached to them—they are achievable at any time during the Program implementation period. In this case, AySA can request disbursement against the achievement of these DLIs at any point after its achievement has formally been verified. Nevertheless, verifications are scheduled to take place on a semi-annual basis (i.e., the timing of disbursements) in order to ensure that AySA has the required budget predictability and flow of funds. DLIs 1.1, 1.2, 2.1, 2.2, 4.1, 4.2, 6.1, 6.2 and 7 are scalable – with the disbursement of the financing proceeds proportional to the progress towards achieving the DLIs, whereas DLIs 3, 5.1 and 5.2 are not scalable. For each DLI with scalable disbursement, the formula to determine the amount of financing proceeds to be disbursed relative to the level of achievement of the DLI is specified in the DLI Matrix in Annex 2.

D. Capacity Building

76. **The Program will be supported by several activities designed to strengthen AySA's capacity and address weaknesses and gaps identified during Program preparation.** The analysis, which led to the identification of these activities, is laid out in the respective Technical, Environmental and Social Systems and Fiduciary Systems assessments. The Program will leverage the global knowledge and expertise from a number of Water Global Practice (GP) initiatives to strengthen AySA's capacity to take on current and new challenges associated with the delivery of safely managed water and sanitation services, as well as with the post-COVID-19 recovery phase. This Technical Assistance program will include support in initiating a transition towards becoming a Utility of the Future – a global Water GP initiative, with particular emphasis on: (i) improving connectivity to sewerage networks in low-income areas, including behavioral and communication aspects, supported by the "Citywide Inclusive Sanitation" global initiative; (ii) incorporating digital solutions and associated institutional aspects, with support from global experts; and (iii) addressing gender gaps, with support from the gender experts, and the "Equal Aqua" global collaborative platform.

⁶⁶ The Municipalities forward the certified works certificates to AySA, for AySA to transfer the required funds to the Municipalities, who then make payment to the work cooperatives



IV. ASSESSMENT SUMMARY

A. Technical (including program economic evaluation)

77. **The Program's strategic relevance is high, as it makes clear contributions to key areas of AySA's core expansion and improvement plans as laid out in the PMOEM.** The Program is aligned with the PMOEM's three Sub-programs and focuses on essential objectives at the core of AySA's mission, that is, to speed up the provision of safely managed WSS services in a sustainable and efficient manner to the peri-urban areas of Buenos Aires, where a large share of the population lives under the poverty line. The Program acquires even greater relevance in the context of the COVID-19 pandemic, where the impacts of the lack of access to safe water and sanitation in vulnerable neighborhoods have become more evident than ever, as these are basic barriers needed to slow down the spread of the virus.

78. **From a long-term perspective, the Program is also designed to increase the operational and financial sustainability of AySA in order to decrease its dependence on federal budget transfers.** The results sought by the Program will serve as steps in the right direction in a challenging macro-economic context. Setting up solid paths to reduce large water leaks and energy consumption, and to digitalize payment methods will reduce costs and increase revenues, and thereby help AySA climb the financial sustainability ladder, allowing it to better utilize subsidies to expand services faster to those who most need them. Moreover, the Program also supports essential modernization processes to make the company more agile in its capacity to respond to shocks.

79. **The technical soundness of the Program is considered solid.** The selection of activities to achieve expected results is adequate. AySA has a long track record of good operation and maintenance of facilities, giving no reason to consider substantial risks in that regard. Moreover, the designs of infrastructure will take into account adaptation measures to limit climate-related risk.⁶⁷

80. **The A+T and C+T programs have long track records of building networks and connecting households in informal settlements.** AySA's proposed evaluation of the potential beneficiaries is comprehensive as it addresses structural and socio-demographic aspects of the population. The nine pre-defined water and sanitation technical connection typologies are appropriate for informal settlements and ensure that the completed infrastructure will comply with existing regulations. An estimated 1,200 workers are expected to be employed in 30 cooperatives in low-income neighborhoods, which is well justified in the context of post COVID-19 economic recovery. Importantly, AySA will provide training to the cooperatives to ensure they are equipped to handle the socio-technical features of the program, including consideration of gender aspects.

81. **It is not uncommon to find households in these informal settlements that are located in flood-prone areas.** This risk will be assessed carefully before providing formal services. This consideration will be included in the DLI verification criteria, which will not recognize household connections in flood-prone areas, unless adequate flood mitigation plans are adopted. This will also incentivize AySA to build connections in climate-safer areas contributing to the adaptation agenda.

82. **The Bank assessed the technical designs for the optimization and extension of the two existing Jagüel and Norte WWTPs** and determined that the proposed technology (activated sludge process) and the quality of the designs is adequate, confirming the proposed modality of contract (works) and the proposed 2-year construction timeline. However, AySA may be able to further optimize designs by performing a more rigorous analysis of technological alternatives during the preparation stage.

⁶⁷ A typical intervention is the elevation of treatment plants and pumping stations.



83. **AySA's program to reduce real water losses is consistent with the characteristics of the water network.** More than half of the water loss reduction will be attained through pipe replacement, which is appropriate given the combination of materials (asbestos cement and cast iron under 100 mm diameter) and age of the network (more than 100 years in many cases). Leaks will also be detected using satellite detection techniques. The methodology was developed following internationally adopted criteria and the actions to achieve the results are proven including previous cost-effective interventions implemented by AySA. AySA intends to measure progress in water savings from upgraded pipes by using past pilot studies relating length of pipes replaced with volume of water recovered. This linkage will be re-assessed during Program implementation. The actions related to distributed energy monitoring in deep wells and sewage pumping stations are the fourth and fifth stages of an ongoing program that has already shown good results, although AySA expects better results from these stages due to the higher power consumption of the targeted facilities.
84. **The investments associated with Results Area 3 to shift the mix of billing methods through the development of digital solutions and a mobile app for citizen engagement constitute an adequate "modernization" package.** These actions are comparable to measures implemented in other regional utilities of similar size undergoing systems and customer management modernization. The implementation of these digitalization processes will require coordination under a technical and commercial leadership. Furthermore, social and communication dialogue with users is as important as the technical challenges, and AySA is adopting the right approach to engage with them through digital channels.
85. **AySA's Prevention and Emergency Plan (PPE) was first implemented thirty years ago and has evolved significantly since then.** The proposed contingency plan for epidemiological risks is anchored in the three main phases of the PPE: (i) plan and prepare; (ii) emergency management; and (iii) recovery to normal activity. The PPE was activated as a response to the COVID-19 outbreak, and applied in line with good regional practices, as well as with the World Health Organization's Technical Guidance on "Water, sanitation, hygiene, and waste management for SARS-CoV-2" of July 2020. AySA implemented technical, commercial and organizational measures to protect staff while maintaining and even expanding services in emergency areas with higher risks of transmission. The contingency plan to be prepared will recognize the possibility of a similar risk materializing, by considering the experience gained and lessons from the current emergency.
86. **Data submitted by AySA through the World Bank's Human Resource Questionnaire on gender diversity in water utilities shows that women are underrepresented in managerial positions at AySA.** Given the differences in the overall number of female and male staff, the target set in DLI 7 is high. The Program plans to monitor achievements on a yearly basis and will benefit from setting achievable annual improvements in promotion rates to drive steady change, considering current challenges posed by the COVID-19 pandemic, which might prevent women from seeking more responsibilities during the first year of the Program.
87. **The level of readiness of Program activities is high in Results Area 1, as well as for some activities related to Results Areas 2 and 3.** In addition to the prior results achieved as contingency measures in response to the COVID-19 pandemic, AySA is starting the bidding process for the optimization of the existing modules of the Norte WWTP. The final design for the plant's module 3 has been finalized, and the engineering designs for Jagüel will be ready in March 2021. Several works contracts for the corresponding sewerage networks are also underway. As for intra-household connections, eligible municipalities, as well as vulnerable settlements within them, have already been identified, and designs for water and sewerage connections are also ready. AySA has also started the development of the Citizen Engagement Strategy, and the first version of the mobile app is expected to be completed by mid-2021. Policies to address gender and equity issues have been outlined as part of AySA's Action Plan for Gender Equality. Capacity building activities for the utility's personnel as well as other actions to achieve a greater representation of women are already under implementation.



88. **Program Expenditure Framework. Regarding the transfers of funds to AySA from the National Government, disaggregation for program expenditure monitoring is made by AySA using its processes and systems.** At the national level, no distinction is made of the financing being provided for PMOEM as PMOEM is not a separately identified line item in the National Budget. AySA's budgeting and accounting information systems are able to track the expenditures incurred under the Program using accounting policies consistent with national public sector policies/standards. However, close implementation support will need to be provided by the Bank to help establish the arrangements to produce the Program Annual Financial Statements. Regarding the Expenditure Categories by Program Results Area, the nature of the expenditures in Results Area 1 to be incurred is predominantly civil works contracts in specific municipalities of AySA's jurisdiction with the associated operating costs of the main WTP and WWTPs. Results Areas 2 and 3, in addition to works contracts and corresponding operation costs, also include goods contracts. The expenditure framework table indicating main expenditure categories is included in Annex 3.
89. **The expenditure's performance is considered adequate regarding the execution rates proposed for the different activities, which fall within the implementation period of the Program.** The activities which take longer to implement are the new modules of the Jagüel and Norte WWTPs, (two years) which is within normal ranges for works of this magnitude. There is a risk of delays in the execution of the intra-household connectivity program since part of the works need to be conducted inside the households and dwellers may feel reluctant to allow workers inside during the pandemic. The main mitigation measures are the adoption of safety measures following AySA's health protocols, the promotional campaigns conducted in the beneficiary areas before the start of the works, and the extension of the timeline to execute these works by 6 months, from 1.5 to 2 years.
90. **Program expenditures are efficient for both the infrastructure activities under Results Areas 1 and 2 (pipe replacement, leak reduction and energy efficiency) but also for soft activities in Results Area 3.** For the larger Program expenditure corresponding to the water and sewerage infrastructure investments (US\$312 million), unit costs are rational with US\$430 per capita, considering the full capacity of the newly built facilities in the long term.
91. **The financial sustainability of AySA is being strengthened to improve operating cost coverage (including maintenance, rehabilitation and depreciation) and reduce its dependency on national funding, which may be subject to the country's macroeconomic situation.** Through this Program, AySA is seeking efficiency gains as a means to improve its financial sustainability and cost-recovery ratio via the implementation of an adequate asset management strategy to prioritize rehabilitation works, lower water losses and improve energy efficiency.
92. **The PMOEM is the vehicle through which AySA plans its investments and has been funded by the National Government since 2010.** The PMOEM is financed mostly by source 11 (government transfers) and partly financed from Source 22 financing (external aid), which evidences support of donors' aid flows to the Program. The unplanned incorporation of nine municipalities in AySA's provision area between 2016 and 2018 coupled with the macroeconomic situation during this period increased operational expenditures and budgetary pressures, further reducing AySA's capacity to invest and fully fulfil the PMOEM's target during that period. However, WSS is a National Government priority, with AySA having the largest share of water and sanitation investments in Argentina with 0.08 percent of the GDP over 2017-2019. The government has reiterated its commitment to make transfers readily available, ensuring that fiscal constraints will not interrupt social subsidies targeted to vulnerable populations for basic services such as water and sanitation.
93. **AySA's M&E arrangements discussed in the previous sections are deemed adequate.** Smooth coordination will need to be ensured among the different AySA departments responsible for the different Program activities. The implementation of the GEMS tool for both Program supervision and verification of results is expected to bring agility in monitoring Program performance and will require a series of capacity building sessions to ensure its full adoption.



94. **The economic and financial analysis focused on eight projects that represent 91 percent of the investments in this Program.** For both the economic and financial analysis, the Net Present Value (NPV) is calculated with a discount rate of 6 percent and with a 30-year horizon.

95. **For the economic analysis, the NPV of the Program is positive (US\$1,433 million) with an Economic Internal Rate of Return (EIRR) of 40 percent.** A cost benefit analysis was carried out and the costs were estimated at economic prices (market prices were adjusted with shadow price coefficients). Results Area 1 shows the highest benefits, which can be partially explained by existing sunk costs from prior investments in the network expansion. The benefits of the expansion projects are also considerable because they benefit densely populated areas with significant environmental problems. The beneficiary areas are low-income compared to AySA's area of coverage and these users, relative to their income, value the water service more highly.

96. **Despite the conservative assumptions used for benefit estimation, projects in Results Area 2 have an economic rate of return above the discount rate.** In addition to the quantified benefits in terms of savings in water production, energy consumption and system maintenance, other benefits include an increase in the useful life of the assets, costs savings from avoiding reactive maintenance and the reduction of interventions in public roads due to water losses. Given the current levels of operational efficiency, these projects are expected to strengthen AySA's financial sustainability in the long term due to operational cost savings leading to a reduction in its dependence on subsidies.

97. **A GHG analysis was carried out on the Program's expansion of sewerage service through the rehabilitation and renovation of Planta Norte y Jagüel WWTPs.** The improvement in NRW and energy efficiency actions in networks, will lead to reductions in GHG emissions due to efficiency gains. The NPV of the Program includes US\$66 million from GHG emissions avoided.

98. **The results of the Program's financial analysis reveal a positive NPV of US\$88.4 million,** largely driven by the operational efficiency improvements and the NRW reductions supported by the Program. This analysis takes AySA's latest financial statement as a baseline and projects future cashflow, taking into account the expected change in revenue and costs resulting from the Program's implementation. It considers operating flows since capital costs are directly subsidized by the National Government through transfers.

B. Fiduciary

99. **The Procurement and Financial Management (FM) systems' capacity and performance, with the implementation of the proposed mitigating measures and agreed actions to strengthen the systems (which are reflected in the PAP), are adequate to provide reasonable assurance that Program funds will be used for the intended purposes,** with due attention to the principles of economy, efficiency, effectiveness, transparency, and accountability.

100. **There are no potential high-value contracts identified under the Program.** The proposed Program is not expected to finance any contract at or above prevailing Operations Procurement Review Committee (OPRC) thresholds considering the estimated activity costs under Substantial risk, which at the time are at minimum of (i) US\$75 million for works; (ii) US\$50 million for goods, information technology and non-consulting services; and (iii) US\$20 million for consulting services.

101. **In order to implement the Bank's Anticorruption guidelines for PforR operations, it has been agreed that AySA will immediately inform the Bank of any complaint or claim related to fraud and corruption which AySA either receives or of which AySA becomes aware.** AySA will provide the Bank with all the records, documentation and information that the Bank may request with respect to such issues. In the event that the Bank decides to conduct its own investigation, the Bank may request the government and/or AySA to exercise its legal rights and remedies (under the relevant contract/s) so as to obtain all information, records and documentation that the Bank may request, and



provide these to the Bank. This process does not limit the rights of the Bank to also make direct requests for information from individuals or firm who are recipients of Bank financing. In line with the obligations arising under the Anticorruption Guidelines, the government and AySA should ensure that individuals or contractors who are recipients of Bank financing are aware that the Bank may decide to exercise this option.

102. **If the Bank determines that it has not been able to receive the documentation, records or information requested by the Bank directly and/or through AySA, the Bank may declare the relevant expenditure ineligible for Bank financing under the Program.** Furthermore, should the Bank conclude that a sanctionable offense has occurred, it may decide to pursue sanctions against the individual or firm in line with the Bank procedures.

103. In addition, the external auditors' TOR will include a requirement to review Program expenditure to confirm that contracts are not awarded to debarred or suspended individuals or firms. AySA will immediately report any allegation of fraud and corruption to the Bank, through exchange of letter, and every semester (together with the Program Monitoring Reports) a report will be prepared containing all alleged cases, with an updated status of the respective actions taken.

104. Based on the specific needs identified to take the responsibility for the fiduciary functions under this Program, certain actions are recommended in the PAP to reduce risk and facilitate Program's implementation.

C. Environmental and Social

105. **The overall environmental and social impact of the Program is expected to be positive.** The Program's Environmental and Social System Assessment (ESSA) found that no significant adverse environmental impacts are expected as a result of the proposed interventions under the Program to support the increase of the supply and quality of drinking water and the expansion of sanitation services in targeted areas. The main positive environmental effects are related to the adequate collection, conduction, treatment and final disposal of a greater volume of sewage, with the consequent reduction of soil and groundwater contamination; the improvement of health conditions for the new population served; the environmental enhancement of the intervention areas from the blocking of septic tanks; among others. The reduction of water losses and energy consumption also entails environmental benefits. The Program would likely have overall positive social impacts on the lives and livelihoods of large vulnerable communities through the expansion of safety managed water and sanitation services, which will also improve the quality of life and reduce WASH-related health risks in targeted neighborhoods. The Program will also likely increase local employment through participation in cooperatives, thus generating income opportunities and promoting social inclusion. The Program will also support AySA's citizen engagement and beneficiary feedback mechanisms, which will help to improve the quality of the service, and AySA's readiness for emergency response.

106. **Environmental risks.** The overall environmental risks and potential impacts of the Program are considered Substantial, basically in connection with interventions in existing WWTPs given the risk and potential impacts associated to civil works in the construction stage (soil movements; materials transportation and handling; noise, dust, debris generation; among others) and, once treatment capacity is expanded, to the increased volume of treated effluents to be discharged to water bodies during the operation stage. However, the proposed works are of a limited scale, will increase the level of treatment of effluents, and will take place within AySA's premises. Furthermore, they will not involve new discharge pipelines or discharge points, which means that the risks and potential environmental impacts to the water bodies where the treated effluents are discharged have been taken into account and, accordingly, evaluated in the original Environmental Impact Assessments (EIAs) of the WWTPs. The original EIAs considered the modular design of the plants (i.e., to be built in stages) and the maximum volume of effluents once all modules are constructed and fully operational. The original EIAs will be updated and presented to the environmental authorities, as part of the renewal of the environmental licenses. The main specific risks that would need to be managed under the Program include: (i) increased health risks due to worker and community exposure to COVID-19; (ii) community and occupational health and safety risks – especially during construction and particularly associated



with the handling of materials with asbestos content; (iii) risks due to interventions in areas with potential physical cultural resources chance findings; and (iv) risks related to potential interventions in zones with conservation values within AySA's concession area. Although AySA's service area has high exposure levels to floods, the possible impacts are already mitigated by AySA and will be further reduced by the actions proposed in this Program, thus the climate and disaster risk rating is *Moderate*.

107. **Social risks.** The overall social risks and potential impacts of the Program are considered *Moderate*, and basically related to the incorporation of new low-income users who would not be able to pay for the WSS services. Other probable negative impacts are associated with the potential temporary displacement of informal economic activities, the possible temporary restriction of access to entry to commercial business during construction (already covered by the Contractors' obligations included in the bidding documents), the short-term interruption of pedestrian traffic, and the potential impact on community services and/or infrastructure. Although there are some urban indigenous communities present in the GBA, the Program's interventions will not generate any negative impacts on these communities nor will they involve any exploitation of their natural resources or knowledge.

108. **National / local / AySA's system to address environmental and social risks.** The ESSA concluded that Argentina has well-established systems to address these environmental, health and safety risks, including several strong central, provincial and municipal level legislations and institutions, as well as an effective regulatory sectoral framework that govern environmental, occupational and community health & safety issues. These are complemented by AySA's corporate environmental, social and health & safety systems. In particular, AySA, through its Directorate for Sustainability, has developed Environmental and Social Technical Specifications (*Especificaciones Técnicas Ambientales y Sociales*, ETAS) that are part of the bidding documents and are included in the contracts once the works are awarded. The ETAS establish good practice standards and procedures as requirements for the contractors to address environmental and social issues during the execution of the works. Also, AySA has developed a series of documents regarding Occupational Health and Safety (OHS) following Argentina's sound legal framework on the matter, which details the requirements for the contractors including the need for a risk assessment of their tasks and the development of a risk management plan. Specific measures recommended in the ESSA to manage identified environmental risks under the Program include: (i) ensuring the implementation of the Argentinian COVID-19 related regulations and corresponding AySA protocols for contracted workers and own staff, following the evolving pandemic and post-pandemic situations; (ii) ensuring the implementation of the national regulation on health and safety and the corresponding AySA's systems, improving them for the proper handling of materials and waste with asbestos content (including potential chance findings during excavation and demolition); and, (iii) strengthening AySA's risk management related to physical cultural resources and sensitive areas located within the urban zone of intervention. Specific measures to manage social risks include: (i) the adaptation of practices and procedures to the realities of low income neighborhoods to guarantee their sustainable access to WSS services (e.g. through the implementation of differentiated programs); (ii) the incorporation of processes to handle potential impact of works on community services and/or informal economic activities in the Program's ETAS; and (iii) the active involvement of municipal government environmental and social teams.

109. AySA also has systems and mechanisms in place to address the social risks identified above. It has a Social Tariff scheme to support low-income populations in its concession area, supports debt cancellation for low-income households unable to pay outstanding water bills and provides subsidies for low-income households to connect to the network, including a 50 percent discount in new connection costs to incentivize their connection to the network and additional subsidies of up to 100 percent of the tariff and free network connections to households in extreme poverty. The ESSA recommends measures to address social risks. Regarding Indigenous Communities, AySA does not have a management instrument specifically targeting indigenous communities but includes them as vulnerable people in general. The ESSA identified opportunities to increase information gathering and analysis about indigenous communities and potential barriers to access water and sanitation services; and ensuring that they have adequate



information to access the Communal Tariff Scheme⁶⁸. In addition, as a result of the analysis, it would be important to implement the necessary adjustments in communications and participation processes that will include indigenous peoples, to make sure that these are conducted in a culturally appropriate manner. The ESSA identified the need for AySA to formally ensure that contractors prepare a code of conduct to manage labor influx risks. Finally, the ESSA recommends that ETAS incorporate measures to manage any potential temporary displacement of informal economic activities. The draft ESSA was disclosed on January 5, 2021 and consulted on January 11, 2021 with two events held with key institutional stakeholders, cooperatives and representatives of civil society organizations. The draft ESSA was also available to a larger group of stakeholders for virtual comments and feedback. A report on the consultations, the feedback received and how it was addressed is included in the final ESSA, which is disclosed prior to Board approval.

110. Communities and individuals who believe that they are adversely affected as a result of a Bank supported PforR operation, as defined by the applicable policy and procedures, may submit complaints to the existing program grievance redress mechanism or the WB's Grievance Redress Service (GRS). The GRS ensures that complaints received are promptly reviewed in order to address pertinent concerns. Affected communities and individuals may submit their complaint to the WB's independent Inspection Panel which determines whether harm occurred, or could occur, as a result of WB non-compliance with its policies and procedures. Complaints may be submitted at any time after concerns have been brought directly to the World Bank's attention, and Bank Management has been given an opportunity to respond. For information on how to submit complaints to the World Bank's corporate Grievance Redress Service (GRS), please visit <http://www.worldbank.org/GRS>. For information on how to submit complaints to the World Bank Inspection Panel, please visit <http://www.inspectionpanel.org>.

D. Climate Change, Gender and Citizen Engagement

111. **Climate Change.** Buenos Aires metropolitan area is naturally exposed to floods, which may increase in frequency and intensity due to climate change. Vulnerability levels in low-income neighborhoods are compounded by different factors, including exposure to flood events, which can cause significant disruptions in the form of loss of assets and well-being. By providing access to safely managed water and sanitation services, and incentivizing the adoption of flood mitigation plans in flood prone areas, the Program will contribute to reduce overall vulnerability levels in these neighborhoods. In addition, AySA's main assets are resilient, as the utility has clear criteria on flood-proof engineering designs to protect the infrastructure both during the construction and the operation phases. The digitalization activities proposed will further increase AySA's adaptive capacity through improved network monitoring and analysis and operational flexibility. From the operational perspective, the energy efficiency and reduction of NRW activities will also lead to GHG mitigation outcomes that will contribute to Argentina's national climate agenda.

112. The Program has been designed taking into account the following commitments related to Climate Change:

- (i) climate and disaster risk screening: the Program has been screened considering (i) the exposure levels in the Program areas and possible impacts in the beneficiaries, and (ii) the mitigation measures defined by AySA and incentivized by this Program for Results, through the corresponding DLIs (1, 2, 3 and 4.1). The consequent climate and disaster risk rating is considered moderate.
- (ii) greenhouse gas (GHG) accounting: the Program GHG accounting was carried out highlighting savings for 871,950 tCO₂-eq over the lifetime of the process. In particular, savings are divided between

⁶⁸ The Communal Tariff Scheme Program is aimed at civil society organizations. It is available to (i) public good entities that prove that they are civil associations; (ii) simple associations that do not pursue profit-making directly or indirectly; (iii) non-profit community organizations; (iv) universities; (v) public schools; (vi) neighborhood sports entities; (viii) theatrical halls; (ix) public hospitals; (x) soup kitchens; and (xi) other users, according to the ERAS evaluation. The types of benefits include (a) access to the service for Institutions that are not connected to the water and / or sewer service (payment of percent the value of the price per cubic meter); (c) debt restructuring, to enable Institutions with debt to cancel their commitments according to their economic possibilities (payment of 50 percent of the original debt, without interest or surcharges, in 12 monthly installments).



activities related to wastewater treatment (710,100 tCO₂-eq), non-revenue water (111,850 tCO₂-eq), and energy efficiency (50,000 tCO₂-eq).

- (iii) shadow price of carbon application (to GHG emissions and reductions). The shadow price of carbon was applied in the economic analysis, accounting for benefits estimated at US\$66 million through the lifetime of the Program.
- (iv) climate adaptation and mitigation co-benefits tracking. The climate mitigation and adaptation benefits were assessed at Program preparation. PDO indicators 2 and 3 are considered the climate-related indicators, as they capture the majority of the co-benefits for the Program. Moreover, intermediate indicators on the number of wells where energy monitoring equipment will be installed, the number of large consumers tele-metered, and number of micro-meters installed can be considered climate-informed indicators as they are also contributing to the mitigation effort.

113. **Gender.** The Program has identified a key gender gap in terms of women's representation in leadership positions at AySA. Numerous factors contribute to this gap including lower rates of female promotion, and gender norms deterring female employees from applying to technical and managerial positions. AySA is taking a series of actions to address the gender gaps in managerial positions under its holistic 2020-2023 Gender Action Plan. These include setting gender targets for staff promotions, launching a targeted communication campaigns to motivate women to apply to management and technical positions, and offering targeted training to fill existing capacity gaps in women's technical and leadership skills. The impact of these actions will be assessed through DLI number 7, which will measure the increase in share of women in leadership positions at AySA. With the support of the Program, AySA hopes to increase women's representation in leadership positions to 28 percent. This in turn, is expected to contribute to improving AySA's capacity to respond to the differential needs of women in its emergency response efforts.

114. **Citizen Engagement.** AySA already regularly conducts and (through the regulator) publishes satisfaction surveys, and through its programs to extend services in low-income areas has integrated approaches to engage communities up-front. However, AySA has recognized the importance of improving and updating its capacity for two-way engagement with citizens, particularly under the constraints posed by COVID, and of investing in the internal systems that will facilitate better tracking, management, analysis and therefore accountability and responsiveness around citizen complaints and feedback. As noted above, AySA will develop a digital channel for community engagement, introduce systems that allow for more open feedback on an ongoing basis (PQRS) including suggestions, reform its internal systems to better track and analyze complaints, and improve response times to customer complaints. These efforts will be tracked through PDO indicator 4, and DLI 6.2, which will disburse progressively as AySA achieves these milestones.

V. RISK

115. **The overall risk of the Program is rated as Substantial.** The key risks are summarized as follows:

- (i) **The macroeconomic risk is considered High.** Despite the recent signs of growth and the agreement between the Government of Argentina and its bondholders to restructure its external commercial debt, the high-risk assessment relates to possible scenarios of exchange rate volatility, high inflation, and overall macroeconomic uncertainty. These could lead to changes in relative prices and in Program costs expressed either in US\$ or ARS, constraints on co-financing requirements and budgetary transfers. There is also uncertainty about future AySA tariff increases—and hence the need for continued government transfers to cover a portion of operating costs—which was considered with conservative assumptions in the financial assessment. To mitigate macroeconomic risks to the Program, disbursements in the DLI matrix have been structured to ensure adequate cashflow to the program. In addition, a considerable amount of AySA's-funded



investments is included in the Program as eligible expenditures. This will facilitate the estimation of Program expenditures equivalent to, at least, the amount of the Bank loan at the end of the Program. Furthermore, the WSS sector is a priority for this government, which is committed to finance additional investment to speed up access to water and sanitation in the vulnerable areas. Notwithstanding these mitigation measures, the team's assessment is that the residual risk remains *High* while the rating may be reassessed during project implementation according to macro-economic conditions and the continued commitment of budgetary transfers to the sector.

(ii) **Environmental and Social risk is assessed as *Substantial*.** As explained above, although the social risk is rated *Moderate*, the *Substantial* risk rating is driven by the overall environmental risks and potential impacts of interventions in existing WWTPs, given the risk and potential impacts associated to civil works in the construction stage, and, once treatment capacity is expanded, to the increased volume of treated effluents to be discharged to water bodies in the operation stage. However, the construction and operational environmental aspects of these works will be managed applying the EIA provisions. Some of the main risks that would need to be managed under the Program are: (i) increased health risks due to worker and community exposure to COVID-19; (ii) community and occupational health and safety risks – especially during construction and particularly associated with the handling of materials with asbestos content; and, (iii) risks associated with the potential temporary displacement of informal street vendors. Argentina has well-established systems to address these social, environmental, health and safety risks, complemented by AySA's corporate systems. In addition, the Program provides for the adoption of a Program Action Plan (PAP) that includes additional measures that are needed to strengthen AySA's systems to manage and mitigate these risks.

(iii) **The overall integrated fiduciary risk rating is considered *Substantial*.** The key fiduciary risks to the development outcomes of the Project that underpin the *Substantial* risk rating are as follows: (i) unfamiliarity of the Borrower and the main implementing entity, AySA with the PforR Instrument; (ii) the risk of awarding a contract to firms and/or individuals debarred or suspended by the Bank; (iii) lack of an effective procurement-related complaints procedure; (iv) exchange rate pressures that pose a risk that the amount of Program Expenditures may be insufficient to cover the withdrawn loan balance; (v) AySA's dependence on transfers from the National Government for the execution of the PMOEM; and (vi) the need to put in place specific arrangements to ensure the Program Annual Financial Statements will be able to be produced.

(iv) The proposed systems-and capacity-strengthening and/or mitigation measures, to address the above risks include the following: (i) close implementation support to be provided by the Bank's fiduciary teams, coupled with the preparation of an Operations Manual; (ii) the external auditors' ToRs will include a requirement to review Program expenditure for ineligible contracts; (iii) eligibility verification to ensure that any person or entity debarred or suspended by the Bank or by another multilateral institution with which the Bank has signed a cross debarment agreement, is not awarded a contract under or otherwise allowed to participate in the Program during the period of such debarment or suspension; (iv) preparation and implementation of a clause in the bidding documents on the handling of complaints, which includes a detailed description of the procedure through which a timely and fair review of the complaint is ensured; (v) review of the semi-annual IFRs to ensure Program Expenditures are on track; (vi) close monitoring and coordination to ensure that necessary budgetary support is provided by the National Government; and (vii) close implementation support to be provided by the Bank to help establish the arrangements to produce the Program Annual Financial Statements.



ANNEX 1. RESULTS FRAMEWORK MATRIX

Results Framework

COUNTRY: Argentina

BUENOS AIRES WATER SUPPLY AND SANITATION WITH A FOCUS ON VULNERABLE AREAS PROGRAM

Program Development Objective(s)

The Program Development Objectives are to increase access to safely managed water and sanitation services with a focus on vulnerable areas of Buenos Aires, and improve AySA's efficiency and capacity to respond to water and sanitation emergency needs

Program Development Objective Indicators by Objectives/Outcomes

Indicator Name	DLI	Baseline	End Target
Increase in access to safely managed water services			
Number of households gaining access to safely managed water services (Number)	DLI 1	0.00	35,000.00
Increase in access to safely managed sanitation services			
Number of households gaining access to safely managed sanitation services (Number)	DLI 2	0.00	76,359.00
Improve AySA's efficiency			
Annual savings from operational and commercial efficiency measures (\$/year) (Number)	DLI 4	0.00	9,528,566.00
Improve AySA's capacity to respond to WSS emergency needs			



Indicator Name	DLI	Baseline	End Target
Number of AySA's emergency response mechanisms strengthened (Number)	DLI 5, 6	0.00	2.00
-			



Intermediate Results Indicator by Results Areas

Indicator Name	DLI	Baseline	End Target
Expansion of WSS services			
Increase in wastewater treatment capacity (m3/s) (Number)		2.26	3.98
Increased pumping capacity in Belgrano WTP to deliver service to Florencio Varela (m3/h) (Number)		0.00	4,000.00
Number of cooperatives trained in the Agua / Cloaca + Trabajo programs (Number)		0.00	30.00
Increase AySA's efficiency			
Length of water supply pipes replaced (Kilometers)		0.00	510.00
Number of sewage pumping stations where energy monitoring equipment will be installed (Number)		69.00	215.00
Number of wells where energy monitoring equipment will be installed (Number)		0.00	800.00
Number of large consumers tele-metered (Number)		0.00	440.00
Number of micro-meters installed (Number)		0.00	71,400.00
Increase AySA's capacity to respond to water supply and sanitation emergency needs			
Prevention and Emergency Plan strengthened (Yes/No)	DLI 5.2	No	Yes
Number of digital solutions implemented (Number)	DLI 6.1	0.00	5.00
Mobile app for citizen engagement in operation (Yes/No)		No	Yes
CRM system for citizen engagement strategy built (Yes/No)	DLI 6.2	No	Yes
Increased share of female staff in management level positions at AySA (Percentage)	DLI 7	22.00	28.00



Monitoring & Evaluation Plan: PDO Indicators

Indicator Name	Definition/Description	Frequency	Datasource	Methodology for Data Collection	Responsibility for Data Collection
Number of households gaining access to safely managed water services	Households that will gain access to safely managed water supply either by the construction of the internal household connections or by improving water quality	Biannual	Verification Report provided by the VA	Specified in Verification Protocol	AySA and VA
Number of households gaining access to safely managed sanitation services	Households that will gain access to safely managed sanitation by the construction of the internal household connections or by the expansion of primary wastewater infrastructure.	Biannual	Verification Report provided by the VA	Specified in the Verification Protocol	AySA and VA
Annual savings from operational and commercial efficiency measures (\$/year)	AySA's annual savings from reduction of water losses, energy efficiency, and improvements the efficiency of bill collection. The targets corresponding to these three efficiency factors are 4,953,466, 2,375,100, and 2,200,000 US\$/year, respectively. The methodology for value calculation is included in the Operations Manual	Biannual	Verification Report provided by the VA	Specified in the Verification Protocol and Operations Manual	AySA and VA
Number of AySA's emergency response mechanisms strengthened	These are 2 mechanisms : (i) the strengthening of AySA's Emergency	Biannual	Verification Report provided by	Included in the Verification Protocol	AySA and VA



	Response and Recovery Plan with the COVID-19 Contingency Plan; and (ii) the implementation of AySA's digitalization strategy, which includes the digitalization of internal systems, namely: 1) consolidation of human resources systems; (2) enabling the remote work modality ; (3) enabling the predictive maintenance of assets; (4) digitalization of commercial processes; (5) optimization of the customer complaint management system ; and (6) development of AySA's Citizen Engagement Strategy.		the VA		
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Monitoring & Evaluation Plan: Intermediate Results Indicators

Indicator Name	Definition/Description	Frequency	Datasource	Methodology for Data Collection	Responsibility for Data Collection
Increase in wastewater treatment capacity (m3/s)	Cumulative potential volume of wastewater treated by the 2 WWTPs under the Program (Jaguel WWTP 0.36 m3/s and Norte WWTP 1.36 m3/s) according to discharge quality regulation. Works in Norte WWTP include the optimization of module 1-2 (0.36 m3/s) and the construction of module 3 (1 m3/s). Therefore, the additional treatment capacity is 1.72 m3/s	At the time of completion for each intervention in the two plants	Audited Technical Report of AySA's operation	AySA will record the flow volumes with the meters installed at the outlet of the WWTPs	AySA
Increased pumping capacity in Belgrano WTP to deliver service to Florencio Varela (m3/h)	Flowrate capacity of the new pumping station to be built under the Program, located in the Belgrano water treatment plant, in the municipality of Florencio Varela, measured in m3/h	At completion of construction	Audited Technical Report of AySA's operation	AySA will record the flow volumes with meters installed at the exit of new pumping station	AySA
Number of cooperatives trained in the Agua / Cloaca + Trabajo programs	The cooperatives must be constituted according to the national institute of social economy (INAES) and registered in the official Social Registry. The cooperatives need to have	Biannual	AySA	AySA will record the number of Cooperatives by reports from FUTRASAFODE's trainings.	AySA



	passed the certification exam run by FUTRASAFODE, Sanitation Workers' Foundation for Training and Development.				
Length of water supply pipes replaced	Length of rehabilitated or renewed pipes in water distribution networks. Rehabilitated pipes can be defined as pipes in which repairs or replacements have been made	Biannual	Certificates of contractors working on rehabilitation tasks and internal reports of network rehabilitation works from AySA	AySA will collect information from contractors and their own crews to prepare the report.	AySA
Number of sewage pumping stations where energy monitoring equipment will be installed	Number of sewerage pumping stations where monitoring equipment will be installed to detect inconsistencies between billing and energy readings that will be fixed during Program implementation to achieve energy savings. The Program will install such equipment in 146 pumping stations, reaching 215 from the existing 69 where this equipment is already installed.	Biannual	Certificates of contractors working on the installation of monitoring equipment	AySA will collect information from contractors and their own crews to prepare the report.	AySA
Number of wells where energy monitoring equipment will be installed	Numbers of wells where monitoring equipment will	Biannual	Certificates of contractors	AySA will collect information from	AySA



	be installed to detect inconsistencies between billing and energy readings that will be fixed during the Program to achieve energy savings.		working on the installation of monitoring equipment and Verification Report	contractors and their own crews to prepare the report.	
Number of large consumers tele-metered	Number of large customer metered and billed whose consumption is over 500 m3/month	Biannual	Certificates of contractors working on the installation of monitoring equipment and Verification Report	AySA will collect information from contractors and their own crews to prepare the report.	AySA
Number of micro-meters installed	Number of micro-meters to be installed as part of the rehabilitation of the water supply network to reduce water losses.	Biannual	Certificates of contractors working on rehabilitation tasks, AySA's technical reports and Verification Report	AySA will collect information from contractors and their own crews to prepare the report.	AySA
Prevention and Emergency Plan strengthened	Preparation of the COVID-19 contingency plan and incorporation in PPE	Once, at the completion of the activity	AySA technological development (DT) department and	Submission of the updated PPE by AySA	AySA



			Verification Report		
Number of digital solutions implemented	(1) consolidation of human resources systems; (2) enabling the remote work modality on a continuous basis; (3) enabling the predictive maintenance of main assets; (4) digitalization of commercial processes in AySA's network of commercial offices; and the (5) optimization of the customer complaint management system through the adoption of mobile and digital solutions. The methodology to measure the achievement of each solution is determined in the Verification Protocol.	Biannual	Report prepared by AySA Commercial Management and Verification Report	AySA will record the electronically the commercial procedures	AySA
Mobile app for citizen engagement in operation	Development and implementation of a new digital channel - mobile app -for community engagement. The app will constitute a new communication channel between users and the utility. Features of the App, to consider it "in	Once at the completion of the activity	AySA's IT Department's technical report and Verification Report	AySA will verify that the app is available to users via IOS or Android platforms	AySA



	operation" are included in the Verification Protocol				
CRM system for citizen engagement strategy built	Development of a Customer Relationship Management system to centralize all communication channels with users. Details to consider it completed are included in the Verification Protocol	Once at completion of the activity	AySA's IT Department's technical report and Verification Report	AySA will verify that the CRM is implemented and complies with the predefined functionalities	AySA
Increased share of female staff in management level positions at AySA	Percentage of women in management positions compared to the total number of employees at the same level	Biannual	Report prepared by AySA HR department and the Verification Report	AySA's internal reports from HR department submitted to the Bank	AySA



ANNEX 2. DISBURSEMENT LINKED INDICATORS, DISBURSEMENT ARRANGEMENTS AND VERIFICATION PROTOCOLS

Disbursement Linked Indicators Matrix				
DLI 1	Number of HH gaining access to safely managed water services			
Type of DLI	Scalability	Unit of Measure	Total Allocated Amount (USD)	As % of Total Financing Amount
Outcome	Yes	Number	35,998,375.00	12.00
Period	Value		Allocated Amount (USD)	Formula
Baseline	0.00			
Prior Results			0.00	
April 2021-June 2026	35,000.00		35,998,375.00	See Sub-DLIs 1.1 and 1.2
DLI 1.1	Number of households gaining access to safely managed water services living in fragile neighborhoods			
Type of DLI	Scalability	Unit of Measure	Total Allocated Amount (USD)	As % of Total Financing Amount
Outcome	Yes	Number	16,867,750.00	5.64
Period	Value		Allocated Amount (USD)	Formula
Baseline	0.00			
Prior Results			0.00	
April 2021-June 2026	13,625.00		16,867,750.00	Disbursement = HHs connected_water in fragile neighborhoods * C where: C: 1238 USD / connection



DLI 1.2	Number of households gaining access to safely managed water services living non-fragile / formal neighborhoods			
Type of DLI	Scalability	Unit of Measure	Total Allocated Amount (USD)	As % of Total Financing Amount
Outcome	Yes	Number	19,130,625.00	6.39
Period	Value		Allocated Amount (USD)	Formula
Baseline	0.00			
Prior Results			0.00	
April 2021-June 2026	21,375.00		19,130,625.00	Disb = HH connected_water in no fragile neighborhoods * C where: C: 895 USD / connection
DLI 2	Number of households gaining access to safely managed sanitation services			
Type of DLI	Scalability	Unit of Measure	Total Allocated Amount (USD)	As % of Total Financing Amount
Outcome	Yes	Number	86,994,236.00	29.00
Period	Value		Allocated Amount (USD)	Formula
Baseline	0.00			
Prior Results			0.00	
April 2021-June 2026	76,359.00		86,994,236.00	See Sub-DLIs 2.1 and 2.2
DLI 2.1	Number of households gaining access to safely managed sanitation services living in fragile neighborhoods			
Type of DLI	Scalability	Unit of Measure	Total Allocated Amount (USD)	As % of Total Financing Amount
Outcome	Yes	Number	37,839,400.00	12.60
Period	Value		Allocated Amount (USD)	Formula



Baseline	0.00			
Prior Results			0.00	
April 2021-June 2026	27,400.00		37,839,400.00	subDLI2.1 = HH connected_sewege in fragile neighborhoods * C where: C: 1381 USD / connection
DLI 2.2	Number of households gaining access to safely managed sanitation services living in non-fragile / formal neighborhoods			
Type of DLI	Scalability	Unit of Measure	Total Allocated Amount (USD)	As % of Total Financing Amount
Outcome	Yes	Number	49,154,836.00	16.40
Period	Value		Allocated Amount (USD)	Formula
Baseline	0.00			
Prior Results			0.00	
April 2021-June 2026	48,959.00		49,154,836.00	sub DLI 2.2 = HH connected_sewege in non-fragile neighborhoods * C where: C: 1004 USD / connection
DLI 3	Preparation and adoption of AySA's Efficiency Improvement Action Plan (EIAP)			
Type of DLI	Scalability	Unit of Measure	Total Allocated Amount (USD)	As % of Total Financing Amount
Output	No	Yes/No	20,000,000.00	6.70
Period	Value		Allocated Amount (USD)	Formula
Baseline	No			
Prior Results			0.00	



April 2021-June 2026	Yes		20,000,000.00	DLI 3 will disburse 100% of the allocated amount when all conditions (1) to (5), identified in the VP, are met.
DLI 4	Implementation of AySA's Efficiency Improvement Action Plan (EIAP)			
Type of DLI	Scalability	Unit of Measure	Total Allocated Amount (USD)	As % of Total Financing Amount
Outcome	Yes	Percentage	100,250,639.00	33.50
Period	Value		Allocated Amount (USD)	Formula
Baseline	0.00			
Prior Results			0.00	
April 2021-June 2026	100.00		100,250,639.00	See Sub-DLIs 4.1 and 4.2
DLI 4.1	Increase in Operational Efficiency (US\$/day)			
Type of DLI	Scalability	Unit of Measure	Total Allocated Amount (USD)	As % of Total Financing Amount
Outcome	Yes	Number	90,250,610.00	30.20
Period	Value		Allocated Amount (USD)	Formula
Baseline	0.00			
Prior Results			0.00	
April 2021-June 2026	20,078.00		90,250,610.00	Disb = c1 * A (m3/day) + c2 * B (kWh/day))x Unit value. Unit value is 4.495 days



DLI 4.2	Increase in Digital Payment share			
Type of DLI	Scalability	Unit of Measure	Total Allocated Amount (USD)	As % of Total Financing Amount
Intermediate Outcome	Yes	Percentage	10,000,029.00	3.30
Period	Value		Allocated Amount (USD)	Formula
Baseline	0.00			
Prior Results			0.00	
April 2021-June 2026	51.00		10,000,029.00	[(Total percentage points achieved in a given Period minus 43) divided by 43]x100 x (196,079 \$/cumulative perc point increase
DLI 5	Implementation and Strengthening of AySA's Prevention and Emergency Plan (PPE)			
Type of DLI	Scalability	Unit of Measure	Total Allocated Amount (USD)	As % of Total Financing Amount
Output	No	Yes/No	20,006,750.00	6.70
Period	Value		Allocated Amount (USD)	Formula
Baseline	No			
Prior Results	Yes		15,000,000.00	See Sub-DLI 5.1
April 2021-June 2026	Yes		5,006,750.00	See Sub-DLI 5.2



DLI 5.1	COVID-19 Contingency measures implemented			
Type of DLI	Scalability	Unit of Measure	Total Allocated Amount (USD)	As % of Total Financing Amount
Output	No	Yes/No	15,000,000.00	5.00
Period	Value		Allocated Amount (USD)	Formula
Baseline	No			
Prior Results	Yes		0.00	
April 2021-June 2026	Yes		15,000,000.00	DLI 5.1 will disburse 100% of the allocated amount when all conditions identified in VP are verified.
DLI 5.2	Preparation of the Epidemiological Risk Contingency Plan			
Type of DLI	Scalability	Unit of Measure	Total Allocated Amount (USD)	As % of Total Financing Amount
Output	No	Yes/No	5,006,750.00	1.70
Period	Value		Allocated Amount (USD)	Formula
Baseline	No			
Prior Results			0.00	
April 2021-June 2026	Yes		5,006,750.00	DLI 5.2 will disburse 100% of the allocated amount when all conditions identified in VP are verified.



DLI 6	Implementation of Digital Solutions			
Type of DLI	Scalability	Unit of Measure	Total Allocated Amount (USD)	As % of Total Financing Amount
Outcome	Yes	Number	33,000,000.00	11.00
Period	Value		Allocated Amount (USD)	Formula
Baseline	0.00			
Prior Results			0.00	
April 2021-June 2026	6.00		33,000,000.00	See Sub-DLIs 6.1 and 6.2
DLI 6.1	Implementation of Digital Solutions for Internal Systems			
Type of DLI	Scalability	Unit of Measure	Total Allocated Amount (USD)	As % of Total Financing Amount
Intermediate Outcome	Yes	Number	26,000,000.00	8.70
Period	Value		Allocated Amount (USD)	Formula
Baseline	0.00			
Prior Results			0.00	
April 2021-June 2026	5.00		26,000,000.00	Disbursement = number of measures implemented x Unit value. Unit value is \$5,200,00
DLI 6.2	Digital Strategy for Citizen Engagement			
Type of DLI	Scalability	Unit of Measure	Total Allocated Amount (USD)	As % of Total Financing Amount
Outcome	Yes	Number	7,000,000.00	2.30
Period	Value		Allocated Amount (USD)	Formula



Baseline	0.00			
Prior Results			0.00	
April 2021-June 2026	5.00		7,000,000.00	Disbursement = number of milestones x Unit Value.. Unit value is \$1,400,000
DLI 7	Increase in share of women in leadership positions in AySA			
Type of DLI	Scalability	Unit of Measure	Total Allocated Amount (USD)	As % of Total Financing Amount
Outcome	Yes	Percentage	3,000,000.00	1.00
Period	Value		Allocated Amount (USD)	Formula
Baseline	22.00			
Prior Results			0.00	
April 2021-June 2026	28.00		3,000,000.00	Disbursement=Achievement x (500,000 \$ per each cumulative incremental percentage point above the initial baseline of 22%)



Verification Protocol Table: Disbursement Linked Indicators

DLI 1	Number of HH gaining access to safely managed water services
Description	Households in fragile and non-fragile neighborhoods that will gain access to safely managed water supply either by i) the construction of the internal household connections in areas of 16 municipalities of peri-urban Buenos Aires where the water network and service pipes already exist, or ii) by improving water quality in the municipality of Florencio Varela
Data source/ Agency	See Sub-DLIs 1.1 and 1.2
Verification Entity	Verification Agency
Procedure	See Sub-DLIs 1.1 and 1.2
DLI 1.1	Number of households gaining access to safely managed water services living in fragile neighborhoods
Description	Households in fragile neighborhoods that will gain access to safely managed water supply either (i) through the construction of the intrahousehold connections in areas of 16 municipalities of peri-urban Buenos Aires where the water network and service pipes already exist, or (ii) by improving water quality in the municipality of Florencio Varela.
Data source/ Agency	AySA will provide: Shapefiles including information on (i) fragile neighborhoods database: UREM (including RENABAP) and ABC, and location of beneficiary neighborhoods and eligible connections, (ii) water quality tests, (iii) information on floodable areas. Details are included in the Operations Manual
Verification Entity	VA
Procedure	HHs must be within the preidentified fragile areas of the municipality (UREM, RENABAP or ABC). HH connections are completed after loan signature date. The VA will define a representative random sampling size with confidence level 95% and the maximum permissible error (e): 5%, and ensure that the connections are working. Distributed water to new connections complies with water quality regulations specified in the OM. Full details for the verification procedure are included in the Operations Manual
DLI 1.2	Number of households gaining access to safely managed water services living non-fragile / formal neighborhoods
Description	Households in formal / non-fragile neighborhoods that will gain access to safely managed water supply by improving water quality in the municipality of Florencio Varela.
Data source/ Agency	AySA will provide: Shapefiles including information on (i) location of eligible area in Florencio Varela and eligible connections, (ii) water quality tests, (iii) general information on floodable areas. Details are included in the Operations



	Manual
Verification Entity	VA
Procedure	HHs must be <u>outside</u> the preidentified fragile areas of the municipality (UREM, RENABAP or ABC). HH connections are completed after loan signature date. The VA will define a representative random sampling size with confidence level 95% and the maximum permissible error (e): 5% and ensure that the connections are working. Distributed water to new connections complies with water quality regulations specified in the OM. Full details for verification are included in the Operations Manual
DLI 2	Number of households gaining access to safely managed sanitation services
Description	Households in fragile and non-fragile / formal neighborhoods that will gain access to safely managed sanitation through the connection to a sewerage system that is linked to a functioning WWTP
Data source/ Agency	AySA
Verification Entity	VA
Procedure	See Sub-DLIs 2.1 and 2.2
DLI 2.1	Number of households gaining access to safely managed sanitation services living in fragile neighborhoods
Description	Households in fragile neighborhoods that will gain access to safely managed sanitation through the construction of intrahousehold connections to a sewerage system linked to a WWTP
Data source/ Agency	AySA will provide shapefile of: (i) Project eligible areas, (ii) fragile neighborhoods database: UREM (including RENABAP) and ABC to verify HH connection eligibility, (iii) data to certify that the wastewater treatment plant complies with regulatory effluent standard, (iv) general information on floodable areas. Details included in the Operations Manual.
Verification Entity	VA
Procedure	HHs must be within the preidentified fragile areas of the municipality (UREM, RENABAP or ABC). HH connections are completed after loan signature date. The VA will define a representative random sampling size with confidence level 95% and the maximum permissible error (e): 5%, and ensure that the connections are working. The WWTPs comply with effluent discharge regulations. Full details for verification are included in the Operations Manual.



DLI 2.2	Number of households gaining access to safely managed sanitation services living in non-fragile / formal neighborhoods
Description	Households in non-fragile / formal neighborhoods that will gain access to safely managed sanitation through the expansion of primary wastewater infrastructure (wastewater treatment and sewerage).
Data source/ Agency	AySA will provide shapefile of: (i) Project eligible areas, (ii) fragile neighborhoods database: UREM (including RENABAP) and ABC to verify HH connection eligibility, (iii) data to certify that the wastewater treatment plant complies with regulatory effluent standard, and (iv) general information on floodable areas. Details included in the Operations Manual.
Verification Entity	VA
Procedure	HHs must be <u>outside</u> the preidentified fragile areas of the municipality (UREM, RENABAP or ABC). HH connections are completed after loan signature date. The VA will define a representative random sampling size with confidence level 95% and the maximum permissible error (e): 5%, and ensure that the connections are working. The WWTPs comply with effluent discharge regulations. Full details for verification are included in the Operations Manual.
DLI 3	Preparation and adoption of AySA's Efficiency Improvement Action Plan (EIAP)
Description	The EIAP will include a minimum content acceptable to the Bank entailing the identification of gaps and weaknesses in the existing systems and processes of AySA that are affecting its overall efficiency, with a focus on operational and commercial areas, establishing a link with the company's financial performance. This is a one-time non-scalable DLI.
Data source/ Agency	AySA will provide the EIAP, copy of all the documentation that is cited in the planning document cited by AySA (particularly other plans, directives, regulations, sources of technical data, budget data), as well as the evidence that demonstrates the official adoption of the EIAP.
Verification Entity	VA
Procedure	The VA will verify the EIAP contents and that it complies with all of the following conditions for disbursement: i) an in-depth diagnostic taking into account the technical, economic, cultural, social, legal and environmental restrictions impacting AySA's performance ii) a clear and precise identification of initiatives and actions contributing to improving operating costs coverage with internally-generated funds or to reducing operating costs per functional unit (in both cases, as demonstrated by a "with and without project" comparison) iii) the prioritization of initiatives and actions, by weighting the contribution of each of them to the above-mentioned operating costs-related objectives iv) a feasibility-level definition of the selected initiatives and actions, in the case of works, and an analysis of alternatives in the cases of "soft" actions, the replacement of equipment or other situations not requiring a feasibility study v) a study to improve the general collection rate (both current billing and past-due revenue). The EIAP will have to be adopted by AySA under one of the modalities included in the



	Operations Manual.
DLI 4	Implementation of AySA's Efficiency Improvement Action Plan (EIAP)
Description	This DLI measures gains in operational efficiency through Sub-DLI 4.1, and in commercial management efficiency through Sub-DLI 4.2
Data source/ Agency	See Sub-DLIs 4.1 and 4.2
Verification Entity	VA
Procedure	See Sub-DLIs 4.1 and 4.2
DLI 4.1	Increase in Operational Efficiency (US\$/day)
Description	Measure AySA's improvements in NRW (in saved m3/day) and energy efficiency (in saved kWh/m3). The DLI is normalized in order to homogenize both components (water and energy savings), so it is measured in US\$/day. AySA expects to reach a target value "A" of 123,374 m3 / day and a target value "B" of 92,960KWh/day / day. C1 and C2 are normalizing values transforming savings in water and energy into monetary savings of \$/day (c1 = 0,11 USD / m3 c2 = 0,07 USD/ kWh). (B target of 92,960KWh/day is equivalent to 93.93GW/year)
Data source/ Agency	For factor "A", NRW, AySA will provide: (i) a shapefile of the intervention area specifying length of the renovated and rehabilitated water networks, (ii) historical data of statistical occurrence of different kind of breakages associated to typical loss rates for each of them to define the average leak in m3/day per repair, (iii) shapefiles that geo-localize the leak intervention with satellite technology and complement sound correlation with its typology and complementary data. For factor "B", energy efficiency: (i) shapefiles that geo-localize all the 800 wells and 146 wastewater pumping where AySA will install Energy Monitoring Equipment, (ii) baseline of energy consumption and water production of each of the wells and the sewage pumping stations, including pump station Saavedra and "Old Pump Station" in water treatment San Martin, (iii) monthly data on energy consumption (kWh) in such locations (iv) Location and monthly data on energy production (kWh) from renewable energy. Details included in the Operations Manual



Verification Entity	VA
Procedure	For factor "A", NRW, the VA will measure in m3/day the water saving resulting from: a) the renovation of pipes, b) water leak repair using satellite information from pilot studies linking "length of pipes replaced" with "volume of water saved". Works in both areas (renovation pipes and leak repair by satellite detection) will be randomly verified in the field using Kobo ToolBox. For factor "B", energy efficiency, the VA will measure in kwh / day that results from the savings due to: a) The Distributed Consumption Monitoring Plan, b) The Rehabilitation of the Saavedra water pumping station, c) Distributed self-generation of energy - 200KW ; and d) The remodeling of the raw water pumping station of the San Martín WTP. The kWh / day saved will be measured by: the difference between a baseline energy consumption before the intervention and the energy consumption after the intervention trough readings in energy meters. The VA will define a representative random sampling size with confidence level 95% and the maximum permissible error (e): 5%. The e VA will verify the amount of savings in US\$, using the following equation: $c1 * A [m3/day] + c2 * B [kWh/day]$ Full verification details are included in the Operations Manual.
DLI 4.2	Increase in Digital Payment share
Description	Percentage increase of the use in AySA of Efficient (Digital) Payment Channels compared to less efficient payment channels. Efficient payment channels are those that imply lower commissions charged to AySA, lower costs in logistics resources and shorter times and costs for users.
Data source/ Agency	AySA will provide a monthly amount and the number of transactions carried out by each of the following payment channels: 1. Banks and Non-Bank Entities 2. Automatic Debit 3. AySA offices 4. Internet and Online Payment 5. Phone-payment and Pagophone 6. Interdepositos Fiscales
Verification Entity	VA
Procedure	The VA will calculate the percentage of digital payment by adding the percentage of payments done with: automatic debit, internet or online payment, and phone payment. Achievement is determined according to the percentage increase with respect to the maximum value of the indicator achieved in previous verification cycles. Full verification details are included in the Operations Manual
DLI 5	Implementation and Strengthening of AySA's Prevention and Emergency Plan (PPE)
Description	This DLI consist of two Sub-DLIs. Sub-DLI 5.1 measuring the implementation of the PPE through the measures related to the COVID-19 pandemic, and Sub-DLI 5.2 related to the strengthening of the PPE with a Contingency Plan with lessons learned



	from the pandemic.
Data source/ Agency	See Sub-DLIs 5.1 and 5.2
Verification Entity	VA
Procedure	See Sub-DLIs 5.1 and 5.2
DLI 5.1	COVID-19 Contingency measures implemented
Description	This sub-DLI will monitor the implementation of measures related to the COVID-19 response, to be recognized as prior results, including (i) regular operation of a crisis committee; (ii) 60,163 m3 of water and 369 m3 of disinfectant delivered to 23 municipalities within the PBA between July 16, 2020 and December 30, 2020, (iii) remote working arrangements made for at least 2,500 AySA staff, (iv) 6 awareness campaigns delivered among AySA's employees about precautions during the COVID-19 pandemic outbreak, (v) implementation of safety protocols for AySA's water +works and sewerage+works programs, (vi) implementation of the wastewater-based epidemiology approach to monitor the presence of SARS-Cov-2 in sewage and complement the Ministry of Health's surveillance system, and (vii) preventive measures for COVID-19 transmission implemented for AySA's staff, reorganization of human resources in AySA's operations department, while maintaining service levels as per details included in the Operations Manual. Non scalable
Data source/ Agency	AySA will provide evidence on the implementation of the 7 measures contained in the DLI
Verification Entity	Third party verification agent to be agreed between AySA and the Bank
Procedure	The VA will verify the evidence provided by AySA on the implementation of the 7 measures as per the details included in the Operations Manual
DLI 5.2	Preparation of the Epidemiological Risk Contingency Plan
Description	PPE updated with a contingency plan to deal with epidemiological risks, including a systematization of lessons learned during the COVID-19 crisis; and the organization and delivery of an international seminar on COVID-19 to share lessons learned on dealing with epidemiological risks, as further detailed in the Verification Protocol
Data source/ Agency	AySA will provide the Epidemiological risk contingency plan and related documentation as per details included in the Operations Manual
Verification Entity	VA



Procedure	The VA will verify that: i) the lessons learned document covers the main issues that affected each relevant area of the service provider, with the corresponding analysis and conclusions; ii) the contingency plan includes, at least, the minimum content listed Operations Manual and iii) the international seminar fulfilled the pursued objective of disseminating experiences of several institutions dealing with the COVID-19 emergency and that AySA shared its progress developing molecular biology technics to identify pathogens in wastewater. Details are included in the Operations Manual
DLI 6	Implementation of Digital Solutions
Description	This DLI consists in two sub-DLIs: sub DLI 6.1 with the implementation of five digital solutions for internal systems and sub DLI 6.2 with the Digital strategy for citizen engagement
Data source/ Agency	See Sub-DLIs 6.1 and 6.2
Verification Entity	VA
Procedure	See Sub-DLIs 6.1 and 6.2
DLI 6.1	Implementation of Digital Solutions for Internal Systems
Description	Implementation of 5 digitalization measures: (i) digital integration of human resources systems, including automatization of 24 interphases in AySA's staff control systems, (ii) mobile devices for remote work provided to an additional 360 AySA staff, from a baseline of 900, (iii) predictive maintenance of main assets, including at least 278 assets (from the list in the Operations Manual) installed in AySA's internal database and operational, (iv) digitalization of commercial processes in AySA's commercial offices, including 14 commercial processes digitalized in all of AySA's commercial offices, and (v) mobile claims system operational, including all of AySA's field working teams using mobile devices for claims and registry problems
Data source/ Agency	AySA will provide technical documentation on the 5 activities associated with the digital solutions
Verification Entity	VA
Procedure	The VA will verify the implementation of the five actions, according to details included in the Operations Manual
DLI 6.2	Digital Strategy for Citizen Engagement
Description	Implementation of mobile app and CRM system to improve AySA's customer relations and citizen engagement, by broadening the digital communication channels, incorporating feedback from users, providing tailored information and improving reception and management of claims, questions and suggestions, ending in the improvement of response times. The disbursement is scalable related to 5 milestones of the strategy



Data source/ Agency	AySA will provide technical documentation on the five stages of the digital strategy for citizen engagement
Verification Entity	VA
Procedure	The VA must verify that in a gradual implementation process AySA carries out the following 5 stages, yearly. For stages 1,2 and 3, respectively: 1. New digital channel for contact with the community (mobile app) that includes typified PQRS and also the possibility of conducting open consultations. Target achieved when mobile app, including predefined functionalities is launched and operational. (Yes / No) 2. Integrated Customer Relationship Management (CRM) system. Target achieved when CRM system, including predefined functionalities is launched and operational. (Yes / No) 3. PQRS assessment and classification according to frequency and average response times. Target achieved PQRS assessment complied with defined requirements. (Yes / No) For stages 4 and 5, the improvement in response time for the 10 most frequent questions/claims/suggestions is assessed. In stage 4, the target will be achieved when at least a reduction of 15 % in the average response time is achieved. Similarly for stage 5, when at least an accumulated 20 % reduction in the average response time is achieved. Details are included in the Operations Manual.
DLI 7	Increase in share of women in leadership positions in AySA
Description	The DLI will reflect efforts made by AySA in promoting equal career advancement opportunities for women and improving working conditions for all AySA employees, with the goal of increasing in the percent of women's representation at management level positions, including in technical departments.
Data source/ Agency	AySA will provide: (i) detailed report of the company's payroll in the eligible leadership positions (Directors, Managers and Chiefs), identifying for each position: the functional title, organizational dependency, identification of the gender of the person in charge, and seniority in said position. (ii) Administrative evidence that support the existence of each of the dependencies identified in the detailed report
Verification Entity	VA
Procedure	The VA will verify the representation of women in leadership positions (Director, Managers and Chiefs), by calculating the percentage of women from the total personnel in those positions. The verification process will be done based on the biannual assessment of women's representation from AySA's official reports. Full verification details are included in the Operations Manual.



ANNEX 3. TECHNICAL ASSESSMENT SUMMARY

A. OVERALL RATIONALE FOR THE PROGRAM

1. **The Program was designed to increase access to WSS services in vulnerable areas of AySA's jurisdiction, while at the same time increase AySA's resilience in times of the COVID-19 pandemic, as well as its efficiency, contributing to reducing its dependency on federal budget transfers and increasing its financial sustainability.** By designing a strong system of incentives through well-defined DLIs, the Program will allow the achievement of challenging results such as expanding connectivity in the poorest areas through community participation, reducing water losses and energy consumption, increasing revenues, streamlining managerial processes and speeding up the agility of internal procedures, as well as promoting innovation and transparency. Moreover, the PforR will acknowledge AySA's ongoing activities to mitigate the shocks derived from the COVID-19 crisis by strengthening the government's and AySA's own systems to recover in a timely manner, as well as prepare for other potential shocks. The Program is well aligned with the government's commitment to prioritize investments in water and sanitation services, with particular focus on vulnerable areas.

B. TECHNICAL SOUNDNESS

Results Area 1: Expansion of WSS services with a focus on vulnerable areas

(A) Household connections in *fragile* neighborhoods

2. **AySA will increase access to safely managed WSS services to households (HH) that, despite their proximity to already installed secondary networks, have not yet connected.** Concurrently, this activity will promote local employment by implementing the corresponding works through local cooperatives, and will support and deepen social intervention strategies that communicate socio-environmental and technical information to encourage HH to connect to and care for WSS networks. Through the Program, AySA will build on the A+T and C+T models to deliver 10,000 intra-HH water connections and 20,000 intra-HH sewerage connections, respectively. Within the eligible municipalities, the Program will target beneficiaries in *fragile* neighborhoods identified under this Program as those falling within the following classified categories by AySA: (i) ABC; (ii) UREM; and (iii) RENABAP. These areas were selected based on their sanitary and socio-economic conditions. *Fragile* neighborhoods are usually located in areas with complex terrain conditions: flood-prone land, high levels of water tables and limited access to basic services and health infrastructure.

3. **AySA's proposed evaluation of the potential beneficiaries is robust as it addresses the population's structural and sociodemographic aspects.** From a technical perspective, the proposed rate of installation of 100 water or 36 sewerage connections per cooperative per month is reasonable based on similar previous pilots in vulnerable areas and the nine pre-defined water and sanitation technical connection typologies are appropriate for the households to benefit from these interventions. Finally, A+T and C+T also have a long track record of mobilizing local workers and promoting jobs in low-income neighborhoods, which is well justified in the context of a post COVID-19 economic recovery phase.

(B) Expansion of primary infrastructure – Florencio Varela (water), Norte and Jagüel (wastewater)

Water Supply Infrastructure

4. **AySA will provide safely managed water services to the municipality of Florencio Varela, by reducing the amount of water coming from existing deep wells, which is of low quality, and substituting it with treated surface water from the Belgrano WTP.** The number of beneficiaries is estimated at 100,000 (equivalent to 25,000 HH) of which 14,500 live in fragile areas. Beneficiaries in this area already have household connections. The proposed solution is a technically reasonable approach to reduce the concentration of nitrates in water down to safe levels as per drinking water quality standards. Investments include civil works at the pumping station of 4000 m³/s capacity and the initial



trunk main of 1600 m. Moreover, AySA selected the most suitable source to minimize the cost of complying with water quality standards, as the Belgrano WTP is the closest to the target area, minimizing capital and operational costs.

Wastewater Infrastructure

5. AySA will build wastewater infrastructure to serve 225,000 people at the end of the Program timeline.

Beneficiaries from these interventions are in the municipalities of Escobar, Esteban Echeverría, Ezeiza, San Fernando, San Isidro, San Martín, and Tigre through infrastructure investments in wastewater treatment modules in existing WWTPs, extension of sewerage networks, including wastewater pumping stations, and household connections to be implemented through regular construction contracts.

- (i) The Norte WWTP – which receives wastewater from the networks of the Tigre, San Fernando and part of San Isidro districts – has two existing parallel treatment modules that treat 0.9 m³/s each, using the activated sludge process. The objective is the optimization of these two modules to first reach a total capacity of 2.16 m³/s, and the construction of a third module with a capacity of 1.0 m³/s. The total capacity of 3.16 m³/s is expected to be reached in 2030 and is equivalent to serve circa 1 million people.
- (ii) The Jagüel WWTP located in the Esteban Echeverría district within the Matanza Riachuelo Basin has two modules with different processes and treatment capacities. Module I includes pre-treatment, biological treatment by the activated sludge process (extended aeration), clarifiers, as well as sludge concentration and drying facilities. It can treat the wastewater of up to 30,000 people but is currently out of service. Module II includes pre-treatment, primary and secondary biological treatment through an attached-growth process (trickling filters), as well as sludge concentration, digestion and drying facilities and has a capacity for serving 150,000 people (0.46m³/s). The project foresees an expansion of Jagüel, Module I to serve 100,000 people (0.36m³ /s), reaching the total capacity of the plant of 250,000 people or 0.82m³/s)
- (iii) The expansion of the sewerage network is done through traditionally used materials such as PVC. A total of 410 km of pipe will be installed and connected to the Norte WWTP network, as well as 83 km (including 5km for the main collector) for the Jagüel WWTP, using pipejacking, a commonly used technique for installing underground pipelines and ducts. AySA has a policy of not allowing connections of HH to sewerage networks until a functioning WWTP can treat the collected sewage, and it will provide users with access to safely managed sanitation services

6. The need for expansion of these WWTPs is justified by the planned expansion of the sewage collection networks, some of which is already ongoing. The proposed treatment process is adequate to ensure compliance with more stringent discharge standards to be enforced in the near future.

7. Technical engineering designs regarding large Program infrastructure prepared by AySA are solid. Notably both WWTPs are designed to incorporate flood protection, with an average elevation of 0.5 m on top of the 100-year flood water levels, increasing infrastructure resilience. Bidding documents are sound and present detailed specifications of works and installations. Along the same line, the 2-year construction timeline for this type of technology is considered to be reasonable (not considering procurement). The technology selected is considered to be standard and if properly designed, constructed and operated, the WWTPs will present high removal efficiencies of pollutants. Works will be contracted using a "turnkey" modality with detailed engineering designs and construction under the contractors' responsibility, ensuring a smooth transition towards the operational phase. A series of aspects where efficiency in the design could be improved have been identified by this assessment and included in the recommendations section.

Results Area 2: Improve AySA's efficiency

(A) Reduction in NRW



8. **AySA aims at reducing physical water losses as a way to increase water availability and improve service quality and extend coverage, thus delaying investments in increasing production capacity.** According to its 2018 water balance, the utility's physical losses stood at 690,767,610 m³/year, equivalent to 36.7 percent of the year production of 1,884,127,810 m³. Building on actions implemented since 2007, AySA proposes to recover a total of 123,374 m³/day through: (i) the replacement of 510 km of networks via trenchless pipe burst technology and high density polyethylene (HDPE) pipes (59,174 m³/day); (ii) the renovation of 30 km of pipelines through internal mechanical cleaning and polyurethane resin coating (2,000 m³/day); (iii) the implementation of 10 district metered areas (DMAs) (1,200 m³/day); (iv) conducting active leakage detection and repair supported by satellite detection technology (51,000 m³/day); and (v) the renovation of variable speed drives (VSD) in the Saavedra pumping station (10,000 m³/day).

9. **AySA's proposal follows sound practices to identify the magnitude of key components of water losses and the proposed actions proved to be successful in projects already implemented by the utility and in other countries in the region.** Pipe replacement is a reasonable approach to reduce physical losses in low-pressure systems and AySA will intervene the oldest pipes in its network or those most prone to breakage such as asbestos cement ones. The implementation of 10 DMAs including consumption meters is a continuation of previous efforts and a move to gain control of the system. Satellite leak detection improves the efficiency of traditional noise correlation in low-pressure systems, as confirmed by AySA in a pilot test. Water recovery targets from pipe replacement were set using data from similar previous activities and will be validated through pilots to be performed in two DMAs. Targets for satellite leak detection and repair were set applying typical loss rates from the burst and background estimates (BABE) model to a statistically probable mix of different types of breakages.

(B) Energy Efficiency Improvements

10. **AySA is seeking to reduce energy consumption to improve its environmental and financial sustainability, reducing energy and maintenance costs.** The total annual energy savings target is 33.93 GWh/year (92,960KWh/day), with active monitoring of consumption at 800 deep wells and 146 sewage pumping stations contributing to savings of 27.14 GWh/year (through a reduction in contracted power, and balancing supply and demand), distributed self-generation providing savings of 1.75 GWh/year (through pilot solar power plants in the San Martin and Belgrano WTPs and the Berazategui WWTP), and the installation of new motors with VSDs at the raw water pumping station at the San Martin WTP and the Saavedra water pumping station contributing 5.04 GWh/year.

11. **AySA has been implementing distributed energy consumption monitoring for several years now, with savings already achieved, and expects better results from actions contemplated under the Program, given that the interventions will be undertaken in larger facilities.** The experience gained by AySA staff during the previous three stages, provides assurance that these higher savings will be achieved. The installation of solar panels has become quite reliable during the last decade and AySA's estimation of 200 kW effective power out of a 1 MW facility is considered conservative given the plants' geographical location. These kinds of pilots and scale-up operations are being successfully implemented in several LAC countries. Replacing old motors at pumping stations with new ones provided with VSDs is a well-known practice in the sector and AySA has already installed VSDs in other pumping stations, and is therefore familiar with these devices, the selection of the right equipment, and the potential results.

(C) Promoting digital payment channels

12. **AySA seeks to improve its commercial efficiency through an increase in the collection volume of the three existing digital collection channels that have the lowest transaction costs (Automatic Debit, AySA's website and digitized Pagophone), which currently adds up to only 43 percent of the total amount collected.** AySA aims to reach 65 percent payments digitalization in terms of amounts collected, in line with other large regional providers through: (i) upgrading of the customer care center (*Centro de Atención Integral al Usuario, CAIU*); (ii) upgrading of the data warehouse; (iii) modernizing the contact center; and (iv) implementing telemetry technology for large customers.



13. **These four actions constitute an adequate package for the digital transformation of AySA's commercial area.** Implementation of this digitalization process will be carried out through the purchase of equipment and various internal software developments. The procurement of servers for the CAIU is already underway, while the other activities are technically well defined. Concurrently, these actions will reduce the occupancy loads inside banks and AySA's commercial offices, which is very important during the COVID-19 crisis. These actions are comparable to modern processes existing in comparably sized water utilities.

Results Area 3: Strengthen AySA's capacity to respond to water supply and sanitation emergency needs

(A) COVID-19 contingency measures implemented

14. **AySA adopted a number of actions to confront the COVID-19 health emergency and the associated mandatory preventive social isolation measures and its consequences on the utility's operations.** These actions have already been implemented, as per AySA's crisis management and coordination mechanisms, and are to be recognized as prior results. The response to the COVID-19 crisis by AySA is a subset of its broader PPE and is a testament to AySA's capacity to expand the scope of the PPE to incorporate new risks and hazards and activate its emergency response decision-making processes and coordination structure. These measures include: (i) the operationalization of AySA's Crisis committee; (ii) the delivery of water and bleach to isolated and underserved areas; (iii) the facilitation of remote working arrangements for AySA staff; (iv) the development of awareness campaigns for AySA employees about precautions to be taken during the COVID-19 pandemic outbreak; (v) the implementation of safety protocols for AySA's A+T and C+T programs; (vi) the use of the WBE method to monitor the presence of COVID-19 in sewage, and (vii) preventive measures for COVID-19 transmission implemented for AySA's staff and reorganization of human resources in AySA's operations department, while maintaining service levels.

15. **AySA's response through this COVID-19 package is in line with good regional practices, and with the WHO's Technical Guidance on "Water, sanitation, hygiene, and waste management for SARS-CoV-2" of July 2020.** In this document, the technical, commercial and organizational measures implemented focus on three key areas, namely: (i) ensuring access, continuity, and quality, particularly critical for handwashing; (ii) providing direct support to cover service fees; and (iii) ensuring employee safety. These sector response measures seen across the region reveal signs that the unforeseen health crisis has presented utilities with opportunities for enhanced connectivity, a crosscutting theme also observed across AySA's response, as demonstrated by its efforts to reach out to unserved populations, to enhance its relationship with customers and other stakeholders, and to accelerate its transition to digital solutions to make informed decisions throughout the lifecycle of its water and wastewater systems.

16. **On the particular topic of WBE, AySA's development and use of this approach is in line with best practices, such as described in the "Wastewater Surveillance of the COVID-19 Genetic Signal in Sewersheds.** Recommendations from Global Experts" Paper published by the Water Research Foundation, as a result of the "International Water Research Summit on Environmental Surveillance of COVID-19 Indicators in Sewersheds" from April 27 to 30, 2020.

(B) Implementing and Strengthening AySA's Prevention and Emergency Plan (PPE)

17. **AySA has developed its PPE which consists of a set of guidelines.** These aimed at: (i) preventing or reducing the possibility of a risk occurring (Proactive Stage); or (ii) providing a quick and efficient response to an ongoing crisis (Reactive Stage). It involves actions such as diagnostics, evaluation, prevention, mitigation, preparedness, response and recovery, all of which are directly linked to the expectations, interests of, and impacts on the different stakeholders. The PPE applies to all activities performed by AySA as part of its mandate of providing WSS services.

18. **AySA's PPE follows the HACCP methodology (Hazard Analysis and Critical Control Points) and its risk identification and emergency management procedures match its operational processes.** This, coupled with the fact that all operational procedures at AySA are ISO certified, provide a high degree of confidence that the PPE is relevant to AySA's operations and reasonably up to date. AySA has an established methodology to update risks already



identified and included in the PPE and the corresponding guidelines and actions associated with them. It is being used regularly throughout the organization to keep the plan relevant and effective. Each area within AySA reviews the relevant aspects of each risk affecting them and reports on changes in the key parameters conditioning the response to each risk.

19. **AySA's PPE was triggered when faced with the impacts of the COVID-19 pandemic and the utility now aims to capture lessons learned through its experience in dealing with an ongoing crisis and to develop a contingency plan to deal with epidemiological risks to be incorporated into the PPE.** This will be accompanied by the development of molecular biology tools, studies and procedures to allow for the detection, analysis and reaction to those threats, without which, the contingency plan would lack its early warning potential.

(C) Implementation of Digital Solutions, for internal systems and citizen engagement

20. **AySA began its transition to digital solutions before the COVID-19 emergency as shown in its IT plan, *Plan de Sistemas*.** The plan targets a wide range of areas within the company, and the Program will focus on: (i) the integration of its three human resources (HR) systems; (ii) solutions to enable the remote work modality on a continuous basis; (iii) predictive maintenance of main assets; (iv) digitalization of commercial processes in AySA's network of commercial offices; and (v) optimization of the customer complaint management system through the adoption of mobile and digital solutions. The development of these activities is led by AySA's Department of Technology and the majority of these activities are ready to be bid.

21. **An assessment of the soundness and readiness of each activity was performed by a digital solution's consulting firm.** The proposed activities were found to be adequate in terms of quality and strategic relevance, with the estimated costs in the correct order of magnitude. These activities will represent a quantitative and qualitative leap in the management of the processes they affect, specifically all those related to client relationship and digitalization of administrative procedures. Mobile solutions which affect field work, as well as the promotion of teleworking, will also be very effective and will increase the performance of operations. As for the activity related to a more proactive asset management approach, it will result in greater service availability and a smarter asset renewal strategy. It should also be noted that timeliness of results deriving from these actions will differ greatly from one activity to another. Some impacts derived from the implementation of these measures may not be immediate, as results for digital solutions often take time to materialize. For example, the implementation of teleworking will bring almost immediate benefits, however, predictive asset management will require significant historical information for the associated algorithm to quantify the effects.

22. **As part of its digital strategy, AySA will be building a functional mobile app and CRM to enhance easy and timely two-way communication and engagement, which will require consolidating internal processes and management of a multitude of media sources and internal databases.** The digital strategy will allow for: (i) rationalized customer data management; (ii) improved customer communications; (iii) timely two-way customer engagement; (iv) organized customer feedback and complaints; (v) greater transparency on public works and service requests, and (vi) greater ease of use for the consumer/citizen. Functionalities related to other technical claims are expected to be integrated in the app's second version, to be launched late 2021. This could be justified considering the background work needed to link AySA's current PQRS systems with the CRM. Nevertheless, the implementation of such functionality is key to guarantee two-way communications between user and utility. Regional initiatives are including in their digital channels the possibility of anonymously reporting fraudulent activities of the utility's employees or contractors, as well as simple geo-referenced report of complaints and PQRS status, open to all app users (whether a utility client or not). An internal interdisciplinary group is working on the development of the mobile app and CRM. Teams are coordinated by the communications officer and areas such as IT, communications department, CRM, technical and infrastructure areas, collaborate in the initiatives. Costs have been estimated for the apps first and second versions.



(D) Promoting gender diversity at AySA

23. **AySA has already adopted actions to respond to the needs of its employees during the COVID-19 crisis, thus ensuring the continuity of its services.** Among these actions are campaigns to encourage a greater division of labor at home, or protocols and communication campaigns to respond to the increased gender-based violence. The Program aims to support AySA in its Inclusion, Gender, Equal Opportunities and Treatment and Workplace Violence Program (“Gender and Inclusion Program”), particularly by supporting AySA in setting targets for gender diversity in leadership positions, in order to increase women’s representation among AySA’s managerial staff.

24. **AySA’s Gender and Inclusion Program is the result of the utility’s and the union’s joint effort to promote respectful and fair treatment, diversity, inclusion and equal opportunities.** These efforts resulted in a series of actions that are already under implementation including two new protocols to address sexual harassment in the workplace and domestic violence, a Gender and Diversity Action Plan for the 2020-2023 period, campaigns to encourage a more equitable division of labor in the household and which question traditional gender roles. Two axes from AySA’s gender action plan focus on promoting the participation of women in decision making positions, particularly in technical areas, where they have traditionally been underrepresented (salaries and career and promotions). AySA is also part of the World Bank’s Equal Aqua Global Platform, which seeks to deepen gender diversity and inclusion in water institutions providing a wealth of resources to those ends.

25. **Under its Gender and Diversity Action Plan, AySA completed a preliminary diagnostic which entailed collecting gender disaggregated data on women and men’s participation at different hierarchical levels in the utility as well their access to career growth opportunities such as advanced education programs.** AySA has also completed a quantitative assessment using the World Bank’s HR questionnaire and is thus eligible to receive a tailored scorecard on their comparative performance on gender diversity which will allow for a more precise assessment of the success of the policies already implemented. Data submitted by AySA through the World Bank’s HR Questionnaire confirms the underrepresentation of women in managerial positions, with only 22 percent of female managers in the company, which is way below the regional average of 35 percent⁶⁹. Actions taken by AySA to reduce gender gaps at the management level, including reviewing their promotion practices, launching targeted communications campaigns to encourage women to apply for promotion opportunities and providing training to fill women’s technical or other capacity gaps, are considered an integrative approach to address the identified gender gap.

26. **AySA’s work to promote gender equity is extensive and focused on creating better employment opportunities for women both at the utility and community levels, adopting a gender perspective in its A+T and C+T programs⁷⁰.** The utility considered other options for promoting gender equity but determined that improving its internal diversity at decision making level would have a more significant long-term impact than short-term project specific initiatives.

C. PROGRAM EXPENDITURE FRAMEWORK

27. **For government transfers to AySA, disaggregation for program expenditure monitoring is made by AySA using its processes and systems.** At the national level, no distinction is made of the financing being provided for the PMOEM as PMOEM is not a separately identified program line item in the National Budget. AySA’s budgeting and accounting information systems are able to track the expenditures incurred under the Program using accounting policies consistent with national public sector policies/standards. The Program’s budgeting transactions will also be

⁶⁹ World Bank Water Data. <https://wbwaterdata.org/breakingbarriers/tool/>. December 17, 2020. (Based on data from 9 LAC water utilities collected through the World Bank’s HR Utility Survey including data submitted by AySA.

⁷⁰ Cooperatives that participate in these programs have committed to provide equal recruitment opportunities for women. Activities carried out under the *Agua más Trabajo* and *Cloaca más Trabajo* programs include training in technical fields for women and men as well as workshops aimed at creating a respectful and equitable workplace and culture, free of gender violence.



processed through e-SIDIF (and UEPEX with respect to Source 22 financed expenditures) after having been processed through AySA's own budgeting and accounting information systems, under the general budget code/line specified above.

Review of the Program Composition by Expenditure Categories / Type of activity

28. **The following Table 7 details the Program's expenditures by sub-program and also by Results Area.** A more detailed version of this table, including budgeting and or accounting lines/codes is included in the Operations Manual, indicating each Program expenditure in detail. This more detailed version will be used to track the Program expenditures during implementation as well as to facilitate the audit of the Program's Financial Statements (and ultimately the overall Program expenditure reconciliation). The nature of the expenditures in Results Area 1 to be incurred is predominantly civil works contracts in specific municipalities of AySA's jurisdiction with the associated operating costs of the main WTP and WWTPs. Results Area 2 and 3, in addition to works contracts and corresponding operation costs, also includes goods contracts.



Table 7: Program Expenditure Framework

Title of the Program or sub-Program		PforR Results Area	Total Program	Program's expenditure projection						
				2020	2021	2022	2023	2024	2025	2026
sub-program 1: Improvements and Expansion of Services										
Intrahousehold connections	Water	1	\$ 3,676,000	\$ -	\$ 662,000	\$ 3,014,000	\$ -	\$ -	\$ -	\$ -
	Sewerage	1	\$ 26,061,000	\$ -	\$ 1,955,000	\$ 16,288,000	\$ 7,818,000	\$ -	\$ -	\$ -
Water Basic Infrastructure	Florencio Varela	1	\$ 49,232,000	\$ -	\$ 3,726,000	\$ 33,531,000	\$ -	\$ 2,927,000	\$ 9,048,000	\$ -
Sewerage Basic Infrastructure	Planta Norte optimization Module III	1	\$ 72,600,000	\$ -	\$ -	\$ 7,284,000	\$ 16,754,000	\$ 48,562,000	\$ -	\$ -
	Planta Norte optimization Module I and II	1	\$ 1,827,000	\$ -	\$ -	\$ 1,827,000	\$ -	\$ -	\$ -	\$ -
	Planta Norte sewerage networks	1	\$ 94,438,000	\$ 1,448,000	\$ 6,565,000	\$ 24,766,000	\$ 19,410,000	\$ 23,523,000	\$ -	\$ -
	El Jagüel plant optimization	1	\$ 12,100,000	\$ -	\$ 1,210,000	\$ 3,025,000	\$ 7,865,000	\$ -	\$ -	\$ -
	El Jagüel sewerage networks	1	\$ 52,071,000	\$ 2,666,000	\$ 4,228,000	\$ 5,495,000	\$ 7,627,000	\$ 16,085,000	\$ 786,000	\$ 524,000
Total sub-Program 1			\$ 312,005,000	\$ 4,114,000	\$ 18,346,000	\$ 95,230,000	\$ 59,474,000	\$ 91,097,000	\$ 9,834,000	\$ 524,000
sub-program 2: Improvements and Maintenance										
Leak reduction Program		2	\$ 106,743,000	\$ 927,000	\$ 6,541,000	\$ 40,758,000	\$ 39,530,000	\$ 18,479,000	\$ -	\$
Energy efficiency: Renovation of pumping station and other energy equipment		2	\$ 16,698,000	\$ -	\$ 5,844,000	\$ 8,349,000	\$ 2,505,000	\$ -	\$ -	\$ -
Total sub-Program 2			\$ 123,441,000	\$ 927,000	\$ 12,385,000	\$ 49,107,000	\$ 42,035,000	\$ 18,479,000	\$ -	\$ -
sub-program 3: Operations										
Energy efficiency: Energy monitoring and solar pannels		2	\$ 8,047,000	\$ -	\$ 908,000	\$ 908,000	\$ 5,324,000	\$ 908,000	\$ -	\$ -
Development of Digital Payment Channels Program		2	\$ 3,751,000	\$ -	\$ 908,000	\$ 1,936,000	\$ 908,000	\$ -	\$ -	\$ -
Implementation of Digital Solutions including Citizen Engagement Strategy		3	\$ 22,500,000	\$ 288,000	\$ 6,580,000	\$ 6,436,000	\$ 4,598,000	\$ 4,598,000	\$ -	\$ -
COVID-19 Assistance and Contingency Plan		3	\$ 1,111,000	\$ 1,111,000	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
Total sub-Program 3			\$ 35,409,000	\$ 1,399,000	\$ 8,396,000	\$ 9,280,000	\$ 10,830,000	\$ 5,506,000	\$ -	\$ -
Other Expenditures										
Program's operational expenditures (WWTP and WTP OPEX)			\$ 100,000,000	\$ 14,286,000	\$ 14,286,000	\$ 14,286,000	\$ 14,285,714	\$ 14,286,000	\$ 14,286,000	\$ 14,286,000
Evaluation and Verification costs			\$ 5,000,000	\$ 750,000	\$ 750,000	\$ 750,000	\$ 750,000	\$ 750,000	\$ 750,000	\$ 500,000
Front-end fees IBRD			\$ 750,000	\$ 750,000	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
Total other expenditures			\$ 105,750,000	\$ 15,786,000	\$ 15,036,000	\$ 15,036,000	\$ 15,035,714	\$ 15,036,000	\$ 15,036,000	\$ 14,786,000
Total Program			\$ 576,605,000	\$ 22,226,000	\$ 54,163,000	\$ 168,653,000	\$ 127,374,714	\$ 130,118,000	\$ 24,870,000	\$ 15,310,000



29. **For the larger Program expenditure corresponding to the water and sanitation infrastructure investments, unit costs are efficient due to the large economies of scale of the Program.** These costs are estimated at US\$430 per capita, considering full capacity of the newly built facilities in the long term (720,000 beneficiaries by 2030, and an investment of 312 million – full investment in RA1). By the end of the Program, when some of the facilities such as the Norte WWTP will still be working at partial capacity, this ratio would be around US\$700 per capita, which is still considered within reasonable limits⁷¹. Within these costs, the ones associated to the connectivity program are the lowest given the high sunk costs in the form of already existing infrastructure (US\$91 per person in the case of water and US\$325 for sewerage connections). The main reason why sewerage connections are more expensive is due to the need to bypass and seal the different typologies of existing onsite solutions that each household may have. For the large infrastructure, the cost of the two WWTP expansions (72 million and 12 million) is also considered efficient for the size and treatment capacity and characteristics (with nutrient removal) of these facilities. In Florencio Varela, investments for water supply infrastructure are estimated at US\$49 million, including civil works at the pumping station and initial trunk main, and are considered reasonable when compared with similar infrastructure in the LAC region of the same size.

30. **For Results Area 2, the costs of pipe replacement per kilometer (117,650 US\$/km) and value of water recovered of (0.11US\$/m³), and the cost/benefit ratios of energy saving interventions (0.07 US\$/kWh) have also been compared with those in similar utilities and are considered logical.** The investments related to the digitalization program in Results Area 3 are relatively small compared with the benefits and savings they can bring to AySA in terms of capacity gained to respond fast and in a more tailored manner to customers, translated into higher service quality standards and customer satisfaction. These activities also imply a hidden cost in the form of a steep learning curve for AySA's technical staff since they will be implemented for the first time, and they have a strong *learning by doing* characteristic. The expenditure framework includes quantifiable costs related to concrete measures implemented during the COVID-19 crisis, as a consequence to apply AySA's Prevention and Emergency Plan. However, AySA's response to the pandemic implied a broader set of measures not so easily monetized, like the internal reorganization of staff, implementation of remote work modalities, quarantines, prevention campaigns, etc. Lastly, AySA's costs associated with the achievement of DLI 7 on gender, relate to internal activities linked to staff costs (such as training sessions, workshops to sensitize staff, etc.), which are difficult to quantify in monetary terms.

31. **The execution rates proposed for the different activities rationally fall within the implementation period of the Program.** The activities which take longer to implement are the new modules of the Jagüel and Norte WWTPs, (two years) which is within normal ranges for works of this magnitude. There is a risk of delays in the execution of the intra-household connectivity program since part of the works need to be undertaken inside the households and dwellers may feel reluctant to allow workers inside during the pandemic. The main mitigation measures are the adoption of safety measures following AySA's health protocols, the promotional campaigns conducted in the beneficiary areas before the start of the works, and the extension of the timeline to execute these works by 6 months, from 1.5 to 2 years.

Adherence of Program Expenditures to Government Priorities

32. **Investments in Water and Sanitation are a priority for this government and the PMOEM is the vehicle through which the government has funded AySA's infrastructure needs since 2010.** AySA primarily finances its operations and the PMOEM through transfers from its majority shareholder, the National State. Law No. 27467 on the National Budget for the year 2019, regulated by Resolution No. 12/19 issued by the Head of the Cabinet of Ministers allocated transfers to AySA for the fiscal year ending on December 31, 2019 for works (costs of acquisition or

⁷¹ A recent analysis being conducted by the Bank indicates that full costs of providing full water and sewerage coverage in Argentina (greenfield projects), as a national average are about US\$ 1000 and US\$1300 per capita respectively.



construction of assets used to provide the service) in the amount of ARS15,562 million (including loans from multilateral entities belonging to Source 22). During the year ending December 31, 2019, AySA received transfers from the National State amounting to ARS21,737 million (ARS10,181 million related to Source 11, ARS11,507 million to Source 22 and ARS49 million from other sources). From the start of the concession and up to December 31, 2019, the Company has received transfers from the National State totaling ARS85,689 million.

33. **The unplanned incorporation of nine municipalities in AySA's provision area between 2016 and 2018 coupled with the macroeconomic situation during this period increased operational expenditures and budgetary pressures, further reducing AySA's capacity to invest and fully fulfill the PMOEM's target during that period.** However, WSS is a National Government priority, with AySA having the largest share of water and sanitation investments in Argentina with 0.08 percent of the GDP over 2017-2019. The government had reiterated its commitment to make transfers readily available, ensuring that fiscal constraints will not interrupt social subsidies targeted to vulnerable populations for basic services such as water and sanitation. Therefore, while government transfers to AySA have been and could be unstable in times of macroeconomic turbulence, the risk of sustainability to Program investments is considered to be minimal given the priority standing assigned to the Program by the government and AySA alike, including their shared interest in both addressing the pressing needs of unserved and low income areas hit hardest by the COVID pandemic, and addressing efficiency in service delivery. Other reasons to assess this risk as low include the small size of the Program to be financed by the World Bank compared to the entire PMOEM, and the prioritization of operation and maintenance (O&M) costs in government transfers to AySA.

Allocation of Budget Across Activities and Budget Execution

34. **The PMOEM is prepared every five years based on works completed/service requirements/information from census, etc. The process for developing the next version of the PMOEM will normally start 1 to 1.5 years before the end of the current PMOEM.** The PMOEM is prepared based on water standards or statistics and specific studies. For example, 1km of water and sewage connections normally requires x number of connections, y km number of pipes, z number of pumps with standard costs/measures for each of these elements to derive the total cost of the works. The PMOEM is adjusted as investments/works are further designed, as well as the impact of changes in the prices. There are processes in place to ensure that up to date listing of prices/costs is maintained (monthly updates, based on for example the prices quoted by suppliers on contracts or the results of recent studies), so that the budgeting of works is maintained as realistic as possible, taking into account both inflationary and exchange rate pressures. The unit responsible for planning and coordinating the execution of the PMOEM is adequately and competently staffed and has vast experience. There are policies and procedures in place to guide the preparation of the PMOEM from a technical perspective, including guidelines on how to cost works, monitor prices, seek explanations in case of variances greater than 15 percent or less than 20 percent of the expected costs/prices. An annual report is also prepared each August and submitted to APLA, which analyzes the execution of the PMOEM, more from a technical perspective as opposed to a financial perspective. AySA does, however, submit monthly reports to APLA detailing the physical and financial progress of each activity included in the PMOEM.

35. **AySA has a high degree of autonomy in budget execution within the parameters of the approved PMOEM and has a high institutional capacity.** The Program is not protected by revenue earmarking mechanisms and, consequently, there is no protected share of the Program.

D. ECONOMIC AND FINANCIAL EVALUATION

36. **Capital expenditures in water and sanitation by the National Government of Argentina represented 0.13 percent of GDP over 2017-2019, and expenditures are expected to increase to 0.30 percent in 2020.** AySA has the largest share of water and sanitation investments in Argentina with 0.08 percent of the GDP over 2017-2019. In 2020, investments by AySA are also expected to increase and reach 0.19 percent, 75 percent of which had already been executed by September 2020. To achieve SDGs 6.1 and 6.2 by 2030, it is estimated that Argentina needs to expand its



annual rate of investment in water and sanitation by 2.7 times, increasing the rate of expansion of by 2.6 times for water networks and 4.8 times for sewerage networks.

37. **The Program aims at supporting a sustainable path leading to more efficient operations at AySA.** First, with respect to the levels of water losses, AySA is implementing an asset management strategy to prioritize rehabilitation works and to lower technical losses with network renewal and loss reduction targets explicitly mentioned in the PMOEM 2019-23. Second, with respect to metering levels, AySA aims to increase metering significantly by 72 percent between 2019 and 2023, which will result in a total of 875,000 meters installed by 2023. To take advantage of the potential efficiency benefits of metering, AySA has recently changed the tariff structure applicable to metered connections by reducing the fixed charge and increasing the share that corresponds to metered consumption.⁷²

38. **The economic and financial analysis focused on eight specific projects that represent 91 percent of investments in this Program.** While the Program includes investments of US\$389 million (without Value Added Tax, VAT), the quantitative economic and financial analysis considered investments that total US\$ 355 million. The benefits of the remaining projects are considered qualitatively. The analysis for each project considers all capital and operating costs required to achieve the objectives. The Net Present Value (NPV) is calculated with a discount rate of 6 percent and with a 30-year horizon.

Table 8: Program Investments Considered in the Financial and Economic Analysis

Projects	Investments (Constant Prices without VAT)		
	Total (US\$ thousands)	Unit Costs (US\$)	Unit
Intrahousehold Water Connections	3,011	75	Person
Intrahousehold Sewerage Connections	21,538	269	Person
South System to Florencio Varela	47,000	551	Person
North Plant	157,000	432*	Person
Jagüel Plant	46,000	319*	Person
Renovation of Water Networks	69,700	1.66	m ³
Saavedra Pumping Station	7,600	2.07	m ³
San Martín Plant	3,000	2.13	kWh
Total for Selected Projects	354,849	n/a	n/a
Total for Program	389,136	n/a	n/a

* Alternative estimations for the Norte and Jagüel WWTPs result in investments of US\$768 per capita (estimated by IADB) and in US\$1340 per capita [estimated by the World Bank (2018)], which are based on different assumptions on total and sunk costs.

Economic analysis

39. **The economic NPV of the Program is positive (US\$1.4 billion) with an Economic Internal Rate of Return (EIRR) of 40 percent.** In the economic analysis, the costs were estimated with economic prices, i.e. market prices adjusted with shadow price coefficients. Operating costs of the project were estimated by AySA and were adjusted to consider the maintenance requirements over the analysis horizon, in consultation with AySA. The evolution over time of investments, number of beneficiaries, and other benefits over time, together with the methodologies used to quantify them are shown in Table 9.

⁷² AySA (2018) Plan De Mejoras, Operación, Expansión y Mantenimiento De Los Servicios 2019-2023. Revisión Quinquenal.



Table 9: Results of the Economic Analysis

Project	Economic Analysis			Method for Assessing Benefits
	EIRR	NPV <i>US\$ Thousands</i>	Benefit-Cost Ratio	
Intrahousehold Water Connections	393%	123,067	23.37	CA and SUR
Intrahousehold Sewerage Connections	100%	208,907	10.22	HP
South System to Florencio Varela	18%	68,463	3.00	CA
Norte WWTP	41%	778,086	5.77	HP
Jagüel WWTP	55%	227,813	7.22	HP
Reduction in water losses	10%	26,668	n/a	CS
Saavedra Pumping Station + San Martín WTP	7%	888	n/a	CS
Total	40%	1,433,892	n/a	

Method: 1) CA and SUR: Costs avoided plus consumer surplus based on non-parametric demand, 2) HP: Hedonic prices and propensity score matching; 3) CA: costs avoided⁷³; 4) CS: Cost savings. Although in the case of the Plants they were also evaluated through the Willingness to Pay, in these cases lower EIRRs were obtained but always above 6 percent.

40. **The projects on intra-household water and sewerage expansion have a positive and sizable economic return.** The results of 393 percent and 100 percent for the EIRR for these projects can be explained by sunk costs from prior investments in the water and sewerage network. In other words, the marginal benefit of connecting additional households is significantly higher than the marginal cost because, other than the intra-household connections, other parts of the water and sanitation systems are already in place. First, the benefits quantified for water connections include avoided costs from purchasing water from other sources, such as bottled water, and the consumer surplus as a result of higher levels of water consumption, which were calculated comparing water consumption patterns of users that are connected and that are not connected to the network. Second, the benefits associated with sewerage intra-household connections was calculated using hedonic prices, i.e., calculating the differential between property values for identical houses, only varying sewerage availability. Lastly, by improving water quality, the benefits to Florencio Varela are also significant because they target densely populated areas. In addition, the beneficiary areas are low-income compared to AySA's area of coverage and these users, relative to their income, place a higher value on receiving service⁷⁴.

41. **The assessment of sewerage projects (Norte and Jagüel WWTPs) also revealed a positive and significant economic return under alternative scenarios.** The benefits were determined using hedonic prices based on the differential between property values for identical houses, varying access to sewerage. In these projects, the costs of the residential connections were included in the total costs. In all cases, only the benefits accrued to the direct beneficiaries were considered, while other external benefits, for example, those perceived by neighbors and those indirect benefits such as savings in the cost of public health were not.

⁷³ CA and SUR: Costs avoided plus consumer surplus based on non-parametric demand, estimated by the World Bank Group for Argentina based on data from Argentina's National Institute of Statistics and Censuses (INDEC), including the 2012 Household Income and Expenditure National Survey, the 2010 Census, and the Continuing Survey of Households from 2010 to 2018. HP: Hedonic prices and propensity score matching, using a database constructed by the World Bank Group for Argentina with property values according to their characteristics with source data from INDEC.

⁷⁴ Same sources as previously but considering benefits in terms of per capita income, differentiated by quintile.



42. **The economic analysis for projects that seek to reduce NRW resulted in a positive economic return despite conservative assumptions used for the estimation of benefits.** The benefits that were considered include the cost savings from water production, electricity, use of chemicals, and the reduction of maintenance needs based on operational costs projections by AySA. It is reasonable to assume that the cost savings associated with these interventions are higher than estimated due to additional benefits that were not quantified, such as an increase in the useful life of the assets, the costs savings from avoiding reactive maintenance, the reduction of interventions in public roads due to water losses, and the consequences of the overexploitation of aquifers.

43. **A sensitivity analysis was carried out considering alternative cost and benefit scenarios for the sewerage treatment plants, as well as for alternative shadow prices of carbon emissions.** First, with regard to the reduction in GHG emissions originating from the Norte and Jagüel WWTPs from energy efficiency improvements and from NRW cost savings, varying the social cost of emissions between low and high scenarios does not significantly affect the EIRR of the Program which remains with an overall EIRR between 39 percent and 40 percent (Table 10). Second, a sensitivity analysis of the results for the interventions involving sewerage treatment plants was carried out with 6 different scenarios with alternative levels of benefits and investments. This considered alternative per capita investment costs (lowest of \$319 and highest of \$1,340), including a scenario in which no sunk costs are assumed. Two scenarios with alternative benefit estimations were also considered. The resulting economic rates of return were higher than a 6 percent discount rate in all scenarios and reached 55 percent in the highest return scenario. The scenario with higher investment costs and lowest per capita benefits for the Norte and Jagüel WWTPs had an economic return of 8.1 and 8.5 percent, respectively.

Table 10: Sensitivity Analysis Using Alternative Shadow Price of Carbon Emissions

US\$ Thousands	Average Price of Carbon			High Price of Carbon			Low Price of Carbon		
	EIRR	NPV	B/C Ratio	EIRR	NPV	B/C Ratio	EIRR	NPV	B/C Ratio
Norte WWTP	41%	778,086	5.77	41%	794,132	5.87	40%	762,039	5.67
Jagüel WWTP	55%	227,813	7.22	56%	232,233	7.34	55%	223,392	7.10
Reduction in water losses	10%	26,668	n/a	11%	27,975	n/a	10%	25,360	n/a
Saavedra Pumping Station + San Martín WTP	7%	888	n/a	7%	1,165	n/a	7%	612	n/a
Total	40%	1,433,892		40%	1,455,943		39%	1,411,841	

Note: GHG emission reductions were estimated for all projects except for intrahousehold connections and the South System to Florencio Varela

44. **By focusing on the most significant benefits expected from the Program, a small percentage of the investments were assessed qualitatively due to challenges in quantifying their benefits.** These investments will positively impact operational performance, transparency and a better relationship with users, increasing the perceived value of the company by the suppliers and by the community at large, which will result in additional private and social benefits.

45. **An increase in the proportion of bills paid with digital methods is expected to increase total revenue collection and generate social and user benefits.** The Program will support an investment of US\$3 million to increase the use of digital payment methods. Roughly 51 percent of AySA's revenue collection currently corresponds to in-person payments, representing more than 13 million annual transactions in banks, non-banks financial institutions and AySA offices. During the COVID-19 pandemic, the fact that most bills are usually paid in-person affected revenue collection. Improved access to digital payments would have partially offset this impact. Digital payments also imply



lower travel and time costs for users and reduce the costs of unwanted late payment fees. It eliminates the risk of carrying cash and the inconvenience of planning payments ahead. Facilitating automatic payment from a bank account is likewise expected to improve the collection ratio.

46. **The Program's support to gender equality in AySA is also expected to generate significant benefits.** These benefits are however challenging to quantify in a cost-benefit analysis. Gender diversity allows greater innovation, better decision-making and governance, increased customer satisfaction and service quality, better employee retention, greater sustainability, operational security and community relations. The social impact of these benefits is considerable, and it can be noted that companies with greater diversity have a 21 percent greater probability of being more profitable and 27 percent greater in creating value.⁷⁵

47. **In addition to the economic benefits highlighted, as the economy enters an unprecedented crisis caused by the COVID-19 pandemic, investment in this Program has a high potential to support jobs.** In the short term, given that construction works are expected to begin as early as 2021, this Project can support job creation during the economic recovery period. A World Bank analysis of job generation potential of alternative infrastructure investments reveals that WSS projects in Argentina can support between 1,088 and 2,687 direct job-years per US\$100 million invested in activities such as water network expansion and primary and secondary sewerage lines. Based on this estimate, the projects of this Program can support a total of 6,102 to 15,070 job-years, after accounting for direct and indirect jobs, *i.e.* indirect jobs are supported within the suppliers to the production process. In addition, about 75 percent of jobs in WSS construction require low- and semi-skilled workers, which are likely to be filled by low-income workers. To support the jobs agenda, AySA implements the A+T and C+T programs which seek to create new jobs through cooperatives. The cooperatives are associations of autonomous workers that provide the labor force for the execution of projects and receive compensation according to the degree of completion. The Programs also offer training and tools for workers. The works associated with expanding intrahousehold water connections of the Program will be implemented through cooperatives.

Financial analysis

48. **An assessment of AySA's financial sustainability and an estimation of the financial impact of the Program was conducted.** This analysis involved a review of recent audited financial statements and AySA's financial projections over 2020-2025. While the economic analysis evaluated the Program from a social point of view, considering benefits to users and externalities, the financial analysis focused only on AySA's financial perspective.⁷⁶

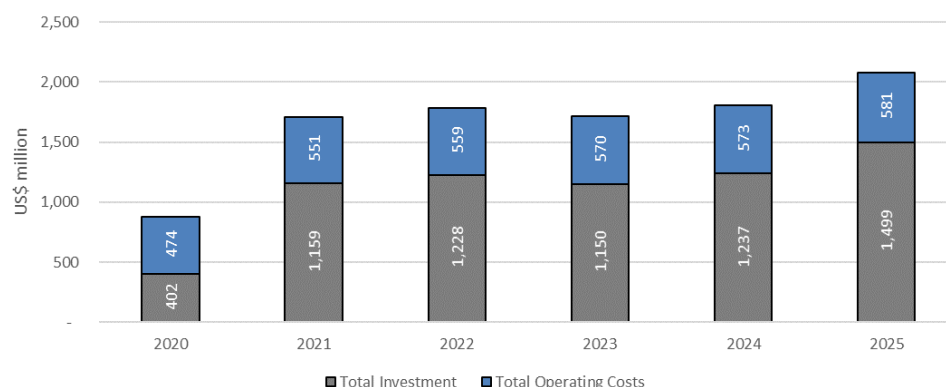
49. **In 2020, AySA's capital costs represented 46 percent of its total costs, and financial projections show that investment will increase significantly over the next few years (Figure 3).** This expected growth in investment shows the Government of Argentina's commitment to pursue the goal of universal access to water and to achieving 75 percent coverage in access to sewerage by 2030. The 2021 National Budget shows that the Government will continue to provide support to cover AySA's operational costs that are funded with tariff revenues, while increasing investment levels. AySA projects an increase in investment from \$401 million in 2020 to \$1.16 billion in 2021, reaching \$1.5 billion in 2025, of which 69 percent will be dedicated to network expansion and 31 percent to improvement and renovation. A focus on investment, especially by expanding assets into poor communities that are hard to reach, can be a more progressive use of resources than supplying subsidies to cover operating costs. Often populations without access to water services are low-income and the expansion of service coverage can generate benefits to newly connected households.

⁷⁵ World Bank. 2019. Women in Water Utilities, Breaking Barriers

⁷⁶ The financial analysis used market prices, regardless if that prices represent the true cost to society.



Figure 3: AySA's Projections for Operating Costs and Total Investment, Constant 2020 Prices



Note: Original projections in Argentine Pesos converted to US\$ at the rate of \$83.54.
Source: AySA Projection 2021-2025.

50. **Following an upward trend in the cost recovery ratio over 2015-2019, in 2019 AySA covered 75 percent of operational costs with tariff revenue, and the remaining 25 percent were covered with transfers from the National Government.** The share of operating revenues to operating costs continuously increased from 41 percent in 2015 to 75 percent in 2017, as a result of tariff increases above inflation rates. In 2018 and 2019 cost recovery levels stood at 72 percent and 75 percent respectively (Figure 4). Financial transfers from the Government of Argentina cover the O&M costs that are not funded with tariff revenue.⁷⁷

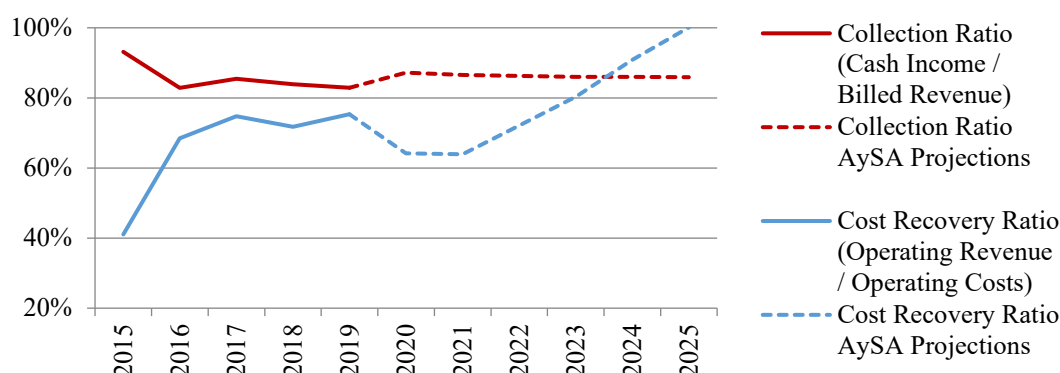
51. **AySA's subsidy scheme is progressive, and includes targeted subsidies, cross-subsidies and supply-side subsidies, as well as subsidies for network expansion.** First, the National Government subsidizes capital costs, which are mainly investments to expand water and sanitation services to low-income areas. Subsidies for network expansion are highly progressive since the main beneficiaries are low-income households, which are the largest proportion of users without access. Second, the subsidies that cover AySA's operating deficit benefits all users, but it is progressive in relative terms (the poorest 20 percent, while having only 3.6 percent of total income, receive 4.7 percent of subsidies). Third, the tariff structure is based on a cross-subsidy scheme, with the tariff varying according to location, area, quality and age of the property that is being served, which is used as a proxy for the income level of users. Finally, there are targeted subsidies whose beneficiaries are determined based on income and are largely progressive because they benefit directly low-income households.

52. **AySA's financial projections for the period 2021-2025 show a commitment from the Government of Argentina to subsidize the financial deficit in the short run, with a view to reducing the operational subsidy over the medium term by reducing supply-side subsidies.** In 2020 and 2021, with no expected increase in tariffs during the COVID-19 crisis, the cost recovery ratio is expected to decrease from 75 percent in 2019 to 64 percent in 2020 and 2021 (Figure 4). However, the projections show that an increase in tariffs in constant prices is expected for the years 2022-2025 will lead to an increase in operational revenues above operating costs until reaching full cost recovery by year 2025.

⁷⁷ US\$ 121.9 million related to Source 11, US\$ 137.7 million to Source 22 and US\$ 0.6 million from other sources, based on a conversion rate of \$83.54.



Figure 4: AySA's Cost Recovery Ratio and Collection Ratio (Realized and Projected), 2015 to 2025



Source: AySA Projection 2021-2025 and AySA Audited Financial Statements

53. **During 2020 and 2021, AySA's financial results will be affected by the COVID-19 pandemic as a result of a decline in revenue collection.** In April 2020, a 37 percent year-over-year decline in revenue collection was estimated, after controlling for a change in tariffs from 2019 to 2020. This can be explained not only as a result of the immediate economic impact of COVID-19, but also because of the mandatory lockdown in effect from March 20, 2020, which led to the closure of in-person payment locations, to the limited mobility of individuals due to stay-at-home orders, and to restrictions for AySA to conduct water meter readings.⁷⁸ Once GBA lifted some mobility restrictions between May and July 2020, the initial decline in revenue collection was reverted, since users were then able to pay their full balance due in person, leading to a full rebound in revenue collection in the months of May, June and July, 2020. A decline of 10 percent year-over-year in the revenue collection for August 2020. Considering the collection ratio, there was a decrease in the proportion of bills paid by the due date from 68 percent to 63 percent, comparing the months of August in 2020 and in 2019. Moreover, the collection ratio (cash income over billed revenue) decreased from 91 percent in 2015 to 84 percent in 2019.⁷⁹ It is also important to highlight that as a policy response to the economic crisis, AySA's suspended service shutoffs for low-income households kept tariffs unchanged.⁸⁰

54. **The Government of Argentina will continue to subsidize AySA's capital investments through transfers, expanding the network to low-income communities and helping ensure a sustainable growth of the water and sanitation network.** Figure 5 shows that the government will continue to support the increase in capital costs from AySA and that the O&M subsidies will be phased out in 5 years. Investments from the Program will be executed by AySA and fully subsidized with transfers from the National Government. Benchmarking studies show that AySA is among the majority (86 percent) of water utilities worldwide that do not generate sufficient operating revenue to cover capital and O&M costs, since it is rather unusual for utilities to simultaneously expand services into the hardest places to reach and collect long run marginal costs recovery tariffs from the poor.⁸¹

⁷⁸ 51 percent of payments were received in AySA's locations and bank's branches in 2019.

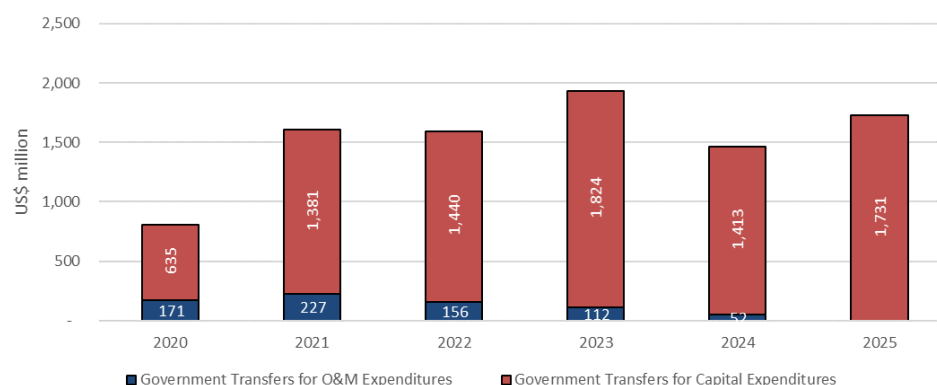
⁷⁹ Measured at 90 days past due date.

⁸⁰ Water tariffs in 2020 were kept constant in nominal terms despite an annual inflation rate of 42.8 percent (in June 2020) and service shutoffs due to nonpayment were suspended as of March 24, 2020. These temporary measures have been instrumental in protecting vulnerable populations during the pandemic and during the recovery stage, offsetting the impact on low-income households. The Government has announced that they will be effective at least until end-2020.

⁸¹ Andres, L. et al. (2019) Doing More with Less: Smarter Subsidies for Water Supply and Sanitation.



Figure 5: AySA's O&M and Capital Transfers from the National Government (Projected)



Note: Original projections in Argentine Pesos converted to US\$ at the rate of \$83.54.

Source: AySA Projection 2021-2025.

55. **The results of the Program's financial analysis reveal a positive NPV of US\$88.4 million, largely driven by the operational efficiency improvements and the NRW reductions supported by the Program.** This analysis takes the latest AySA's financial statement as a baseline and projects future cashflow, taking into account the expected change in revenue and costs as a result of the Program's implementation. It considers operating flows since capital costs are directly subsidized by the National Government through transfers.⁸² This financial analysis also calculated the ring-fenced NPV for each individual project. First, the reduced costs associated with the operational efficiency improvements lead to the highest financial NPV of US\$52.2 million among all project.⁸³ Second, no financial return is expected for projects associated with network expansion in hard-to-reach low-income communities, which can be explained by the fact that tariffs (with targeted subsidies and cross-subsidies) are relatively low in the areas of intervention. These interventions, however, have the largest economic returns since the network expansions will benefit currently unconnected, low-income and vulnerable households.⁸⁴ It is important to highlight that positive financial NPV for the construction of new assets in a poor community when revenues are circumscribed by those users is unusual among water utilities worldwide. Third, the NPV of the South System to Florencio Varela is also largely positive, considering that production from surface water has lower operating costs compared to underground water. The estimations for this Project also considered an increase in the collection ratio (Table 11).

⁸² According to the World Bank (2014) IPF Economic Analysis Guidance Note, the financial analysis may include explicit cash subsidies in the financial returns of an agency that operates in commercial terms.

⁸³ This estimation is conservative since it was not possible to quantify the benefits associated with an increase in the useful life of the assets, the costs savings from avoiding reactive maintenance, the reduction of interventions in public roads due to water losses.

⁸⁴ The change in cashflow from the project was estimated based on the current tariffs and collection ratio in the areas the project will have an impact on, which are lower than average due to the low-income population served.



Table 11: Results of the Financial Analysis

Projects	NPV (US\$ Thousands)
Intrahousehold Water Connections*	-569
Intrahousehold Sewerage Connections*	-865
South System to Florencio Varela	2,104
Norte WWTP	33,602
Jagüel WWTP*	-5,056
Reduction in water losses	52,176
Saavedra Pumping Station + San Martín WTP	7,030
Total	88,421

Note: Financial analysis based on operating cashflow.

* Considers the currently subsidized water tariffs (targeted subsidies and cross-subsidies) for low-income households in the estimation of the additional operating revenue from network expansion projects.

E. RISK AND MITIGATION MEASURES

56. The overall risk is from a technical perspective is “Moderate”, given that the rating is expected to reflect risks prior to the mitigation activities contemplated under the Program and to be documented in this Program Appraisal Document. Table 12 identified the main risks and mitigation activities proposed that will be implemented via recommendations, technical assistance or binding actions contained in the PAP.

Table 12: Risk Analysis

RA	Risk / Assumption	Mitigation Measures	Risk level
All	AySA has difficulties to collect timely monitoring information on the Program status, given the variety of activities and number of departments in AySA involved in Program implementation	Reinforce the Program Coordination Unit with technical staff in charge of monitoring and evaluation activities, as per Program Action Plan (Action 1)	Moderate
1	COVID situation affecting / delaying works to connect households in vulnerable neighborhoods.	Adoption of safety measures following AySA's health protocols, promotional campaigns conducted in the beneficiary areas before the start of the works, and the extension of the timeline to execute these works by 6 months, from 1.5 to 2 years	Moderate
1	The design of the new modules in the Norte and Jagüel WWTPs may present opportunities for greater efficiency in terms of CAPEX and OPEX that are not considered.	AySA can accept the support of Technical Assistance to review proposed designs thus helping AySA in the identification and consideration of such opportunities.	Moderate
1	Risk of uneven uptake of water and sewerage connections within vulnerable neighborhoods due to HH opting out of the program, returning to unimproved water sources. This could reduce the environmental and public health benefits and/or lead to HH feelings of exclusion.	Targeted social outreach and communications campaign that explains the affordable social tariff, the benefits of safely managed intra-HH connections	Low
1	Risks of recurrent flooding damaging newly built household connections made under the Program	Clear communication of the Verification Protocol, which declares household connections located in floodable areas as ineligible unless flood mitigation	Moderate



		plans are adopted	
3	Development of a mobile app and CRM system with focus on commercial management or lack of consideration of users' needs into design, given that it is developed with a conventional approach that limits two-way communication functionalities for the interaction between user and utility.	Technical assistance to incorporate innovative approaches to promote stakeholder inclusive processes, is an opportunity that AySA can consider	Low
3	The new digital tools may provoke internal organizational changes in the medium and long term.	It is recommended that AySA conducts an assessment on how further digitization (process automation) will modify current processes and relationship in the commercial area and other areas of the company to adapt the organization and develop human resources.	Moderate
3	Impossibility to achieve target of equal promotion rates form women and men linked to lack of experienced or sufficient qualifications female personnel, and/or limited pool of candidates as Program evolves.	Communication campaigns and trainings on soft skills should be intensified or targeted to specific to potential promotion candidates, preparing them for future opportunities. Activities such as mentoring by female colleagues in leadership positions can motivate younger women navigate the organizations hierarchy.	Moderate



ANNEX 4. SUMMARY FIDUCIARY SYSTEMS ASSESSMENT

A. CONCLUSIONS

Reasonable Assurance

1. The Procurement and Financial Management (FM) systems' capacity and performance, with the implementation of the proposed mitigating measures and agreed actions to strengthen the systems (which are reflected in the Program Action Plan, PAP), are adequate to provide reasonable assurance that Program funds will be used for the intended purposes, with due attention to the principles of economy, efficiency, effectiveness, transparency, and accountability.

Risk Assessment

2. The overall integrated fiduciary risk rating is considered **Substantial**. The key fiduciary risks to the development outcomes of the Project that underpin the Substantial risk rating are as follows: (i) unfamiliarity of the Borrower and the main implementing entity, AySA with the PforR Instrument; (ii) the risk of awarding a contract to firms and/or individuals debarred or suspended by the Bank; (iii) lack of an effective procurement-related complaints procedure; (iv) exchange rate pressures that pose a risk that the amount of Program Expenditures may be insufficient to cover the withdrawn loan balance; (v) AySA dependence on the transfers from the National State for the execution of the PMOEM; and (vi) specific arrangements that will need to be put in place to produce the Program Annual Financial Statements.

3. The proposed systems-and capacity-strengthening and/or mitigation measures, to address the above risks include the following: (i) close implementation support to be provided by the Bank's fiduciary teams, coupled with the preparation of an Operations Manual (OM); (ii) a) the external auditors' Terms of Reference (TOR) will include a requirement to review Program expenditure for ineligible contracts; b) eligibility verification to ensure that any person or entity debarred or suspended by the Bank, or by another multilateral institution with which the Bank has signed a cross debarment agreement, is not awarded a contract under the Program during the period of such debarment or suspension; (iii) preparation and implementation of a clause in the bidding documents on the handling of complaints, which includes a detailed description of the procedure through which a timely and fair revision of the complaint is ensured; (iv) review of the semesterly IFRs and accounting conventions used to prepare IFRs, to ensure Program Expenditures are on track and; (v) close monitoring and coordination to ensure the necessary budgetary support is provided by the National Government; and (vi) close implementation support to be provided by the Bank to help establish the arrangements to produce the Program Annual Financial Statements.

Procurement Exclusions

4. There are no potential high-value contracts identified under the Program. The proposed Program is not expected to finance any contract at or above prevailing Operations Procurement Review Committee (OPRC) thresholds considering the estimated activity costs under Substantial risk, which at the time are at minimum of (i) US\$75 million for works; (ii) US\$50 million for goods, information technology and non-consulting services; and (iii) US\$20 million for consulting services.

Scope

5. AySA is the concessionaire responsible for providing WSS services for the residents of the CABA and 26 municipalities in the PBA. The Program will support the implementation of highly relevant areas of AySA's PMOEM 2019-2023 and its next version covering the period 2024-2028. The purpose of the PMOEM is to ensure the maintenance, improvement, performance and operation throughout the regulated area of the systems necessary for the provision of the WSS services.



6. **The PMOEM is prepared by AySA and approved by the Secretariat of Infrastructure and Water Policy (*Secretaría de Infraestructura y Política Hídrica*, SIPH⁸⁵) after review by the APLA.** The PMOEM's implementation is subject to annual budget availability, and it is revised every five years.

Program Scope

7. **The current PMOEM covers the period 2019-2023 and comprises the three different sub-Programs: (i) Improvements and Expansion; (ii) Improvement and Maintenance; and (iii) Operations.** The PMOEM also includes operational expenses associated with the implementation of these programs. The Program will support activities according to the Results Areas with well-designed geographical and thematic boundaries from the PMOEM plus the operational expenditures corresponding to the newly built infrastructure under the Program. The overall cost of the Program has been determined. The PforR's budget totals US\$576.6 million, out of which US\$ 300 million (52 percent) will be funded by the Bank. The scope of the PforR Program supports three Results Areas.

Review of the Public Financial Management Cycle

Planning and Budgeting

8. **AySA finances its operations and the PMOEM through transfers from its majority shareholder, the National State, by funds generated by its operations and by obtaining bank loans or issuing debt bonds in financial markets.** Consequently, AySA depends on the transfers from the National State for the execution of the PMOEM and the payment of its liabilities.

9. **The Program will follow the Federal budget cycle and process.** The formal budget process in Argentina consists of four stages. The first stage entails the formulation of the Executive's budget proposal, and the second involves the congressional discussion of the proposal and approval of the budget law. In the third stage, the proposal is implemented, and in the fourth, execution of the budget is evaluated and controlled. The fiscal year runs from January 1 to December 31.

10. **During the formulation stage, based on revenue estimates and forecasts of the main macroeconomic variables, medium-term ceilings on budget aggregates and ministry/agency expenditures are proposed by the National Budget Office (*Oficina Nacional de Presupuesto*, ONP⁸⁶) and approved by the Head of the Cabinet (based on established priorities and public policies).** The line ministries/agencies will then prepare their preliminary draft budgets on this basis. In the case of the PMOEM, there is a specific unit within AySA (*Coordinación y Seguimiento de PMOEM*) that acts as the focal point for preparation of the budget documentation relating to the PMOEM, which is then included in the broader AySA budget proposal to be sent to the ONP for review. Consequently, there is due process between the implementing agency (via the MOPW) and the ONP for negotiating the annual budget.

11. **The Executive must submit the annual budget⁸⁷ to the National Congress⁸⁸ for its approval before September 15, when the approval stage begins.** The National Congress then evaluates the proposal through the Budget Committee (*Comisión de Presupuesto y Hacienda*) of each chamber. The budget proposal is received by the Chamber of Deputies, considered and dispatched, before it arrives in the Senate for analysis and approval. After the National Congress approves the budget, the Executive can introduce line item vetoes.

⁸⁵ The Ministry of the Public Works (MOPW) has the sectoral competence for water and sanitation services policy at the national level. Within the line ministry, the SIPH establishes the national policy and planning for water and sanitation services and decides the financial national resource allocations to and within the sector.

⁸⁶ Every year the National Budget Office disseminates a general Budget Circular (*Manual para la Formulación del Presupuesto de la Administración Pública Nacional*) that provides clear and comprehensive guidelines for the formulation of the budget proposal.

⁸⁷ Argentina's score of 58 on the 2019 Open Budget Index is higher than its score in 2017.

⁸⁸ The Argentine National Legislature is a bicameral institution, with a 257-seat Chamber of Deputies as the lower chamber and a 72-seat Senate as the upper chamber.



12. **During the next stage, the implementation stage, the Executive distributes the expenditure authorizations among the line ministries/agencies and introduces modifications⁸⁹.** The ONP prepares a wide range of in-year budget execution reports. Budget execution reports capture figures for each stage of the budget execution process, as recorded in the Federal Government's Integrated Financial Management Information System (*Sistema Integrado de Información Financiera, or e-SIDIF*) by every budgetary unit. Expenditures made from transfers to decentralized units within the central government are included in these reports. AySA monitors budget execution on a monthly basis, using internal budget reports that provide accurate, comprehensive, and understandable information, and any budget modifications required are usually requested twice a year (in April and October). Consequently, actual expenditures are compared to the budget with reasonable frequency, and explanations are provided for significant variations from the budget.

13. **During the final stage, the evaluation and control stage, the National Congress—through the Government Supreme Audit Institution (*Auditoría General de la Nación, AGN*)—performs ex post controls regarding budget implementation and must evaluate and approve the annual budget execution report and financial statements elaborated by the Executive (*Cuenta de Inversión*).** The AGN provides adequate budget oversight. Under the law, it has full discretion to undertake audits as it sees fit. Moreover, the head of the institution is appointed by the legislature and cannot be removed without legislative approval, which bolsters its independence. Finally, the AGN is provided with sufficient resources to fulfill its mandate.

Budget Classifications

14. **Transfers to AySA are appropriated and allocated based on:**

- (i) an organizational unit classification code of 1.1.1.91.00.000-*Obligaciones a Cargo del Tesoro* (National Public Sector: Non-Financial (1); National Administration (1); National Central Administration (1); Jurisdiction (91) (Jurisdiction 91 in simple terms);
- (ii) an economic classification⁹⁰ of “Transfers” (5) and a sub-function classification of “Transfers to other entities from the National Public Sector” (5)- (Code 55 in simple terms);
- (iii) a source of financing code (National State (Source 11) and transfers from loans obtained by the National State from the multilateral bodies (Source 22);
- (iv) a program code (Financial Assistance to Public Entities and other Entities (Code 95));
- (v) a functional classification of Social Services (Code 3) and Water and Sanitation (Code 8); and
- (vi) distinction between recurrent (Code 21) and capital (Code 22) costs.

15. **In the above-mentioned budget line item or classification, reflecting the Transfers to AySA, disaggregation for program expenditure monitoring is made by AySA using its processes and systems.** AySA’s budgeting and accounting information systems tracks the expenditures incurred under the Program using accounting policies consistent with national public sector policies/standards. Close implementation support will need to be provided by the Bank, to help establish the arrangements to produce the Program Annual Financial Statements. The Program’s budgeting transactions will be also be processed through e-SIDIF under the general budget code/line specified above (and UEPEX with respect to Source 22 financed expenditures) after having been processed through AySA’s own budgeting and accounting information systems.

⁸⁹ The general rules for in-year budget amendments are established in the National Public Sector Financial Management and Control Systems Act.

⁹⁰ There are nine economic classifications (Personnel Costs, Goods to Consume, Non personnel Costs, Goods to Use, Transfers, Increase in Financial Assets, Debt Service and Liability Repayments, Other Costs and Figurative Costs) and 67 subfunctions.



16. The review of the Program Composition by Program/Activity and Expenditure Categories- is included in the Technical Assessment (Annex 3)

Financial Sustainability and Funding Predictability

17. **Law No. 27.467 on the National Budget for the year 2019 allocated transfers to AySA for the fiscal year ending on December 31, 2019 for works (costs of acquisition or construction of assets used to provide the service) in the amount of ARS15,562 million (including loans from multilateral entities belonging to Source 22).** Nine budgetary changes were subsequently made. In summary, during the year ended on December 31, 2019, AySA received transfers from the National State amounting to ARS21,737 million (ARS10,181 million related to Source 11, ARS11,507 million to Source 22 and ARS49 million from other sources). From the start of the concession and up to December 31, 2019, the Company has received transfers from the National State of ARS85,689 million.

18. **The financial sustainability of AySA has to be strengthened to reach at least full operating cost coverage on a routine basis (including maintenance, rehabilitation and depreciation) and reduce dependency with regard to national funding, which may be subject to the country's macroeconomic situation.** AySA's is seeking efficiency gains as a means to improve its financial sustainability and cost-recovery ratio via the implementation of an adequate asset management strategy to prioritize rehabilitation works, lower technical losses and improve energy efficiency.⁹¹ All these targets (network renewal, loss reduction, energy efficiency) are explicitly mentioned in the PMOEM 2019-23.

⁹¹ Due to the flat topography of the provision area, a large number of pumps are required to operate the service, which generates important electricity costs. At the end of 2015, electricity tariffs went up, affecting AySA's production costs and making AySA the fourth largest consumer of electricity in the country.



Adherence of Program Expenditure to Government Priorities and Budget Execution

Table 13: Budget vis-à-vis Actual Expenditures, 2014 - 2019

1000's of AR\$	Initial Allocated Budget					
	2014	2015	2016	2017	2018	2019
Operating Costs	5,059,336	6,960,974	10,919,347	16,032,648	20,004,529	28,500,395
Investments	5,812,913	8,817,212	11,523,520	18,591,306	22,907,880	25,278,604
<i>Improvement and Maintenance</i>	<i>1,638,749</i>	<i>3,094,999</i>	<i>4,352,839</i>	<i>5,573,152</i>	<i>4,565,000</i>	<i>5,500,000</i>
<i>Expansion</i>	<i>4,174,164</i>	<i>5,722,214</i>	<i>7,170,681</i>	<i>13,018,154</i>	<i>18,342,880</i>	<i>19,778,604</i>

1000's of AR\$	Budget Execution					
	2014	2015	2016	2017	2018	2019
Operating Costs	5,043,613	7,013,931	10,468,436	14,550,486	19,736,981	29,890,733
Investments	5,648,673	6,782,796	9,653,481	17,744,759	20,149,482	25,214,065
<i>Improvement and Maintenance</i>	<i>1,628,213</i>	<i>2,271,790</i>	<i>3,472,358</i>	<i>4,487,493</i>	<i>5,856,239</i>	<i>5,678,288</i>
<i>Expansion</i>	<i>4,020,461</i>	<i>4,511,006</i>	<i>6,181,123</i>	<i>13,257,266</i>	<i>14,293,242</i>	<i>19,535,777</i>

Percentage	Budget Outturn					
	2014	2015	2016	2017	2018	2019
Operating Costs	99.69%	100.76%	95.87%	90.76%	98.66%	104.88%
Investments	97.17%	76.93%	83.77%	95.45%	87.96%	99.74%
<i>Improvement and Maintenance</i>	<i>99.36%</i>	<i>73.40%</i>	<i>79.77%</i>	<i>80.52%</i>	<i>128.29%</i>	<i>103.24%</i>
<i>Expansion</i>	<i>96.32%</i>	<i>78.83%</i>	<i>86.20%</i>	<i>101.84%</i>	<i>77.92%</i>	<i>98.77%</i>

19. The unplanned incorporation of nine municipalities in AySA's provision area between 2016 and 2018 coupled with the macroeconomic situation increased operational expenditures and budgetary pressures further reducing AySA's capacity to invest and fulfill the PMOEM's 2014-18 targets. Nevertheless, WSS is a National Government priority and the mobilization of citizen organizations, water users, non-governmental organizations, local governments and environmental groups, has contributed to raise the profile of urban water issues in Argentina, as evidenced by the National Supreme Court ruling passed in 2008, that ordered the government to take the necessary measures to clean up the highly polluted *Matanza-Riachuelo* basin. The PMOEM is also partly financed from Source 22 financing (external aid), which evidences support of donors' aid flows to the Program. AySA has a high degree of autonomy in budget execution, within the parameters of the approved PMOEM, and has a high institutional capacity. The Program is not protected by revenue earmarking mechanisms and consequently there is no protected share of the Program.

20. The Program's budget is realistic and implemented in an orderly and predictable manner. There is a reasonable expectation that the required resources will be appropriated in the financial years when required.



Nevertheless, this is an issue that requires close monitoring during supervision, to ensure the necessary resources are made available to AySA and that the Program is a high enough priority that it survives any budget cuts.

Procurement Planning

21. **For the service expansion works under the Directorate of Contracts and Management Control, AySA determines specific dates for the different bidding processes related to PMOEM activities.** This information is used as a procurement planning tool internally, and is not published on AySA's website. Furthermore, every year a Procurement Plan is prepared by all of the company's management and is consolidated by the Directorate of Purchases and Storage, which also monitors the execution of this Plan. This Plan includes the improvement and maintenance works, goods and services, but does not include the service expansion works contracts and is not made publicly available. The Plan is made available to the company's Board of Directors, together with the company's budget.

Procurement Profile of the Program

22. **Supporting AySA in “rebuilding better” after the COVID-19 outbreak, by increasing access to safely managed water and sanitation services in selected vulnerable areas in Buenos Aires, and improving AySA's efficiency, sustainability and resilience to future shocks, the Program consists of the following activities, to be carried out by AySA:**

- (i) Expansion of WSS services, which consists of, inter alia: supporting studies, design, and implementation of water supply and sanitation intrahousehold connections; capacity building activities and training to local cooperatives; and supporting studies, design and implementation of water supply primary infrastructure.
- (ii) Increase AySA's operational efficiency, which consists of, inter alia: network renovation; installation of pressure management valves; leak detection and repair; installation of energy monitoring tools; and energy audits
- (iii) Build AySA's resilience and sustainability through:
 - Mitigation of COVID-19 shock in AySA's operations, which consists of (i) provision of water through trucks, boats and bottled water and disinfectant to unserved areas with special attention to vulnerable areas; (ii) suspension of service cuts for those in arrears; and (iii) sensitization campaigns on handwashing and hygiene.
 - Improving AySA's financial position, which consists of (i) digitalization of workflow and commercial management processes and purchase of servers; and (ii) informatization of the data warehouse.
 - Improvement of water quality, which consists of the construction of additional modules in two existing wastewater treatment plants.
 - Increased transparency and social participation, which consists of (i) production of user-friendly digital reports on AySA's performance; and (ii) participatory activities in order to establishing a relationship of trust and mutual accountability between low-income communities and the utilities increasing social awareness of the importance of intra-household connections.

Procurement Performance

23. **AySA relies on skilled procurement staff and it is not reasonable to expect any bumps on demand for more procurement work.** AySA's staff working on procurement has solid capacity and knowledge of local procurement rules and procedures. The Bank team reviewed data of all contracts for works awarded by AySA from January 2016 to December 2019 and found that 83 percent of all contracts were awarded through open competitive procurement processes whereas 13 percent followed less competitive methods (limited competitive bidding) and 3 percent were awarded without competition. When considering the amount of money spent through procurement processes for works, the use of open competitive methods is even more prominent, at 91 percent.



24. **Competitiveness.** AySA received on average 6.9 bids per bidding when implementing open bidding processes during 2016-2019. When considering limited competitive bidding goes down to three 3.7 bids per open bidding process.

25. **Timeliness.** Data reviewed by the Bank regarding a sample of 15 percent of the open competitive bidding processes for works during 2016-2017 shows that AySA took on average 139.6 days (4.6 months) to award and sign contract. Most of the time spent to turnaround a procurement process goes on (i) bid evaluation and award recommendation (58.8 days), and (ii) contract signing (35.3 days).

26. **General conditions for bidding documents for procurement of works are published in the procedures manual, “Manual de Procedimientos”.** These general conditions apply to all bidding documents published by AySA and it’s a standard bidding document of mandatory use developed by AySA based on many years of experience of company’s procurement. The Bank reviewed the standard bidding documents, and a summary of its content is described in 3.2.3 *Procurement processes and procedures* of the FSA.

Budget Execution

Treasury Management and Funds Flow

27. **There are adequate arrangements for transferring funding from the MOE to AySA, to ensure funds are made available in line with implementation plans and that funds are made available in an orderly and predictable manner.** For National Treasury financing (Source 11 financing) the flow of funds will follow the normal/standard process as for all Source 11 financing.

28. **There will be a total of 12 DLIs.** The financing amount allocated per DLI has been determined based on the relative importance of the indicator, to provide the incentive necessary to achieve overall Program goals and outcomes. There are no DLIs defined as being time bound. The DLIs do not have a firm achievement date attached to them—they are achievable at any time during the Program implementation period. In this case, AySA can request disbursement against the achievement of these DLIs at any point after its achievement has formally been verified. Nevertheless, verifications are scheduled to take place on a semesterly basis (i.e. the timing of disbursements) in order to ensure that AySA has the required budget predictability and flow of funds. DLIs 1.1, 1.2, 2.1, 2.2, 4.1, 4.2, 6.1, 6.2 and 7 are scalable—thus allowing for disbursements to be proportional to the progress towards achieving the targeted DLI value—whereas DLIs 3, 5.1 and 5.2 are not scalable (the indicators relate to actions that are either achieved or not). For each DLI with scalable disbursement, the formula to determine the amount of PforR financing proceeds to be disbursed relative to the level of achievement of the DLI is specified in the DLI Matrix.

29. **Verification Protocols-AySA will contract a firm (to be paid for as part of the Program expenditures) to provide credible verification of the achievement of the DLIs, with the exception of the Prior Results Financing (DLIs met by AySA between the date of the Program Concept Review (July 16, 2020) and the date of the signature of the Legal Agreement)⁹².** The Verification Agent selected, will need to be acceptable to the Bank, and will thus be assessed by the Bank, to ensure it has the necessary independence, experience and capacity to ensure credible verification.

30. **AySA will prepare Program Monitoring Reports to document the status of achievement of the DLIs.** These reports will be verified by the VA, appointed by AySA, as per Terms of Reference (TOR) agreed with the Bank. Upon validation of DLIs by the VA, AySA will communicate the achievement of DLIs and corresponding DLI values to the Bank, supported by the relevant evidence and documentation. Following the Bank's review of the complete documentation, including any additional information considered necessary to confirm the achievement of the DLI results, the Bank will

⁹² Prior Results under DLI 5.1 will be verified before Loan signing by an interim third-party verification agent (but not the consulting firm referred to above) agreed by AySA and the World Bank.



confirm the achievement of the DLI(s) and the level of Program financing proceeds available for disbursement against each DLI.

31. **Disbursement requests or Withdrawal Applications (WA) will be submitted to the Bank by AySA using the Bank's e-disbursement (Client Connection) system.** A copy of the Bank's official communication, confirming the DLI achievement, should be attached to the disbursement requests.

32. **The Operation does intend to use the functionality of PforR Advances.** The Bank may make an advance payment of up to US\$75 million (maximum)⁹³ for one or more DLIs, which have not yet been achieved. In addition, an amount of US\$ 15 million of the PforR operation will be disbursed on account of Prior Results Financing.

33. **The General Conditions state, that if after the Closing Date the Borrower fails to provide the Bank evidence satisfactory to the Bank, that the withdrawn loan balance does not exceed the total amount of Program Expenditures (payments made on or after the Signing Date but before the Closing Date), the Borrower shall upon notice from the Bank promptly refund to the Bank such excess amount of withdrawn balances.** Due to exchange rate pressures⁹⁴, there is a risk that the amount of Program Expenditures once converted to US\$ may be insufficient to cover the withdrawn loan balance in US\$ terms⁹⁵. This risk will be closely monitored during implementation, not only during supervision missions, but also by review of the semesterly IFRs, coupled together with the flow of funds and accounting arrangements described below.

34. **The Bank's loan will be disbursed via the Central Bank of Argentina to a segregated US\$ denominated bank account held by the MOPW (under the control of DiGePPSE) at the *Banco de la Nacion*.** The US\$ will then be converted to ARS as required⁹⁶ (based on funding requests from AySA to DiGePPSE to meet/pay for upcoming/incurred Program expenditures) to meet payment obligations and to be deposited initially in a segregated ARS denominated account also held by the MOPW (under the control of DiGePPSE) (also held at the *Banco de la Nacion*). From this account funds are then finally transferred to a separate ARS denominated AySA physical bank accounts⁹⁷ (also held at the *Banco de la Nacion*) set up for this specific programmatic/financing purposes. From this AySA separate bank account, payments will be made to suppliers/vendors/contractors (including the works cooperatives under the *Agua + Trabajo* and *Cloaca + Trabajo* programs that are implemented in coordination with the respective Municipalities⁹⁸). This arrangement will allow AySA to account for and know the exact exchange rates to use to record Program expenses (based on US\$ received), thus eliminating refund problems at the end of the Program, due to possible exchange rate differences.

35. **The Bank may decide, without formally extending the Closing Date, to disburse or approve the use of proceeds of the loan for WAs received within six months after the Closing Date for DLIs achieved by AySA before the Closing Date.** Upon request, the Bank may decide to extend the period for receipt of such WAs.

⁹³ On June 30, 2020, the Bank's Board approved proposals for adjusted pricing and repayment terms (the "FY21 measures") (to take effect from July 1, 2020 onwards) in respect of certain lending and guarantee operations that use funds from the crisis buffer under IBRD's Financial Sustainability Framework (FSF), as well as certain fast-disbursing operations. The term "fast-disbursing" as it applies to IBRD loans means all DPF operations (except special development policy loans), as well as IPF and PforR operations that have cumulative disbursements equal to or greater than 60 percent of the loan, prior to, or as of, the date that falls two calendar years following the commitment. In the context of this Program, care will need to be taken by the Borrower/AySA to control disbursement requests (including advance requests), to ensure that the IBRD loan does not qualify as a "fast-disbursing" loan that would then be subject to a shorter average maturity (i.e. faster repayment) of 8 years.

⁹⁴ The Argentine peso continued to depreciate to cross 80.3 against the USD, its lowest level on record, amid a grueling domestic economic crisis and the coronavirus pandemic. Despite winning \$38 billion in debt relief from its international creditors, Argentina's economy is projected to shrink by around 12 percent in 2020 due to the COVID-19 outbreak and it is locked in a currency crisis with foreign reserves falling to a four-year low.

⁹⁵ Given that at the end of each semester, the exchange rate normally used to convert local currency expenditures to US\$, may be very different from the exchange rates in effect during the semester, when the payments were made.

⁹⁶ This is normally done based on 2-month expenditure forecasts i.e. payment expected to be made by AySA in the next 2 months.

⁹⁷ AySA does not make foreign currency denominated payments; all of these payments are made in local currency using the equivalent value in ARS.

⁹⁸ AySA does not advance funds to the Municipalities. The Municipalities forward the certified works certificates to AySA for payment to the works cooperatives.



Accounting and Financial Reporting

36. **The Program's transactions will be recorded primarily in AySA's budget and accounting information systems maintained in Oracle.** Every activity/works contract has a specific code (*Código Identificador*⁹⁹), that permits the identification and tracking of the activity/works within AySA's budget and accounting system (depending on the type of activities¹⁰⁰ and source of financing as well). AySA's accounting system also permits the tracking of expenditures following cost center (*Centro de Responsabilidad Estructural*, CRE¹⁰¹) and functional/geographical classifications (*Centro de Responsabilidad Funcional*, CRF¹⁰²) and AySA has a detailed Chart of Accounts. A separate Annex will be included in the OM, detailing the activities that fall within the definition of the Program, to aid in the recording and tracking of the Program's expenditures. This Annex will be updated as and when changes in the underlying composition of the Program activities take place, that will also require the Bank's no objection.

37. **The Program's transactions will also be recorded in e-SIDIF (under the general budget line/item code). This is done by the Ministry of Public Works (MOP), based on information submitted by AySA to MOP.** Furthermore, the Federal Government system specially designed for the execution of multilateral finance operations, the System of Financial Administration and Control for Agencies Executing Foreign Loans (*Sistema de Administración y Control Financiero para las Unidades Ejecutoras de Préstamos Externos*, UEPEX) will be utilized to record the Program's transactions financed from Source 22. UEPEX provides a good ex-ante internal control framework and it is considered adequate for accounting purposes.

38. **AySA's ORACLE based accounting system, e-SIDIF and UEPEX will be used, to generate the Program's Annual Financial Statements using the cash basis, both in the currency of Argentina (ARS) and in United States Dollars (US\$).** AySA has experience in preparing financial monitoring reports for other financiers. AySA will prepare the semesterly IFRs and submit the IFRs, no later than 60 days after the end of each reporting period. At the end of each fiscal year, AySA will prepare the Program Annual Financial Statements that will be audited. The final semester IFRs with accompanying notes, may serve as the Program's Annual Financial Statements. AySA will be responsible for liaising/coordinating with AGN for the audit of the Program's Annual Financial Statements.

Procurement Processes and Procedures

39. **"Manual de Procedimientos de Gestión de Contrataciones de Obras, Bienes y Servicios" is the legal text regulating procurement in AySA and it is available for download on the website.** The "Manual de Procedimientos" regulates procurement of works, goods, and services.

40. **Open competitive bidding, named "Licitación / Concurso Público", applies when the call to participate is directed to an indeterminate number of possible bidders with the capacity to be bound and will be applicable when the estimated amount of the contract exceeds the minimum indicated in the table below.** Requires publication of an invitation for bids at two newspaper of national circulation and AySA's website, at least 20 calendar days before the date for bid opening.

41. **Limited competitive bidding, named "Licitación / Concurso Privado", applies when the call to participate is directed to vendors that have been registered in AySA's Registry or others by activity and backgrounds, and will be applicable when the estimated amount of the contract does not exceed the minimums indicated in the table below.** The list of vendors to be invited are selected taking into account the information provided by different sources: vendors

⁹⁹ As set out in AySA's internal documents labelled R-ABU-BU-001 and R-ABU-BU-003.

¹⁰⁰ These codes are able to distinguish between rehabilitation, expansion or renovation works, as well as other characteristics.

¹⁰¹ For example: 90100-Inv. Compañía, 91000-Inv. Región Norte (sumador), 91010-Inv. Región Norte, 91020-Inv. Dist. San Fernando / Tigre.

¹⁰² For example: SS-Regional Capital-300-399, RR-Comercial-Grandes Usuarios-400-439, EF-Económico Financiera-440-444, RP-Relaciones con la Prensa-445-449, DG-Dir. General-450-454, DJ-Asuntos Jurídicos-455-457, IN-Programa Educativo y Rel. Internacionales-458-459.



registration, market analysis, specific experience and technical capacity by the type of work, economic and financial capacity, etc.

42. **Direct Contracting will be carried out by exception and its use is not limited by any monetary threshold, through the issuance of a purchase order or a contract when any of the 10 cases described in the “Manual de Procedimientos” occurs.** All cases of Direct Contracts must be expressed in a report signed by the Director of the requesting area, explaining the reasons and circumstances that justify them or prevent the application of other procedures in a timely manner and the reasonableness of prices.

43. **Publication.** For the purchase of goods, the contracting of improvement and maintenance works and services (consulting and non-consulting services), the advertisements are published in national newspapers, AySA website and COMPR.AR website, according to the different methods to be applied.

44. **Bidding documents.** Standard bidding documents for procurement of works, goods and services are published within the “Manual de Procedimientos de Gestión de Contrataciones de Obras, Bienes y Servicios” on AySA’s website. The same standard bidding document (Bidding document and General Conditions) applies for consulting and non-consulting services. However, Special Conditions differ according to the contracting of the type of services to be contracted.

45. **Complaints.** There is single clause included in the special conditions of the open and limited competitive bidding documents by which bidders may make written objections to the bidding process. However, there is no complaints procedure in the *Manual de Procedimientos*, nor in other document regarding this matter. The Program Action Plan proposes to develop an effective complaint procedure that regulates the activities to be carried out to manage the claims that the bidders enter in a procurement process in order to question the evaluation of bids and award of contracts, due to inconsistency with or violation of the applicable procurement rules or evaluation criteria, and in general to any matter that is not clear to the bidders.

46. **Procurement information disclosure.** Bidding documents, clarifications and amendments, bid opening record and award recommendation with the results of the procurement processes are available and published on AySA’s website <https://www.aysa.com.ar/>

47. **Inter-American Development Bank-procurement arrangements.** While drawing on the above described systems, some of the IDB’s-financed contracts, included in the Program, will be subject to specific procurement arrangements. Contracts financed by the IDB will be procured on the same basis (similar procurement principles, same team) as any other contracts under the Program and will be subject to IDB’s own procurement policy, including specific bidding documents and procedures. IDB’s operational procurement policies and procedures are consistent with the Bank Core Procurement Principles of the Investment Project Financing (IPF) Procurement Policy, and these rules provide assurance that Program funds will be used for the intended purposes with due attention to the principles of economy, efficiency, effectiveness, transparency and accountability.

Contract Administration

48. **Regarding service expansion works, the Directorate of Contracts and Management Control carries out payment certificates controls as follows:**

- (i) Numerical controls on quantities and amounts: control of accumulated amounts with previous certificates; control of certified quantities; arithmetic control of certified amounts (price x quantity); justification of extension and additional; correspondence of redetermined prices vs. approved prices.
- (ii) Controls on the mandatory documentation: labor obligations, social security, insurance and other records).
- (iii) Control on taxes; validity of the invoice and coincidence with the certificate; purpose and amount of the



securities.

- (iv) Other controls: advance payment deduction verification; compliance of stock material certificates; transfer of certificates and invoices authorization; and return of securities.

49. **On the other hand, there is a specific procedure for the tracking and administration of purchase orders in which different units are involved according to their missions and functions.** The Directorate of Purchases and Storage issues the purchase orders of works, goods or services and communicates them to both the awarded supplier and the requesting unit. Each unit oversees purchase order management and administration and in the case of any breach by the supplier, the unit communicates it to the Directorate, who supports them in every issue related to the application of penalties and sanctions to suppliers, and acts as a link between the requesting unit, the supplier and the legal area. In addition, the result of the supplier's performance evaluation for the procurement of goods, improvement works, and maintenance and services are recorded in the Oracle Quality System.

Procurement Risks

50. **Data shows that procurement processes implemented by AySA are transparent and competitive.** In fact, open competitive bidding is the procurement method to be followed for contracts estimated to cost over US\$1M for goods and US\$1.5 M for works and are published in AySA's open access website and 2 national newspapers. Data for the period of 2016-2019 shows that 91 percent of the money spent for procurement of works followed open competitive bidding disclosed at its website.

51. **Procurement staff are professional, experienced, and knowledgeable on policies and procedures, and although AySA has prior experience in implementing Bank IPF Operations, this will be the first PforR Operation.** It is assessed that Program procurement arrangements relying on AySA's system are acceptable in terms of procedures and present no major risk against result achievements with respect to procurement management.

52. **Nevertheless, as was previously mentioned, the lack of an effective complaints procedure that regulates the procedure to be followed may limit the fairness in procurement procedures and affect equal opportunities and treatment for bidders.** Thus, a credible procedure for addressing procurement-related complaints and providing recourse should be agreed and implemented.

Internal Control Framework

Internal Controls

53. **The internal control environment to be used for the Program is anchored in Argentina's legal and institutional framework as well as AySA's internal approval processes and systems, that provide for reasonable segregation of duties, supervision, quality control reviews and reconciliation.** The processes flows appear to be well understood with adequate control over and stewardship of Program activities and funds. Within the framework of its mission and adopting the best market practices, AySA has also decided to implement a Code of Good Corporate Governance.

54. **All accounting and support documents are retained on a permanent basis, using a system that allows for easy retrieval for the authorized user.** The Program's internal control system will be supported by the OM. The OM should be prepared by AySA and be approved by the Bank and be maintained/updated throughout the Projects' life.

55. **The Access to Public Information Law (Ley 27.275) regulates the right of access to public information.** AySA's "Information Transparency and Disclosure Policy" regulates the mechanism by which the requests for Access to Public Information that AySA receives will be made and treated, and the publication on the corporate website of Active Information, as established by Law 27.275.

56. **In addition, relevant procurement information (bidding documents, clarifications and amendments, bid opening records and award recommendation) is made publicly available through AySA's accessible and widely**



available web site. However, key financial information about the PMOEM (budgets, budget execution reports and audit reports) are not made available to the public by AySA in an accessible form. To access AySA's annual financial statements, users have to request this information by filling out a form on AySA's website.

57. **AySA maintains detailed fixed asset registers, and there is a specific unit/department within AySA responsible for the asset monitoring.** Assets are assigned unique identifier codes, to facilitate identification and tracking, with counts taking place at least once a year (as part of the preparation of AySA's Annual Financial Statements) and then reconciled with the accounting records. The sale or disposal of fixed assets is done throughout the year, following established norms and procedures. There are also detailed depreciation schedules utilized to reflect a truer and fairer picture of the value of these assets and to prepare AySA to be ready to purchase new assets once an asset gets worn out.

58. **Bank reconciliations are prepared in most cases automatically via AySA's Oracle information system but can also be prepared manually.** They are generated/prepared by someone other than those who process or approve payments and then are reviewed and approved by the accounting manager, and then submitted to the treasury manager for final approval. There is clear definition and segregation of functions in procurement processes, where staff and committees participating in the processes work independently and are accountable for their individual contribution.

59. **Code of Best Integrity Practice of AySA.** AySA staff, whatever their rank and hierarchy, as well as any person acting on your behalf and/or representation, shall ensure compliance with the "Code of Best Integrity Practices" (*Código de Buenas Prácticas de Integridad*) and must act professionally, efficiently and diligently, in accordance with the principles of good faith, integrity and impartiality.

60. **Conflict of Interest Policy.** When AySA staff realize that they are or could be facing a conflict of interest, whether current or potential, they must disclose it and apply the procedure established in the "Conflict of Interest Policy" (*Política de Conflictos de Interés*). Any violation or potential violation of the terms of the code must be notified to the Directorate of Integrity and Best Practices who will report to the Board through the Integrity and Best Practices Committee. In each of the procurement procedures AySA will inform the names and positions of the staff with competence or decision-making capacity on the contracting procedure, including Board Members, Directors, Managers, etc., and will not hire natural or legal persons who do not present the Statement of Conflict of Interests provided in the Policy. Through this procedure, bidders must state if they have a judiciary or extra judiciary conflict with the National Government or any relationship with the President, Vice-President, Chief of Cabinet, Ministers or any official involved in procurement processes.

61. **"Complaints, reports, investigations and disciplinary procedures protocol".** It regulates the mechanism by which complaints are received and managed within AySA in order to guarantee an adequate and timely response in the event of a breach of the Code of Best Integrity Practices or any of its policies, protocols, or complementary procedures.

Internal Audit

62. **AySA is subject to internal audit by the General Syndicate of the Nation (*Sindicatura General de la Nación, SIGEN*), which is the National Government's Internal Audit Agency, under the jurisdiction of the Executive branch.** SIGEN is an integral part of the National Government's internal control system providing the following core services: (i) assessing the adequacy and effectiveness of the internal control system (internal audit of the Executive branch); (ii) supervising and coordinating the actions of the various Ministries/Agencies' Internal Audit Units (IAUs) and approving their audit plans; and (iii) compliance auditing of procurement processes and contracts.

63. **SIGEN has conducted internal audit reviews relating to AySA, independent of the audits carried out by AySA's own IAU.** In these cases, the IAU coordinates the audit between SIGEN and the respective AySA areas to be audited.



64. **AySA's IAU¹⁰³ prepares an Annual Internal Audit Work Plan (Audit Plan), based on the guidelines issued by SIGEN.** Once the Audit Plan is approved (including any revision required), the Audit Plan is discussed and approved by AySA's Board of Directors, before being sent to SIGEN for final approval. SIGEN has an information system to record/load the Audit Plan (with the details of all the audits to be carried out in the year). In the same system, the results of the audits are captured (once completed) including observations and follow-up actions required. The Audit Committee (meetings are held once or twice a month), reviews all audit reports. AySA's IAU does annual follow up of its recommendations.

65. **The Bank reviewed the IAU's 2020 Audit Plan and found it comprehensive. This Audit Plan was reviewed/adjusted to take into account the impacts of COVID 19, including the isolation quarantine measures taken (and approved by SIGEN in June 2020) to ensure that whatever activities were planned for the second semester of 2020 were still viable.** One of the activities planned for in the 2020 Audit Plan is a review to ensure that AySA has prepared the 2019-2023 PMOEM in compliance with the Regulatory Framework (Activity # 12-4).

66. **The IAU also plays a coordination role in the event that the Supreme Audit Institution of Argentina (Auditoria General de la Nacion-AGN) initiates an audit.** When initiating an audit, AGN always requests a copy of any available internal audit reports produced by the IAU covering the sector to be audited, which it then relies upon to execute its own audit. To the extent that internal (or external) audit reports become available in the future relating to the Program specifically, the status of the auditors' findings and recommendations will be followed up, during supervision missions.

Program Governance and Anticorruption Arrangements

67. **The Criminal Code (Law No. 11.179) establishes responsibilities, accountabilities, and penalties for officials and citizens in relation to acts of corruption.** Furthermore, Law No. 27.401 establishes criminal liabilities related to bribery, traffic of influences and illicit enrichment, inter alia, for both domestic and foreign companies, state-owned or not.

68. **In Argentina, bribery offences, whether domestic or foreign, are federal crimes. Investigations may be initiated by a complaint made by any person to the police, public prosecutor, or judge.** Allegations of corruption could be filed at the OA either by e-mail, anonymously or with protection of identity, at the *Procuraduría de Investigaciones Administrativas* or at the government agency where the offence took place.

69. **Public officials who become aware of the commission of a crime while performing their duties, must report it to the authorities.** Enforcement authorities are legally required to commence the investigation once becoming aware of a bribery allegation. The investigation is conducted by the investigating judge or can be delegated to a public prosecutor. Once the investigation is completed, the judge may decide whether the matter should move to the oral trial stage, in which case a separate judge or judicial panel conducts the trial until its completion. There is thus institutional capacity to monitor and address governance and corruption issues.

70. **The World Bank's Institutional Integrity Vice-Presidency (INT) may also, jointly with the Borrower and/or AySA or on its own initiative, investigate any allegations or other indications of Fraud and Corruption (as defined in the ACG) in connection with the Program or any part of the Program.** In all such cases the Borrower and AySA will collaborate with INT to acquire all records and documentation that INT may reasonably request from the operation regarding the use of the Program financing. If the Borrower or the Bank determines that any person or entity has engaged in Fraud and Corruption (as defined in the ACG) in connection with the Program, the Borrower will take timely

¹⁰³ The Internal Audit Department of AySA has also implemented a Quality Management System with the purpose of providing a professional internal audit service



and appropriate action, satisfactory to the World Bank, to remedy or otherwise address the situation and prevent its recurrence.

71. **In order to implement the Bank's Anticorruption guidelines for PforR operations, it has been agreed that any expenditures arisen out of contracts given to individuals or firms debarred by the Bank or under suspension by the Bank are not eligible for Bank financing under the Program.** The external auditors' TOR will include a requirement to review Program expenditures to confirm that contracts are not awarded to debarred or suspended firms.

72. **AySA will immediately inform the Bank (through e-mail and official letter) of any complaint, claim or allegation related to fraud and corruption which AySA either receives or of which AySA becomes aware; in addition, every semester (together with the Program Monitoring Reports) a report will be prepared containing all alleged cases, with an updated status of the respective actions taken.** AySA will immediately provide the Bank with all the records, documentation and information that the Bank may request with respect to such issues. In the event that the Bank decides to conduct its own investigation, the Bank may request the government and/or AySA to exercise its/their legal rights and remedies (under the relevant contract/s) so as to obtain all information, records and documentation that the Bank may request, and provide these to the Bank. This process does not limit the rights of the Bank to also make direct requests for information from individuals or contractors who are recipients of Bank financing. In line with the obligations arising under the Anticorruption Guidelines, the government and AySA should ensure that individuals or firms who are recipients of Bank financing are aware that the Bank may decide to exercise this option. If the Bank determines that it has not been able to receive the documentation, records or information requested by the Bank directly and/or through AySA, the Bank may declare the relevant expenditure ineligible for Bank financing under the Program. Furthermore, should the Bank conclude that a sanctionable offense has occurred, it may decide to pursue sanctions against the individual or firm in line with Bank procedures.

Program Auditing

73. **There are no current audits of the PMOEM specifically. AySA's own Annual Financial Statements have, however, been audited by an associate firm (Pistrelli, Henry Martin y Asociados S.R.L) within Ernst and Young's (E&Y) Global Network.** The E&Y associated auditors expressed an unmodified audit opinion on AySA's 2019 Annual Financial Statements but did include an emphasis of matter paragraph indicating that AySA is dependent on the transfers from the National Government, to finance its operations including the PMOEM. Both AGN and SIGEN expressed a similar opinion, in their review of AySA's 2019 Annual Financial Statements.

74. **For purposes of the Program, the external audit of the Program's Annual Financial Statements will be conducted by AGN.** The AGN is the current auditor for Federal-level projects being financed by the Bank and has the adequate capacity to deliver a quality audit.

75. **The auditors will be required to issue an opinion on the Program's Annual Financial Statements and produce a management letter in which any internal control weaknesses are identified, with a view to contributing to the strengthening of the control environment.** The auditor's report will be submitted to the Bank no later than six months after the end of the fiscal year. The Bank will review the audit report and will periodically determine whether the audit recommendations are satisfactorily implemented. The Bank also requires that the Borrower disclose the audited Program's Annual Financial Statements in a manner acceptable to the Bank and following the Bank's formal receipt of these statements from the Borrower, the Bank will also make them available to the public in accordance with *The World Bank Policy on Access to Information*.

Procurement and Financial Management Capacity

76. **AySA has a team of fiduciary staff (FM and procurement) and is responsible for monitoring the implementation of the Program's integrated fiduciary aspects.** Staff are professional, experienced, and knowledgeable on governmental policies and procedures, and although they have prior experience in implementing Bank IPF



Operations, this will be the first PforR Operation.



ANNEX 5. SUMMARY ENVIRONMENTAL AND SOCIAL SYSTEMS ASSESSMENT

1. **An Environmental and Social Systems Assessment (ESSA) was carried out, according to the PforR Financing Policy.** The ESSA examined the scope, risks, benefits and potential impacts of the Program from an environmental and social perspective. In addition, the ESSA examined the extent to which the legal and institutional frameworks pertinent to the Program as well as AySA's corporate systems and processes are materially consistent with the core principles outlined in the Bank's Policy and Directive for PforR Financing. The findings of this assessment are based on a review of existing information and virtual meetings with representatives of key counterparts and other relevant stakeholders. These discussions and the information provided allowed the team to gain detailed knowledge of social and environmental systems that apply to the Program. The draft ESSA was disclosed on January 5, 2021 and consulted on January 11, 2021 with two events held with key institutional stakeholders, cooperatives and representatives of civil society organizations. The draft ESSA was also available to a larger group of stakeholders for virtual comments and feedback. A report on the consultations, the feedback received and how it addressed is included in the final ESSA, which is disclosed prior to Board approval.
2. **The overall assessment concluded that the current systems are materially consistent with the PforR Policy and its core principles** related to the adequacy of environmental and social regulatory and institutional systems, impacts on cultural and natural resources, community and workers health and safety, land management under the Program, cultural appropriateness and risks of potential conflict. The ESSA informed that main risks and adverse environmental and social impacts are sufficiently covered by the regulations and institutional capacities of AySA and the Regulatory Framework, although areas with an opportunity for improvement have been also identified.
3. **The Program will likely have overall positive environmental and social impacts on the lives and livelihoods of large vulnerable communities.** The Program will support an increase of the offer and quality of drinking water and the expansion of sanitation services in targeted areas, with no significant adverse environmental impacts expected due to the proposed interventions. Main positive environmental effects are related to the adequate collection, treatment and final disposal of a greater volume of sewage, with consequent improvements in terms of soil and groundwater contamination; the improvement of health conditions for the new population served; and the environmental enhancement of the intervention areas from the sealing of septic tanks; among others. The reduction of water losses and energy consumption also entail environmental benefits. The Program would likely have overall positive social impacts on the lives and livelihoods of large vulnerable communities through the expansion of safe water and sanitation services, which will also improve the quality of life and reduce WASH-related health risks in targeted neighborhoods, and increase local employment through participation in cooperatives, thus generating income opportunities and promoting social inclusion. The Program will also support AySA's citizen engagement and beneficiary feedback mechanisms, which will help to improve the quality of the service, and AySA's readiness for emergency response.
4. **Environmental Risks.** The overall environmental risks and potential impacts of the Program are considered *Substantial*, basically in connection with interventions in existing sewage treatment plants given the risk and potential impacts associated to civil works in the construction stage (soil movements; materials transportation and handling; noise, dust, debris generation; among others) and, once treatment capacity is expanded, to the increased volume of treated effluents to be discharged to water bodies during the operation stage. However, these proposed works are of a limited scale, will increase the level of treatment of effluents, and will take place within AySA's premises. Also, they will not involve new discharge pipelines or discharge points, which means that the risks and potential environmental impacts to the water bodies where the treated effluents are discharged have been taken into account and, accordingly, evaluated in the original Environmental Impact Assessments (EIAs) of the wastewater treatment plants (WWTPs). The original EIAs considered the modular design of the plants (i.e. to be built in stages) and the maximum volume of effluents once all modules are constructed and fully operational. The original EIAs will be updated and presented to the environmental authorities as part of the renewal of the environmental licenses. The main specific risks that would need



to be managed under the Program include: (i) increased health risks due to workers and community exposure to COVID-19; (ii) community and occupational health and safety risks – especially during construction and particularly associated with the handling of materials with asbestos content; (iii) risks due to interventions in areas with potential physical cultural resources chance findings, and risks related to potential interventions in zones with conservation values within AySA's concession area.

5. **Social Risks.** The overall social risks and potential impacts of the Program are considered *Moderate*, basically related to the incorporation of new low-income users who would not be able to pay for those services. Other probable negative impacts are associated with the potential temporary displacement of informal economic activities, the possible temporary restriction of access to entry to commercial business during construction, the short-term interruption of pedestrian traffic, and the potential impact on community services and/or infrastructure. In addition, although there are some urban indigenous communities present in the GBA, the Program's interventions will not generate any negative impacts on these communities nor will they involve any exploitation of their natural resources or knowledge.

6. **National/local/AySA's system to address Environmental and Social Risks.** The ESSA concluded that Argentina has well-established systems to address these environmental, health and safety risks, including several strong central, provincial and municipal level legislations and institutions, as well as an effective regulatory sectoral framework, that govern environmental, occupational and community health & safety issues. These are complemented by AySA's corporate environmental, social and health & safety systems.

7. **In particular, AySA's Sustainability Directorate has developed Environmental and Social Technical Specifications (*Especificaciones Técnicas Ambientales y Sociales, ETAS*).** These are part of the bidding documents of the company, and, accordingly, are included in the contracts once the works are awarded. The ETAS establish good practice standards and procedures as requirements for the contractors to address environmental and social issues during the execution of the works. Basically, they include prevention, control, and mitigation measures to avoid or minimize the negative effects on the environment/ surroundings. The ETAS describe the scope of the Environmental Management Plan that the contractor must develop with the particularities of the works, with the corresponding plans, programs, and procedures to be considered (e.g. Environmental Monitoring Plan, Management of Hazardous Waste and Spills of Hazardous Substances, Environmental Training Programs and Procedures for Grievance Management, among others), and with details on the aspects to be addressed throughout the cycle of the works. For example, administrative issues related to environmental management; installation of construction camps, with conditions related to vehicles and machinery, supplies and construction materials; management of potential impacts on third parties (people and/or property outside the works); relationship with the community and grievance procedures; waste; protection of the environment; historical and archaeological/ paleontological heritage; and demobilization of work sites and construction camps. ETAS also describe the tools for the monitoring and control of deviations or non-compliance issues to be implemented by the contractor and states the need to maintain an auditable "environmental file" at the site with the required documentation. The Sustainability Directorate periodically reviews and improves, as necessary, the ETAS. The Program will support the improvement of the ETAS in certain aspects that have been found to be reinforced or added for the proper management of the works foreseen under the Program. These actions are included in the combined Environment and Social Action Plan.

8. There is an additional document¹⁰⁴ developed by AySA's Community Development Directorate, with adjusted environmental and social specifications to be applied in the case of activities carried out by Cooperatives and focused on the A+T and C+T programs. The Community Development Directorate is adjusting these specifications for the households interventions to be implemented under the Program.

¹⁰⁴ "Modifications to the general technical specifications and their annexes for the provision of water and sewerage"



9. **Regarding Occupational Health and Safety (OHS), AySA has developed a series of documents, following the sound Argentine legal framework on the matter, which details the requirements for the contractors.** They require for contractors to perform a risk assessment of their tasks and to develop a risk management plan. AySA shares with the contractors a comprehensive set of OHS procedures, operational instructions, and record formats that covers the main aspects of their activities (basic OHS conditions, legal requirements, particular risks, etc.). One of the main OHS procedures describes the mandatory guidelines for a safe work environment for works associated to different possible general risks (electrical; falls from heights; suffocation in confined spaces; collapse of walls during excavations, etc.); and certain particular risks (transport of combustibles, tasks with oxyacetylene, transfer and lifting of loads, etc.). Considering the present circumstances of the COVID-19 pandemic, AySA has also developed a Coronavirus Prevention Protocol for safe work. AySA has specific procedures developed to supervise and control contractors' OHS legal compliance as well as for the planning and carrying out the inspections in the work sites.
10. **AySA also has systems and mechanisms in place to address the social risks identified above.** It has a Social Tariff scheme to support low-income populations in its concession area, supports debt cancellation for low-income households unable to pay outstanding water bills, and provides subsidies for low-income households to connect to the network, including a 50 percent discount in new connection costs to incentivize their connection to the network and additional subsidies of up to 100 percent of the tariff and free network connections to households in extreme poverty.
11. **The identified risks are easy to mitigate and the ESSA proposes a number of recommendations in this respect.** The main recommendations identified in the ESSA to manage environmental risks under the Program include: (i) ensuring implementation of the Argentinian COVID-19 related regulations and corresponding AySA protocols for contracted workers and own staff, following the evolving situations of the pandemic and the post-pandemic; (ii) ensuring implementation of the Argentina regulation on health and safety and the corresponding AySA systems, improving them for the proper handling of materials and waste with asbestos content (including potential chance findings during excavation and demolition); and, (iii) strengthening AySA's risk management related to physical cultural resources and sensitive areas located within the urban zone of intervention. Specific recommendations to manage social risks include: (i) the adaptation of practices and procedures to the realities of low income neighborhoods to guarantee their sustainable access to the water and sanitation services (e.g. through the implementation of differentiated programs); (ii) the incorporation of processes to manage the potential impacts of works on community services and/or informal economic activities in these works' Environmental and Social Management Plans (the technical specifications contain provisions to avoid impacts on formal economic activities but don't include ways to address potential impacts on informal activities); (iii) active involvement of municipal government environmental and social teams; and (iv) implementing actions with a focus on the urban indigenous communities that are present in the Program's implementation area including increasing information gathering and analysis about these communities and potential barriers to access water and sanitation services, and ensuring that they have adequate information to access the Communal Tariff Scheme, in a culturally appropriate manner; and (v) implementation by cooperatives and contractors of a code of conduct to manage labor influx risks.
12. **Gender.** The ESSA has identified that although women's representation among AySA staff has increased consistently since 2006, they only represent 23 percent of overall staff at AySA and 22 percent of managers; and are promoted at significantly lower rates, particularly to managerial level positions. Given these challenges, AySA is taking a systematic approach through Its Inclusion, Gender, Equal Opportunities and Treatment and Workplace Violence Program (IGEOPWV), to promote equal career advancement opportunities for women and improve working conditions for all AySA employees. The Program will support one of the main objectives of this program, focused on increasing women's representation in decision making positions, including in technical departments. AySA intends to achieve an increase in women's representation in managerial positions – from 22 to 28 percent. In addition, to improve women's opportunities for promotion, AySA is launching targeted communication campaigns to encourage women to apply to promotion opportunities, actively encouraging women to apply to positions in job postings and focusing on providing training to fill women's technical or other capacity gaps.



13. **Citizen Engagement:** The ESSA also identified opportunities to strengthen AySA's capacity to engage citizens through a phased implementation of a digital platform that will facilitate a two-way engagement between users/citizens and the utility. The digital strategy for improved citizen engagement will focus on the implementation of the following activities: (i) incorporation of a new digital channel (mobile app) for community engagement; (ii) development of a solid and integrated customer relationship management system (CRM); (iii) classification of PQRS and average response times; and (iv) improvement on response time.
14. **Exclusion of High-Risk Activities.** The ESSA recommends that the following activities be excluded from Program support in view of associated high environmental and social risks: (i) the construction of new Water Treatment Plants, or new WWTPs, nor will it finance works related to new discharge pipelines or discharge points of effluents treated in existing WWTPs; (ii) works that may require the acquisition and/or taking of land that would result in the displacement or loss of housing, assets or access to assets, income or sources of income; whether those affected have to move elsewhere or not.
15. **Environmental and Social Challenges.** There are some environmental and social challenges that the Program may need to address as part of the Program Action Plan. These are associated with the identification of areas with conservation value and areas with archaeological and paleontological sensitivity within AySA's concession area, an update of the EIA procedures and the ETAS on those aspects and on the management of materials and wastes with asbestos content, the implementation of a code of conduct for cooperatives and other contractors working with AySA, and including in the civil works' ESMPs management of temporary impact on informal economic activities and program on spaces for community use.
16. Based on the risks assessed, the combined Environment and Social Action Plan is presented in Annex 6, as part of the Program Action Plan (PAP).
17. **OP/BP 7.50, International Waterways.** Given that the water and wastewater plants associated with the increase of the offer and quality of drinking water and the expansion of sanitation services in targeted areas are related to, directly or indirectly, the La Plata River basin, which Argentina shares with Bolivia, Brazil, Paraguay and Uruguay, the Program triggers OP/BP 7.50 on Projects in International Waterways. Nevertheless, since the interventions constitute works limited to renovation and expansion of existing schemes that will not change the nature of these schemes or impact the quality or quantity of water flows to the other riparians and will not be adversely affected by the riparians' possible water use, an exception to the corresponding notification requirements to the riparian countries was approved by the Regional Vice President on December 8, 2020.



ANNEX 6. PROGRAM ACTION PLAN

Action Description	Source	DLI#	Responsibility	Timing		Completion Measurement
1. Proposal to strengthen PCU for monitoring of program activities and DLIs, ensuring that the PCU has sufficient staff for these tasks, including focal points for each DLI. The proposal should be acceptable to the Bank, prepared and implemented.	Technical		AySA	Other	4 months after the Effective Date	Proposal submitted to the Bank and implemented
2. AySA will ensure that no expenses under the Program arise from a contract awarded to a company or individual debarred or suspended by the WB. (Compliance with the eligibility requirements related to fraud and corruption (according to the WBG ACGs))	Fiduciary Systems		AySA	Other	Throughout Program Implementation	List of contracts awarded to firms and individuals is sent to the Bank annually.
3. Preparation and implementation of a clause in the bidding documents on the handling of complaints, which includes a detailed description of the procedure through which a timely and fair revision of the complaint is ensured.	Fiduciary Systems		AySA	Other	Preparation within the first 12 months from the Effective Date and application thereafter throughout Program Implementation	Clause on the treatment of procurement-related complaints to be included in the bidding documents, prepared and implemented in a satisfactory manner to the Bank
4. Inform the WB of F&C allegations and	Fiduciary Systems		AySA	Other	Biannually throughout	Full information, records and documentation the case



provide information requested. Make sure that persons or contractors benefitting from WB financing are aware that the Bank may decide to apply sanctions against the person or company according to WB procedures					Program Implementation	submitted to the Bank as the Bank may request
5. Map and georeference areas of conservation value within the AySA concession area. Update the map areas with high archaeological and paleontological sensitivity within the AySA concession area.	Environmental and Social Systems		AySA	Other	Within the first three months from the Effective Date	1) Publication on the institutional website 2) Incorporation of the information on the EIAs 3) Incorporation in the environmental and social technical specifications (ETAS) update
6. Improve the procedures and processes for the management of materials with asbestos	Environmental and Social Systems		AySA	Other	Within the first three months from the Effective Date	1) Attach the procedure to the ETAS and incorporate when updated 2) Asbestos management procedure implemented 3) Updated H&S procedure 4) Development of training content and training to the staff provided, with details on training roster
7. Preparation of a code of conduct for cooperatives and a model or list of minimum requirements that will be required of the codes presented by contractors. Prepare a model of the beneficiary adhesion and conformity note.	Environmental and Social Systems		AySA	Other	Within the first three months from the Effective Date, and prior to the start of the Program's works (whichever occurs first)	1) Publication on the institutional website 2) Incorporation into the Environmental and Social Management Plans (ESMPs) 3) Incorporation into the ETAS update
8. Incorporate in the OM the minimum	Environmental and Social		AySA	Other	Within the first three months	Incorporation into the Environmental and Social



requirements to be demanded from contractors in the context of ESMP in cases where temporary effects on non-illegal informal economic activities or temporary effects on spaces for community use are identified	Systems				from the Effective Date, and prior to the start of the Program's works (whichever occurs first)	Management Plans (ESMPs)
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ANNEX 7. IMPLEMENTATION SUPPORT PLAN

1. The PforR operation supporting AySA will require considerable focused support from the Bank. One challenge will be to coordinate the actions agreed in the Program Action Plan with operational activity on the ground, ensuring that information flows effectively between the different departments and technical areas within AySA and the PCU in charge of Program Monitoring and Reporting to the World Bank. As a number of the actions included in the PAP require fulfillment in the months following loan effectiveness, specific support to AySA during the early stages will be critical.
2. The PforR mode of operation requires dedicated efforts to M&E so that the implementing agency can accurately report on results achieved before each verification period. AySA will use an existing system, SARWeb, for Program monitoring of DLI 1 and 2. AySA staff are already familiar with SARWeb, which can flexibly assign tasks, set reminders, and manage documentation such as surveys and photos. For verification of results, the VA will use the KoBoToolbox platform. SARWeb and KoBoToolbox are similar and compatible tools, and the World Bank will ensure that they are able to share data effectively to facilitate the monitoring and verification processes.
3. It is particularly important that the capacity building strategy to be implemented through Technical Assistance activities start in a timely manner, again early in the lifetime of Program implementation. For instance, as AySA is in the process of designing the upgrade of the Jagüel WWTP, the Bank will ensure that timely Technical Assistance is provided to strengthen AySA's capacity in developing efficient and optimized designs of wastewater treatment plants. Similarly, the Bank will provide a show case of best international practices in developing citizen engagement applications, before AySA starts the developing of the digital platform for citizen engagement.
4. The focus of Bank implementation support over and above will therefore be focused on making the results-based incentive system work to its full potential. This will include: (i) reviewing implementation progress (including that of the Program Action Plan) and achievement of Program results; (ii) providing support on resolving emerging Program implementation issues and on building institutional capacity; (iii) monitoring the adequacy of systems performance, and monitoring compliance with legal agreements; and (iv) supporting the Government in monitoring changes in risks.
5. Key to effective implementation support will be the coordination of its timing with critical points in the timelines of planning and verification of results for payment requests to the World Bank. The first implementation support mission will take place as soon as possible after effectiveness to provide direct feedback on the quality of the monitoring system and the likely acceptability of the first results to be presented in the first verification period. It is expected that at that stage considerable progress will have been made towards many of the actions in the PAP and these will also be reviewed at this time. The first mission is therefore expected to include staff predominantly from the environmental, social and fiduciary teams. Subsequent implementation support may have a stronger emphasis on verification/M&E skills and technical implementation expertise.
6. An outline of the implementation support required is provided in the tables below.



Main focus of Implementation Support

Time	Focus	Skills Needed	Resources Estimate	Partner Role
First twelve months	Compliance with PAP actions. Ensuring adequate M&E capacity at AySA and facilitating familiarity with the PforR process (guidance in preparation of Program Monitoring Reports). Possible adjustments in operational procedures (OM). Technical meetings with AySA and the VA before the start of the verification process to provide training with KoBo Toolbox for the VA and ensure common understanding of the Verification Protocol.	Legal; fiduciary; engineering/procurement; environmental and social; water and sanitation monitoring and evaluation; technical (water, sanitation, economics and finance)	2 implementation support missions plus 3 short/technical visits 2 x 10 people x 1 week = 20 weeks Plus 3 x 5 people x 1 week = 15 weeks <u>Total 35 weeks over 12 months</u>	Inputs received from IADB on progress made on sewerage network construction. Technical advice from specialists related to the technical assistance.
12-48 months	Supervising verification processes. Follow up on lessons learned from verification and possible adjustments in the verification protocol. Supervising Technical Assistance, follow up with PAP and continuous support for use of M&E tools. Preparation of MTR	Legal; fiduciary; engineering/procurement; environmental and social; water and sanitation monitoring and evaluation; technical (water, sanitation, economics and finance)	2 implementation support missions plus 2 short/technical visits <u>per year</u> 2 x 10 people x 1 week = 20 weeks Plus 2 x 5 people x 1 week = 10 weeks <u>Total 90 weeks over 36 months</u>	Inputs received from IADB on progress made on sewerage network construction. Technical advice from specialists related to technical assistance
Other	Independent audit/assessment of verification of results	Independent technical expertise	2 people for 4 weeks <u>Total 8 weeks</u>	Independent Observers

Task Team Skills Mix Requirements for Implementation Support

Skills Needed	Number of Staff Weeks	Number of Trips	Comments
Legal	10	NA	
Fiduciary	20	NA	
Social	10	NA	
Environment	10	NA	
M&E	20	10	
Water Supply	15	10	
Sanitation	15	10	
Economics and finance	6	NA	

Role of Partners in Program Implementation

Name	Institution/Country	Role
IADB	Regional LAC	Financier of sewerage networks



ANNEX 8. ADJUSTMENTS TO THE COUNTRY PROGRAM IN RESPONSE TO COVID-19

1. This annex presents adjustments made to the World Bank Group's (WBG's) program in response to the COVID-19 crisis endorsed by the government of Argentina. They follow the four pillars of the WBG's COVID 19 Crisis Response Approach Paper: (i) emergency support to health interventions for saving lives; (ii) social response for protecting poor and vulnerable people; (iii) economic response for saving livelihoods, preserving jobs, and ensuring more sustainable business growth and job creation; and (iv) support for strengthening policies, institutions and investments for sustainable recovery.

Impact of the COVID-19 pandemic on the country and government response

2. **Despite its adoption of strict quarantine measures early on, Argentina ranks among the top ten countries with COVID-19 cases.** Although the GoA moved swiftly in imposing a mandatory quarantine to contain the COVID spread, following more than six months of varying degrees of lockdown in the Buenos Aires City and Province, confinement measures were recently relaxed despite rapid spread, especially in the interior. With almost 1.9 million reported cases of infections and over 45,000 deaths, Argentina now posts one of the highest contagion rates in the world, and COVID cases are spreading to other provinces which have limited hospital capacity.

3. **The economic crisis, worsened by the lockdown, is having deep impacts on employment, incomes and poverty.** Argentina's economy was already in a fragile situation when the pandemic hit, entering into its third year of economic contraction, facing high unemployment and more than a third of workers in the informal sector. The lockdown led to a collapse in domestic demand and resulted in a rapid decline in employment and increase in poverty rates. Based on official statistics employment dropped from 44.2 to 33.4 percent between the first and second quarters of 2020, and partially recovered to 37.4 in the third quarter, as lockdown measures were gradually eased. Following a similar trend, the share of active population in the labor market declined from 47.1 to 38.4 percent and then increased to 42.3 percent in the same period, while unemployment climbed from 10.4 to 13.1 percent in the second quarter, declining to 11.7 percent in the third quarter. This occurred within the context of many firm closures¹⁰⁵, offsetting the effects of the emergency package that temporarily banned termination of employment and offered payroll subsidies and cheap credit to the private sector. In addition to expanding the coverage of existing social protection schemes, emergency transfer programs have been launched to reduce impacts on the most vulnerable groups and have been phased out by end-2020.

4. **Although the economy is slowly recovering as lockdown measures have been lifted and idle production capacity is restarted, the recovery pace is expected to be sluggish.** High uncertainty owing to the pandemic and macroeconomic imbalances, price and capital controls, difficulties in accessing credit and softened consumer demand will hold back new investments while the need for fiscal consolidation and lack of access to international capital markets will limit the scope for demand stimulus. After a projected contraction of 10.6 percent in real GDP in 2020 following two years of recession in 2018 and 2019, the rebound in 2021 is only projected at 4.9 percent.¹⁰⁶ The economy will remain below its potential output and not reach 2019 GDP-levels (already a low base) before 2023.

WBG support for responding to the crisis

5. **Given the financial, economic and health shocks that have rolled through Argentina since August 2019 and altered the foundational assumptions of the current CPF, the WBG team has decided to begin a PLR this fiscal**

¹⁰⁵ The estimated loss in employers registered to social security was 16,500 in January-June 2020, compared to a total loss of 11,000 during 2019.

¹⁰⁶ According to the latest World Bank projections: <http://pubdocs.worldbank.org/en/983991599838744787/Global-Economic-Prospect-Analysis-January-2021-Analysis-LAC.pdf>



year with the new government. Based on early discussions, there is likely to be a strong emphasis on economic recovery, job creation and sustainable development. Through the early months of the crisis, the Bank has pushed ahead a structured response focused on the vulnerable and a COVID response. In parallel, the IMF program is being designed in close collaboration with the Bank, and will likely create positive pressure for reforms, fiscal adjustments and expenditure efficiencies. The World Bank will provide inter alia, analytical support on targeting of social protection measures and public expenditure effectiveness; prioritization of infrastructure investments; and efficiency and management of consumer subsidies. The GoA has endorsed the WBG's proposed adjustments to its program along the four pillars defined in the WBG's COVID 19 Crisis Response Approach Paper.

6. **Support to health for saving lives.** The Bank provided Argentina with \$35m COVID emergency support as part of the first Batch of the Global MPA Program as well as support through Bank-facilitated procurement. The project focuses on strengthening health systems to cope with the pandemic and financing necessary medical supplies and equipment. The Electricity Emergency Response Social Tariff Transition in the Province of Buenos Aires Project is being restructured to include an emergency response for food program to the poorest. Similarly, the Argentina Universal Health Coverage Program has just been restructured to expedite disbursements and provide more liquidity to the system; an Additional Financing of \$250m is being prepared to cover financing gaps. The government has also requested \$300m in Bank financing for its maternity and early childhood program, "*Plan 1000 Dias*."

7. **Social response for protecting poor and vulnerable people.** To expand protection of the poor from the worsening crisis, the Bank team prepared a \$300m Additional Financing for the Universal Childhood Assistance Project to expand the reach of social protection of vulnerable and unemployed families with children which was approved by the Board in March 2019. The team has processed a \$44m restructuring and extension of the Plan Belgrano Water Supply and Sanitation Project to bring safe water to vulnerable *Wichi* indigenous communities in the north, an area affected by a Dengue epidemic followed by the COVID-19 pandemic. The present \$300m Program for Results will expand water, sanitation and hygiene in vulnerable urban and peri-urban communities. Additional financing to expand health and social protection services for the poor is being considered.

8. **Economic response for saving livelihoods, preserving jobs, and ensuring more sustainable business growth and job creation.** For FY21 the Bank is preparing a \$320 million urban railway project in the Buenos Aires Metropolitan Area to provide access to jobs in the COVID-19 Recovery Phase, with strong focus on hygiene and viability of urban transit systems. A Level 1 Restructuring of the Salado Integrated River Basin Management Project is being concluded to provide access to improved housing and local jobs in vulnerable communities in this metropolitan area. A joint Poverty and Infrastructure team is assessing potential job impacts—direct and indirect—from different forms of investment to help inform the government about the relative potential of those urgent public investments to stimulate demand.

9. **Support for strengthening policies, institutions and investments for resilient and sustainable recovery.** The Bank will continue reviewing current public expenditure systems and support improvements through automatization and digital management systems, focusing on improving transparency, efficiency and targeting of social programs. For FY21, the Bank is also preparing an \$80m Additional Financing for the Public Sector Modernization Project with a focus on digital solutions for health and social protection systems and to ensure business continuity at times of crisis. This project will have an important component to support digital inclusion of remote and less connected regions.

Selectivity, Complementarity, Partnerships

10. **The World Bank, IFC and MIGA work collaborative closely in Argentina.** IFC participate in critical meetings on business environment, regulation, contract integrity, taxation, and policies to influence credit availability for enterprises have been conveyed jointly to officials. The World Bank actively supported MIGA's extension of coverage to Banco Santander to protect mandated dollar reserves at the Central Bank. The IFC Manager participates in



management team meetings and staff have reviewed all relevant lending programs under design, such as the SME liquidity facility. The government closely coordinates and prioritizes support to different IFIs in different sectors and regions in Argentina.